

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2021

Analysis Performed by : #

Job File : ...\\172-C.pvdb

Date of Analysis : Apr 21,2026 9:42am

PV Elite 25 SP2, October 2023



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FileName : 172-C

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Vessel Design Summary:

Step: 32 9:43am Apr 21,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2021

Diameter Spec : 1100.000 mm ID
Vessel Design Length, Tangent to Tangent 5897.85 mm
Specified Datum Line Distance 0.00 mm
Shell Side Design Temperature 454 °C
Channel Side Design Temperature 358 °C
Shell Side Design Pressure 4550.000 KPa.
Channel Side Design Pressure 13819.999 KPa.
Shop Shell Side Test Pressure 6268.901 KPa.
Field Shell Side Test Pressure 5915.000 KPa.
Shop Channel Side Test Pressure 18908.748 KPa.
Field Channel Side Test Pressure 17965.998 KPa.
Wind Design Code ASCE/SEI 7-16
Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Normal ized	Impact Tested
Shell	SA-387 11	2	...	K11789	No	No
Shell	SA-387 22	...	...	K21590	No	No
Head	SA-387 22	2	...	K21590	No	No
Flange	SA-336 F11	3	...	K11572	No	No
Nozzle	SA-336 F11	3	...	K11572	No	No
Nozzle	SA-336 F22	...	...	K21590	No	No
Nozzle Flg	SA-182 F11	2	...	K11572	No	No
Nozzle Flg	SA-105	...	...	K03504	No	No
Tubes	SA-213 T22	...	...	K21590	No	No
Tubesheet	SA-336 F22	3	...	K21590	No	No
Flg Bolting	SA-193 B7	...	64 < t <= 100	G41400	No	No
Hrz Bolting	SA-307 A	...	6 <= t <= 100		No	No
Saddles	SA-387 22	2	...	K21590	No	No

- Normalized is determined based on the UCS-66 material curve selection and Figure UCS-66.
- Impact Tested is based on material selection and material data properties.

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
CH COVER	13819.999	101.30	14545.000	3.0000	N/A	No
CH Flange	13827.008	101.30	14674.000	3.0000	N/A	No
Channel shell	13827.026	101.30	15304.000	3.0000	N/A	No
Shell	4550.108	0.00	5624.000	3.0000	N/A	No
Shell head	4550.099	0.00	4822.000	3.0000	No	No

Liquid Level: 1100.00 mm Dens.: 0.000 kg./cm³ Sp. Gr.: 0.010

Element Types and Properties:

Element	"To" Elev	Element	Nominal	Finished	Reqd Thk	Reqd Thk	Long	Circ
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Vessel Design Summary:

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Type	mm	Length mm	Thickness mm	Thickness mm	Internal mm	External mm	Eff	Eff
Body Flg	277.7	270.0	278.0	270.0	263.3	...	1.00	1.00
Body Flg	641.8	361.0	273.0	265.0	260.3	...	1.00	1.00
Cylinder	1641.8	1000.0	64.0	64.0	57.7	...	1.00	1.00
Cylinder	5847.8	4000.0	28.0	28.0	23.1	...	1.00	1.00
Ellipse	5897.8	50.0	28.0	23.8	22.6	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	270	201.828	101.3	No Calc
20	30	265	172.872	101.3	No Calc
30	Tsht	64	6.63544	101.3	7852
Tsht	40	64	No Calc	101.3	No Calc
40	50	28	No Calc	...	2347
50	60	23.8	No Calc	...	1945

External Pressure Calculations:

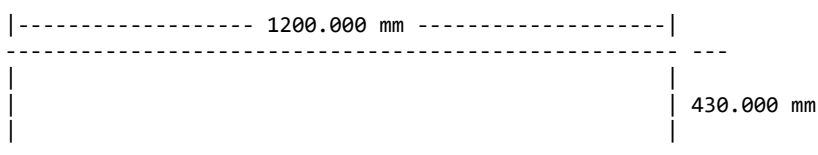
From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	999.97	70421.2	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	50	4141.7	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Saddle Parameters:

Saddle Width 203.000 mm
 Saddle Bearing Angle 120.000 deg.
 Centerline Dimension 1070.000 mm
 Wear Pad Width 250.000 mm
 Wear Pad Thickness 15.000 mm
 Wear Pad Bearing Angle 132.000 deg.
 Distance from Saddle to Tangent 300.000 mm

Baseplate Length 1200.000 mm
 Baseplate Thickness 20.000 mm
 Baseplate Width 430.000 mm
 Number of Ribs (including outside ribs) 3
 Rib Thickness 10.000 mm
 Web Thickness 10.000 mm
 Height of Center Web 306.000 mm
 Number of Bolts in Baseplate 4

Baseplate Sketch



Baseplate Plan View



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Vessel Design Summary:

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| | 20.000 mm

Baseplate Side View

Maximum Tensile Bolt Load 91. Kgf

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load 27430.25 Kgf
Maximum Transverse Saddle Shear Load 702.00 Kgf
Maximum Longitudinal Saddle Shear Load 1404.00 Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load 26491.83 Kgf
Maximum Transverse Saddle Shear Load 108.62 Kgf
Maximum Longitudinal Saddle Shear Load 103.02 Kgf

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N3- Tube inlet	WRC-297	0.892	Shell	Passed
N4- Tube outle	WRC-297	0.892	Shell	Passed
Shell inlet-N1	WRC-297	0.950	Shell	Passed
Shell outlet-	WRC-297	0.950	Shell	Passed

Weights:

Fabricated - Bare W/O Removable Internals 21162.0 kg.
Shop Test - Fabricated + Water (Full) 26650.3 kg.
Shipping - Fab. + Rem. Intls.+ Shipping App. 21162.0 kg.
Erected - Fab. + Rem. Intls.+ Insul. (etc) 23138.5 kg.
Empty - Fab. + Intls. + Details + Wghts. 23138.5 kg.
Operating - Empty + Operating Liquid (No CA) 24396.9 kg.
Field Test - Empty Weight + Water (Full) 28126.8 kg.

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Nozzle Summary:

Step: 30 9:43am Apr 21,2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N3- Tube inlet	15137	OK	...	OK	7.80	OK	Passed
N4- Tube outlet	15130	OK	...	OK	7.80	OK	Passed
D1	15304	...	...	OK	6.42	OK	No Calc[*]
Shell inlet-N1	5292	...	...	OK	16.97	OK	Passed
Shell outlet- N	5292	...	...	OK	16.97	OK	Passed
V1	5624	...	...	OK	8.02	OK	No Calc[*]
D2	5624	...	...	OK	8.02	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 5292 Nozzle : Shell inlet-N1 [Shellside]
 Minimum MAWP Nozzles : 15130 Nozzle : N4- Tube outlet [Tubeside]

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the Spatial Relationship between the Nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N3- Tube inlet	941.850	0.000	210.900
30	N4- Tube outlet	941.850	180.000	210.900
30	D1	1341.850	180.000	182.325
40	Shell inlet-N1	3747.850	180.000	1536.000
40	Shell outlet- N	3647.850	0.000	1536.000
40	V1	2347.850	0.000	148.176
40	D2	5047.850	180.000	148.176

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Step: 29 9:43am Apr 21,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
D1	2.000 in	160	WNF	2.375	8.738	...	...	317.44	1500
V1	2.000 in	XXS	WNF	2.375	11.074	...	...	280.19	900
D2	2.000 in	XXS	WNF	2.375	11.074	...	...	280.19	900
N3- Tube inlet	3.000 in	160	WNF	3.500	11.125	...	...	318.91	1500
N4- Tube outlet	3.000 in	160	WNF	3.500	11.125	...	...	318.91	1500
Shell inlet-N1	30.000 in Actual	None	None	32.520	32.000	...	...	626.58	...
Shell outlet- N	30.000 in Actual	None	None	31.969	25.000	...	...	626.58	...

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
D1	SA-336 F11	60.000	15.000	...	...	...
V1	SA-336 F22	25.000	20.000	...	...	...
D2	SA-336 F22	25.000	20.000	...	...	...
N3- Tube in	SA-336 F11	60.000	15.000	...	...	...
N4- Tube ou	SA-336 F11	56.000	15.000	...	...	...
Shell inlet	SA-336 F22	25.000	20.000	...	...	...
Shell outle	SA-336 F22	25.000	20.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
D1	1341.850	180.0	250.00	0.00	Channel shell
V1	2347.850	0.0	250.00	0.00	Shell
D2	5047.850	180.0	250.00	0.00	Shell
N3- Tube inlet	941.850	0.0	250.00	0.00	Channel shell
N4- Tube outlet	941.850	180.0	250.00	0.00	Channel shell
Shell inlet-N1	3747.850	180.0	250.00	0.00	Shell
Shell outlet- N	3647.850	0.0	250.00	0.00	Shell

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Bill of Material:

Step: 28 9:43am Apr 21,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BODY FLANGE: 278.0mm THK X 1655.0mm OD	SA-336 F11
1	BODY FLANGE: 273.0mm THK X 1655.0mm OD	SA-336 F11
1	ELLIPTICAL HEAD: 2.0 X 1, 28.0mm THK X 1100.0mm ID X 50.0mm	SA-387 22
1	INSULATION: 275mm X 150mm THK	Hot
1	INSULATION: 350mm X 150mm THK	Hot
1	INSULATION: 1000mm X 150mm THK	Hot
1	BAFFLES: 991.0mm DIA X 38.0mm THK	SA-387 22
8	TIE RODS: 14mm DIA X 3499mm	SA-193 B7M
1	TUBESHEET: 1288.0mm X 206.0mm THK	SA-336 F22
750	TUBES: 3650.0mm X 19.0mm DIA X 2.1mm THK	SA-213 T22
2	SADDLE: 203.0mm X 120 DEG	SA-387 22
1	INSULATION: 4000mm X 150mm THK	Hot
1	INSULATION: 300mm X 150mm THK	Hot
1	GASKET: 1180.0mm OD X 1132.0mm ID	...
24	BODY FLANGE BOLTS: 88.9mm DIA	SA-193 B7
48	NUTS FOR BODY FLANGE BOLTS: 88.9mm DIA	...
2	CYLINDRICAL SEGMENT 64.0mm THK X 1372.7mm X 1000.0mm	SA-387 11
1	CYLINDRICAL SEGMENT 64.0mm THK X 710.3mm X 1000.0mm	SA-387 11
2	CYLINDRICAL SEGMENT 28.0mm THK X 1330.5mm X 4000.0mm	SA-387 22
1	CYLINDRICAL SEGMENT 28.0mm THK X 794.8mm X 4000.0mm	SA-387 22
1	NAMEPLATE	...



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MDMT Summary:

Step: 31 9:43am Apr 21,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Shell	[8]	A	22	7		0.735	28.000	1.00	Yes
Shell head	[10]	A	17	10		0.863	23.800	1.00	Yes
Shell head	[7]	A	22	7		0.735	28.000	1.00	Yes
Left Saddle	[24]	A	4	-11		0.735	15.000	1.00	Yes
Right saddle	[24]	A	4	-11		0.735	15.000	1.00	Yes
Shell inlet-N1	[1]	A	24	4		0.642	30.000	1.00	No
Shell outlet-	[1]	A	24	4		0.642	30.000	1.00	No
V1	[1]	A	22	5		0.693	28.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-104					
D2	[1]	A	22	5		0.693	28.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-104					
Nozl. Bolting	[22]		-48			1.000	64.000	1.00	No
Tubesheet: SS	[13]	A	37	-40		0.351	51.500	1.00	No
Warmest MDMT:			37	10					
CH COVER	[11]	B	23	22		0.978	67.500	1.00	No
CH Flange	[11]	B	22	18		0.932	64.000	1.00	Yes
Channel shell	[8]	A	41	38		0.947	64.000	1.00	Yes
N3- Tube inlet	[1]	A	-29	-48		0.534	9.735	1.00	Yes
Nozzle Flg	[5]	B	-18	-46					
N4- Tube outle	[1]	A	-29	-48		0.534	9.735	1.00	Yes
Nozzle Flg	[5]	B	-18	-46					
D1	[1]	A	-29	-48		0.520	7.645	1.00	Yes
Nozzle Flg	[5]	B	-18	-46					
Nozl. Bolting	[22]		-48			1.000	64.000	1.00	No
Tubesheet: CS	[14]	A	41	31		0.820	64.000	1.00	No
Bolting	[21]		-40						
Warmest MDMT:			41	38					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	10.0	-12.0	Fail
Channel/Tube	38.0	-12.0	Fail

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material



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MDMT Summary:

Step: 31 9:43am Apr 21,2026

[23] - Stiffening Ring to Shell Weld

[24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Warnings and Errors:

Step: 0 9:42am Apr 21,2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Warning:

There is a flange in this model connected to an element whose temperature is different. The bolt allowable stress is based on it's parent design temperature. You may need to manually adjust the operating bolt allowable stress to a value based upon a representative maximum temperature the bolts will be subject to.

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.

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Input Echo:

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Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	4550	KPa.
Channel Side Design Pressure	13820	KPa.
Shell Side Design Temperature	454.0	°C
Channel Side Design Temperature	358.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Sour	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No



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Input Echo:

Step: 1 9:42am Apr 21,2026

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	97.2	Km/hr
Surface Roughness Category	C: Open Terrain	
Type of Surface	Moderately Smooth	
Base Elevation	16000	mm
Percent Wind for Hydrotest	50.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	0	mm
Distance Upwind of Crest	0	mm
Distance from Crest to the Vessel	0	mm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm

Seismic Design Code	ASCE/SEI 7-16	
Seismic Load Reduction Scale Factor	1.000	
Importance Factor	1.250	
Table Value Fa	1.600	
Table Value Fv	2.400	
Max. Mapped Res. Acceleration	0.147	[Ss]
Max. Eff. Ground Acceleration	0.065	[S]
Force Modification Factor R	3.000	
Site Class	D	
Component Elevation Ratio	0.000	z/h
Amplification Factor	1.000	Ap
Force Factor	0.000	
Consider Vertical Acceleration	Yes	
Minimum Acceleration Multiplier	0.000	
User Value of Sds (used if > 0)	0.000	
User Value of Sd1 (used if > 0)	0.000	
Moment Reduction Factor Tau	1.000	

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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Input Echo:

Step: 1 9:42am Apr 21,2026

Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios ≤ 0.35 , MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be $< -55F (-48C)$	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element From Node	10
Element To Node	20
Element Type	Flange
Description	CH COVER
Distance "FROM" to "TO"	270 mm
Flange Outside Diameter	1655 mm
Element Thickness	270 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	278 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	13820 KPa.
Design Temperature Internal Pressure	358 °C
Design External Pressure	101.3 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-336 F11
Allowable Stress, Ambient	148 N./mm ²
Allowable Stress, Operating	148 N./mm ²
Allowable Stress, Hydrotest	278.99 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	16.002 mm
Yield Stress, Operating	236.72 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-3
UNS Number	K11572
Class / Thickness / Grade	3
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	10
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	275 mm
Thickness of Insulation	150 mm
Density	0.000352 kg./cm ³

Element From Node	20
Element To Node	30



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Input Echo:

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Element Type	Flange
Description	CH Flange
Distance "FROM" to "TO"	361 mm
Flange Inside Diameter	1100 mm
Element Thickness	265 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	273 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	13820 KPa.
Design Temperature Internal Pressure	358 °C
Design External Pressure	101.3 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-336 F11
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1100 mm
Liquid Density	0.000649 kg./cm ³

Element From Node	20
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	350 mm
Thickness of Insulation	150 mm
Density	0.000352 kg./cm ³

Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	Channel shell
Distance "FROM" to "TO"	1000 mm
Inside Diameter	1100 mm
Element Thickness	64 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	64 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	13820 KPa.
Design Temperature Internal Pressure	400 °C
Design External Pressure	101.3 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11
Allowable Stress, Ambient	148 N./mm ²
Allowable Stress, Operating	148 N./mm ²
Allowable Stress, Hydrotest	278.99 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	16.002 mm
Yield Stress, Operating	229 N./mm ²
UCS-66 Chart Curve Designation	A
External Pressure Chart Name	CS-3
UNS Number	K11789
Class / Thickness / Grade	2
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No



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Input Echo:

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Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1100 mm
Liquid Density	0.000649 kg./cm ³

Element From Node	30
Detail Type	Insulation
Detail ID	Ins: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1000 mm
Thickness of Insulation	150 mm
Density	0.000352 kg./cm ³

Element From Node	30
Detail Type	Nozzle
Detail ID	N3- Tube inlet
Dist. from "FROM" Node / Offset dist	300 mm
Nozzle Diameter	3 in.
Nozzle Schedule	160
Nozzle Class	1500
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	56.624 Kg
Grade of Attached Flange	GR 1.9
Nozzle Matl	SA-336 F11

Element From Node	30
Detail Type	Nozzle
Detail ID	N4- Tube outlet
Dist. from "FROM" Node / Offset dist	300 mm
Nozzle Diameter	3 in.
Nozzle Schedule	160
Nozzle Class	1500
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	56.624 Kg
Grade of Attached Flange	GR 1.9
Nozzle Matl	SA-336 F11

Element From Node	30
Detail Type	Nozzle
Detail ID	D1
Dist. from "FROM" Node / Offset dist	700 mm
Nozzle Diameter	2 in.
Nozzle Schedule	160
Nozzle Class	1500
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	30.177 Kg
Grade of Attached Flange	GR 1.9
Nozzle Matl	SA-336 F11

Element From Node	30
Detail Type	Weight
Detail ID	Pass Partition
Dist. from "FROM" Node / Offset dist	500 mm
Miscellaneous Weight	250 Kg
Offset from Element Centerline	0 mm

Element From Node	40
Element To Node	50



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Input Echo:

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Element Type	Cylinder
Description	Shell
Distance "FROM" to "TO"	4000 mm
Inside Diameter	1100 mm
Element Thickness	28 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	28 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4550 KPa.
Design Temperature Internal Pressure	454 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 22
Allowable Stress, Ambient	148 N./mm ²
Allowable Stress, Operating	127.78 N./mm ²
Allowable Stress, Hydrotest	278.99 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	0 mm
Yield Stress, Operating	230.89 N./mm ²
UCS-66 Chart Curve Designation	A
External Pressure Chart Name	CS-3
UNS Number	K21590
Class / Thickness / Grade	2
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	300 mm
Width of Saddle	203 mm
Height of Saddle at Bottom	1070 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	250 mm
Thickness of Wear Plate	15 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.40000001
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm

Element From Node	40
Detail Type	Saddle
Detail ID	Right saddle
Dist. from "FROM" Node / Offset dist	3550 mm
Width of Saddle	203 mm
Height of Saddle at Bottom	1170 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	250 mm
Thickness of Wear Plate	15 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.40000001
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm



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Input Echo:

Step: 1 9:42am Apr 21,2026

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1100 mm
Liquid Density	0.99840E-05 kg./cm ³

Element From Node	40
Detail Type	Insulation
Detail ID	Ins: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	4000 mm
Thickness of Insulation	150 mm
Density	0.000352 kg./cm ³

Element From Node	40
Detail Type	Nozzle
Detail ID	Shell inlet-N1
Dist. from "FROM" Node / Offset dist	1900 mm
Nozzle Diameter	30 in.
Nozzle Schedule	None
Nozzle Class	None
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	437.99 Kgf
Grade of Attached Flange	None
Nozzle Matl	SA-336 F22

Element From Node	40
Detail Type	Nozzle
Detail ID	Shell outlet- N
Dist. from "FROM" Node / Offset dist	1800 mm
Nozzle Diameter	30 in.
Nozzle Schedule	None
Nozzle Class	None
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	427.9 Kgf
Grade of Attached Flange	None
Nozzle Matl	SA-336 F22

Element From Node	40
Detail Type	Nozzle
Detail ID	V1
Dist. from "FROM" Node / Offset dist	500 mm
Nozzle Diameter	2 in.
Nozzle Schedule	XXS
Nozzle Class	900
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	20.341 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-336 F22

Element From Node	40
Detail Type	Nozzle
Detail ID	D2
Dist. from "FROM" Node / Offset dist	3200 mm
Nozzle Diameter	2 in.
Nozzle Schedule	XXS
Nozzle Class	900
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	GR 1.1



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Input Echo:

Step: 1 9:42am Apr 21,2026

Nozzle Matl	SA-336 F22
Element From Node	40
Detail Type	Weight
Detail ID	Fire proofing
Dist. from "FROM" Node / Offset dist	300 mm
Miscellaneous Weight	250 Kg
Offset from Element Centerline	1000 mm
Element From Node	40
Detail Type	Weight
Detail ID	Fire proofing
Dist. from "FROM" Node / Offset dist	3550 mm
Miscellaneous Weight	250 Kg
Offset from Element Centerline	1000 mm

Element From Node	50
Element To Node	60
Element Type	Elliptical
Description	Shell head
Distance "FROM" to "TO"	50 mm
Inside Diameter	1100 mm
Element Thickness	23.8 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	28 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4550 KPa.
Design Temperature Internal Pressure	454 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 22
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1000 mm
Liquid Density	0.99840E-05 kg./cm ³

Element From Node	50
Detail Type	Insulation
Detail ID	Ins: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	300 mm
Thickness of Insulation	150 mm
Density	0.000352 kg./cm ³



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XY Coordinate Calculations:

Step: 2 9:42am Apr 21,2026

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
CH COVER		277.675	...	270	...
CH Flange		641.85	...	361	...
Channel shell		1641.85	...	1000	...
Shell		5847.85	...	4000	...
Shell head		5897.85	...	50	...

PV Elite includes an 1/8 inch (3.175mm) raised face and gasket thicknesses for girth flanges and tubesheet thicknesses where applicable in the Tangent to Tangent length calculation. The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact.

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Flg Calc [Int P]: New Flange

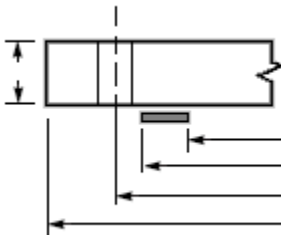
Flng: 5 9:42am Apr 21,2026

Flange Input Data Values

Description: New Flange :

CH COVER

Flange Type	TEMA Channel Cover		
Design Pressure	P	13820.00	KPa.
Design Temperature		358	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	1655.000	mm
Flange Thickness	t	270.0000	mm
Flange Material	SA-336 F11		
Flange Material UNS number	K11572		
Flange Allowable Stress At Temperature	Sfo	148.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	148.00	N./mm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	158.68	N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00	N./mm ²
Perimeter along the Center of the Bolts	L	4633.849	mm
Youngs Modulus for Blind Flange Matl.	Sy	181790.67	N./mm ²
Allowed Channel Cover Deflection	dfAllow	1.3750	mm
Diameter of Bolt Circle	C	1475.000	mm
Nominal Bolt Diameter	a	88.9000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	24		
Flange Face Outside Diameter	Fod	1193.000	mm
Flange Face Inside Diameter	Fid	1100.000	mm
Flange Facing Sketch	1, Code Sketch 1a		
Gasket Outside Diameter	Go	1180.000	mm
Gasket Inside Diameter	Gi	1132.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Column for Gasket Seating	2, Code Column II		
Gasket Thickness	tg	4.5000	mm



ASME Code, Section VIII Division 1, 2021

Gasket Contact Width,	$N = (Go - Gi)/2$	24.000 mm
Basic Gasket Width,	$bo = N/2$	12.000 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	8.729 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	1162.541 mm



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Flg Calc [Int P]: New Flange

Flng: 5 9:42am Apr 21, 2026

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H]:

$$= \pi/4 \cdot G^2 \cdot P_{eq}$$

$$= \pi/4 \cdot 1163^2 \cdot 13820$$

$$= 1495930.125 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.73 \cdot \pi \cdot 1163 \cdot 3 \cdot 13820$$

$$= 269582.406 \text{ Kg}$$

Operating Bolt Load [Wm1]:

$$= \max(H + H_p + H'p, 0)$$

$$= \max(1495930 + 269582 + 0, 0)$$

$$= 1765512.500 \text{ Kg}$$

Gasket Seating Bolt Load [Wm2]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 8.73 \cdot \pi \cdot 1163 + 0 \cdot 0 \cdot 0$$

$$= 224151.312 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max(W_{m1}/S_b, W_{m2}/S_a)$$

$$= \max(1765512/159, 224151/159)$$

$$= 1091.143 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts per App. 2 eq. (3) [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 88.9 + 6 \cdot 267/(3 + 0.5)$$

$$= 635.514 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1475 \cdot \sin(\pi / 24)$$

$$= 192.526 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per App. 2 eq. (7) [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 193 / (2 \cdot 88.9 + 267) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1091.143	1354.681	
Radial Distance between Bolts and Edge:	82.550	90.000	
Circ. Spacing between the Bolts:	180.975	192.526	635.514

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 1355 \cdot 159 / (68.9 \cdot \pi (1180 + 1132))$$

$$= 43.009 \text{ mm} [\text{Note: Exceeds actual gasket width, 24.000 }]$$

Flange Design Bolt Load, Gasket Seating [W]:

$$= S_a (A_m + A_b) / 2$$

$$= 159 (1091 + 1355) / 2$$

$$= 1982694.88 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= W_{m1}$$

$$= 1765512.50 \text{ Kg}$$



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Flg Calc [Int P]: New Flange

Flng: 5 9:42am Apr 21,2026

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1475 - 1163) / 2$$

$$= 156.2293 \text{ mm}$$

Tangential Flange Stress, Flat Head (UG-34), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 1765512 \cdot 156 \cdot 1 / (267^2 \cdot 1163) + 0.3 \cdot 1 \cdot 13820 \cdot 1163^2 / 267^2$$

$$= 140.62 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (UG-34), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 1982694 \cdot 156 \cdot 1 / (267^2 \cdot 1163)$$

$$= 69.64 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 1765512 / 1355$$

$$= 127.81 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (224151 / 1355)$$

$$= 16.23 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	140.62	148.00	69.64	148.00
Bolting	127.81	158.68	16.23	159.00

TEMA Channel Cover Deflection Calculations:

Actual deflection at center per TEMA formula 0.8815 mm

Allowed deflection at center of Cover 1.3750 mm

Reqd. Blind Flange Thickness at Center 263.258 mm

Reqd. Blind Flange Thickness at Gasket 186.155 mm

Min. Required Flange Thickness per TEMA 230.224 mm

Estimated M.A.W.P. (Operating) 14545 KPa.

Estimated Finished Weight of Flange at given Thk. 4501.7 kg.

Estimated Unfinished Weight of Forging at given Thk 4501.7 kg.

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.98, Temperature Reduction per Fig. UCS 66.1 = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 23 °C

Min Metal Temp. at Required thickness (UCS 66.1) 22 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

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Flg Calc [Int P]: CH Flange

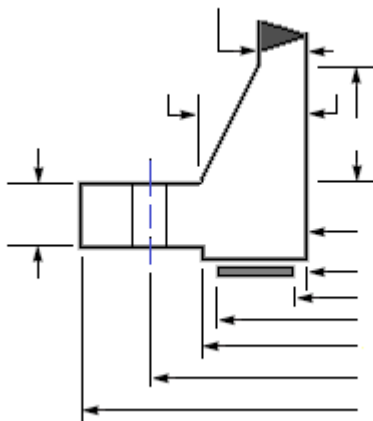
Flng: 6 9:42am Apr 21,2026

Flange Input Data Values

Description: CH Flange :

CH Flange

Flange Type	Integral Weld Neck		
Design Pressure	P	13827.01	KPa.
Design Temperature		358	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	1100.000	mm
Flange Outside Diameter	A	1655.000	mm
Flange Thickness	t	265.0000	mm
Thickness of Hub at Small End	go	64.0000	mm
Thickness of Hub at Large End	gl	80.0000	mm
Length of Hub	h	96.0000	mm
Flange Material	SA-336 F11		
Flange Material UNS number	K11572		
Flange Allowable Stress At Temperature	Sfo	148.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	148.00	N./mm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	158.68	N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00	N./mm ²
Diameter of Bolt Circle	C	1475.000	mm
Nominal Bolt Diameter	a	88.9000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	24		
Flange Face Outside Diameter	Fod	1193.000	mm
Flange Face Inside Diameter	Fid	1100.000	mm
Flange Facing Sketch	1, Code Sketch 1a		
Gasket Outside Diameter	Go	1180.000	mm
Gasket Inside Diameter	Gi	1132.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Column for Gasket Seating	2, Code Column II		
Gasket Thickness	tg	4.5000	mm





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Flg Calc [Int P]: CH Flange

Flng: 6 9:42am Apr 21,2026

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (13827 \cdot (1100/2 + 3)) / (148 \cdot 1 - 0.6 \cdot 13827) + Ca$$

$$= 57.7369 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (148 \cdot 1 \cdot 61) / (553 + 0.6 \cdot 61)$$

$$= 15310.910 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (148 \cdot 1 \cdot 64) / (550 + 0.6 \cdot 64)$$

$$= 16096.667 \text{ KPa.}$$

Corroded Flange ID,	Bcor = B+2*ci	1106.000 mm
Corroded Large Hub,	g1Cor = g1-ci	77.000 mm
Corroded Small Hub,	g0Cor = go-ci	61.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	107.500 mm

Gasket Contact Width,	N = (Go - Gi)/2	24.000 mm
Basic Gasket Width,	bo = N/2	12.000 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	8.729 mm
Gasket Reaction Diameter,	G = Go - 2 * b	1162.541 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H]:

$$= \pi/4 \cdot G^2 \cdot Peq$$

$$= \pi/4 \cdot 1163^2 \cdot 13827$$

$$= 1496688.750 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.73 \cdot \pi \cdot 1163 \cdot 3 \cdot 13827$$

$$= 269719.125 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 1106^2 \cdot 13827$$

$$= 1354643.000 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 1496688 - 1354643$$

$$= 142045.750 \text{ Kg}$$

Operating Bolt Load [Wm1]:

$$= \max(H + Hp + H'p, 0)$$

$$= \max(1496688 + 269719 + 0, 0)$$

$$= 1766408.000 \text{ Kg}$$

Gasket Seating Bolt Load [Wm2]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.9 \cdot 8.73 \cdot \pi \cdot 1163 + 0 \cdot 0$$

$$= 224151.312 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max(Wm1/Sb, Wm2/Sa)$$

$$= \max(1766408/159, 224151/159)$$

$$= 1091.697 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts per App. 2 eq. (3) [Bsmx]:



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Flg Calc [Int P]: CH Flange

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$$\begin{aligned}
 &= 2a + 6t/(m + 0.5) \\
 &= 2 \cdot 88.9 + 6 \cdot 265/(3 + 0.5) \\
 &= 632.086 \text{ mm}
 \end{aligned}$$

Actual Circumferential Bolt Spacing [Bs]:

$$\begin{aligned}
 &= C \cdot \sin(\pi / n) \\
 &= 1475 \cdot \sin(\pi / 24) \\
 &= 192.526 \text{ mm}
 \end{aligned}$$

ASME Moment Multiplier for Bolt Spacing per App. 2 eq. (7) [Bsc]:

$$\begin{aligned}
 &= \max(\sqrt{(Bs/(2a + t))}, 1) \\
 &= \max(\sqrt{(193/(2 \cdot 88.9 + 265))}, 1) \\
 &= 1.0000
 \end{aligned}$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1091.697	1354.681	
Radial Distance between Hub and Bolts:	104.775	107.500	
Radial Distance between Bolts and Edge:	82.550	90.000	
Circ. Spacing between the Bolts:	180.975	192.526	632.086

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned}
 &= Ab \cdot Sa / (y \cdot \pi(Go + Gi)) \\
 &= 1355 \cdot 159 / (68.9 \cdot \pi(1180 + 1132)) \\
 &= 43.009 \text{ mm} [\text{Note: Exceeds actual gasket width, 24.000}]
 \end{aligned}$$

Note: Recommended Min. Width for Sheet and Composite Gaskets per table 2-4 :

$$= 32.000 \text{ mm} [\text{Note: Exceeds actual gasket width, 24.000}]$$

Flange Design Bolt Load, Gasket Seating [W]:

$$\begin{aligned}
 &= Sa(Am + Ab) / 2 \\
 &= 159(1092 + 1355) / 2 \\
 &= 1983143.50 \text{ Kg}
 \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned}
 &= Wm1 - H \\
 &= 1766408 - 1496688 \\
 &= 269719.19 \text{ Kg}
 \end{aligned}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$\begin{aligned}
 &= (C - G) / 2 \\
 &= (1475 - 1163) / 2 \\
 &= 156.2293 \text{ mm}
 \end{aligned}$$

Distance to Face Pressure Reaction [ht]:

$$\begin{aligned}
 &= (R + g1 + hg) / 2 \\
 &= (107 + 77 + 156) / 2 \\
 &= 170.3646 \text{ mm}
 \end{aligned}$$

Distance to End Pressure Reaction [hd]:

$$\begin{aligned}
 &= R + (g1 / 2) \\
 &= 107 + (77 / 2.0) \\
 &= 146.0000 \text{ mm}
 \end{aligned}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	1354643.	146.0000	1.0000	197782.
Face Pressure, Mt	142046.	170.3646	1.0000	24200.



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Flg Calc [Int P]: CH Flange Flng: 6 9:42am Apr 21,2026

Gasket Load, Mg	269719.	156.2293	1.0000	42139.
Gasket Seating, Matm	1983144.	156.2293	1.0000	309831.

Total Moment for Operation, Mop 264121. Kg-m.
Total Moment for Gasket seating, Matm 309831. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 259.742 mm
Hub Ratio, h/h0 = HL / H0 0.370
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.262

Flange Factors for Integral Flange:

Factor F 0.882
Factor V 0.415
Factor f 1.000
Factors from Figure 2-7.1 K = 1.496
T = 1.712 U = 5.483
Y = 4.990 Z = 2.614
e = 0.0034 1/mm d = 0.12774E+08 mm³
Stress Factors Alpha = 1.900
Beta = 2.200 Gamma = 1.110
Delta = 1.457 Lambda = 2.566

Longitudinal Hub Stress, Operating [SHo]:

$= (f \cdot Mop / Bcor) / (L \cdot g1^2)$
 $= (1 \cdot 264121 / 1106) / (2.57 \cdot 77^2)$
 $= 153.90 \text{ N./mm}^2$

Longitudinal Hub Stress, Seating [SHa]:

$= (f \cdot Matm / Bcor) / (L \cdot g1^2)$
 $= (1 \cdot 309831 / 1106) / (2.57 \cdot 77^2)$
 $= 180.54 \text{ N./mm}^2$

Radial Flange Stress, Operating [SRo]:

$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$
 $= (2.2 \cdot 264121 / 1106) / (2.57 \cdot 265^2)$
 $= 28.59 \text{ N./mm}^2$

Radial Flange Stress, Seating [SRa]:

$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$
 $= (2.2 \cdot 309831 / 1106) / (2.57 \cdot 265^2)$
 $= 33.53 \text{ N./mm}^2$

Tangential Flange Stress, Operating [STo]:

$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$
 $= (4.99 \cdot 264121 / (265^2 \cdot 1106)) - 2.61 \cdot 28.6$
 $= 91.67 \text{ N./mm}^2$

Tangential Flange Stress, Seating [STa]:

$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$
 $= (4.99 \cdot 309831 / (265^2 \cdot 1106)) - 2.61 \cdot 33.5$
 $= 107.53 \text{ N./mm}^2$

Average Flange Stress, Operating [SAo]:

$= (SHo + \max(SRo, STo)) / 2$
 $= (154 + \max(28.6, 91.7)) / 2$
 $= 122.78 \text{ N./mm}^2$

Average Flange Stress, Seating [SAa]:

$= (SHa + \max(SRa, STa)) / 2$
 $= (181 + \max(33.5, 108)) / 2$
 $= 144.03 \text{ N./mm}^2$

Bolt Stress, Operating [BSo]:

$= Wm1 / Ab$
 $= 1766408 / 1355$



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Flg Calc [Int P]: CH Flange

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= 127.87 N./mm²

Bolt Stress, Seating [BSa]:

= (Wm2 / Ab)

= (224151/1355)

= 16.23 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	153.90	222.00	180.54	222.00
Radial Flange	28.59	148.00	33.53	148.00
Tangential Flange	91.67	148.00	107.53	148.00
Maximum Average	122.78	148.00	144.03	148.00
Bolting	127.87	158.68	16.23	159.00

Minimum Required Flange Thickness 260.325 mm
 Estimated M.A.W.P. (Operating) 14681 KPa.
 Estimated Finished Weight of Flange at given Thk. 2663.7 kg.
 Estimated Unfinished Weight of Forging at given Thk 3360.0 kg.

Flange Stress results per WRC-538 at computed Sfmax (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	508.65	541.08	0.940	508.65	620.02	0.820
Tangential Flange	St	302.96	405.81	0.747	302.96	465.02	0.651
Tangential Flange	Sto	131.61	270.54	0.486	131.61	310.01	0.425
Combined	Sh/f + St	811.61	811.61	1.000	811.61	930.03	0.873
Combined	Sr + St	397.44	811.61	0.490	397.44	930.03	0.427
Combined	Sh + Sto	640.26	811.61	0.789	640.26	930.03	0.688
Radial Flange	Sr	94.48	541.08	0.175	94.48	620.02	0.152
Maximum Stress Ratio				1.000			0.873



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Flg Calc [Int P]: CH Flange

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Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1, F1num / F1den)

= max(1, 0.94/1.41)

= 1.0000

Hub Stress Correction Factor [f2]:

= max(0.25, fn2) if h/h0 < 0.35, else max (0.25, fn3)

= max(0.25, 0.36/1.43) if h/h0 < 0.35, else max (0.25, 0.2/0.77)

= 0.2587

In this case, the hub ratio h/h0, 0.3696 was ≥ 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	458.5 N./mm ²
Minimum permissible bolt stress	Sbmin	262.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	404.5 N./mm ²
Target assembly gasket stress	SgT	208.0 N./mm ²
Maximum permissible gasket stress	Sgmax	380.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700



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Flg Calc [Int P]: CH Flange

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Root Area of a Single Bolt	Abs	56.4450 cm ²
Single flange rotation at Sfmax	Thetafmax	0.320 deg
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	310.0 N./mm ²
Flange Material Operating Yield Stress	Syo	236.7 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.325 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (1180^2 - 1132^2)$$

$$= 871.6 \text{ cm}^2$$

Target Bolt Stress [Sbsel]:

$$= SGT \cdot Ag / (nb \cdot Ab) \quad (O-1)$$

$$= 208 \cdot 872 / (24 \cdot 56.4)$$

$$= 133.8 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]:

$$= \min(Sbsel, Sbmax) \quad (O-4)$$

$$= \min(134, 458)$$

$$= 133.8 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]:

$$= \max(Sbsel, Sbmin) \quad (O-5)$$

$$= \max(134, 262)$$

$$= 262.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]:

$$= \min(Sbsel, Sfmax) \quad (O-6)$$

$$= \min(262, 404)$$

$$= 262.0 \text{ N./mm}^2$$

Gasket Assembly Stress [GAS]:

$$= Sgmin - S(Ag / (nb \cdot Abs)) \quad (O-7)$$

$$= 140 (872 / (24 \cdot 56.4))$$

$$= 90.1 \text{ N./mm}^2$$

Since Sbsel >= GAS, check passes.

Gasket Operating Stress [GOS]:

$$= (Sgmin - O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb) \quad (O-8)$$

$$= (97 \cdot 872 + \pi/4 \cdot 13827 \cdot 1132^2) / (0.7 \cdot 56.4 \cdot 24)$$

$$= 235.915 \text{ N./mm}^2$$

Since Sbsel >= GOS, check passes.

Gasket Maximum Stress [GMS]:

$$= Sgmax(Ag / (nb \cdot Abs)) \quad (O-9)$$

$$= 380 (872 / (24 \cdot 56.4))$$

$$= 244.5 \text{ N./mm}^2$$

Since Sbsel > GMS, check fails.

Flange Rotation Limit [FRL]:

$$= Sfmax(\text{thetagmax} / \text{thetafmax}) \quad (O-10)$$

$$= 404 (1/0.32)$$

$$= 1264.0 \text{ N./mm}^2$$

Since Sbsel <= FRL, check passes.

Assembly Bolt Torque [Tb]:

$$= Sbsel \cdot K \cdot Abs \cdot \phi b / 12 \quad (O-2)$$



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Flg Calc [Int P]: CH Flange

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$$= 262 \cdot 0.2 \cdot 56.4 \cdot 88.9/12$$

$$= 2681.3 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 82 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 158 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange Rigidity Based on Required Thickness [ASME]:

Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 309831/1 \cdot 9807 \cdot 0.41 / (2.48 \cdot 204092 \cdot 61^2 \cdot 260 \cdot 0.3)$$

$$= 0.447 \text{ (should be } \leq 1)$$

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 264121/1 \cdot 9807 \cdot 0.41 / (2.48 \cdot 181791 \cdot 61^2 \cdot 260 \cdot 0.3)$$

$$= 0.428 \text{ (should be } \leq 1)$$

Flange Rigidity Based on Given Thickness [ASME]:

Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 309831/1 \cdot 9807 \cdot 0.41 / (2.57 \cdot 204092 \cdot 61^2 \cdot 260 \cdot 0.3)$$

$$= 0.432 \text{ (should be } \leq 1)$$

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 264121/1 \cdot 9807 \cdot 0.41 / (2.57 \cdot 181791 \cdot 61^2 \cdot 260 \cdot 0.3)$$

$$= 0.414 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.93, Temperature Reduction per Fig. UCS 66.1 = 4 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 22 °C

Min Metal Temp. at Required thickness (UCS 66.1) 18 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

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Internal Pressure Calculations: Step: 5 9:42am Apr 21,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²
CH COVER		13820	278	3	1655	148
CH Flange		13827	273	3	1100	148
Channel shell		13827	64	3	1100	148
Shell		4550.1	28	3	1100	127.78
Shell head		4550.1	28	3	1100	127.78

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm
CH COVER		13820	14545	14874	270	263.258
CH Flange		13820	14674	15606	265	260.325
Channel shell		13820	15304	16097	64	57.737
Shell		4550	5624	7311	28	23.123
Shell head		4550	4822	6376	23.8	22.6216

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 4822 KPa.
 Shell Side MAPnc = 6376 KPa.
 Channel Side MAWP = 14545 KPa.
 Channel Side MAPnc = 14874 KPa.

Elements Suitable for Design Internal Pressure.

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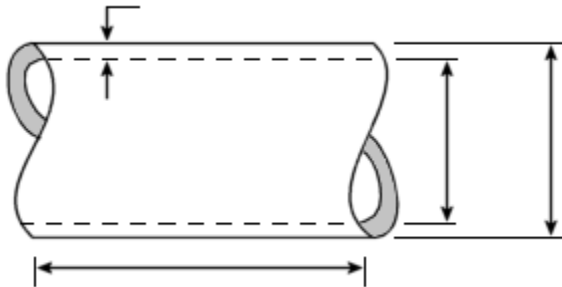
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Element: Channel shell

Step: 6 9:42am Apr 21,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Channel shell

Cylindrical Shell Design Information:			
Design Pressure P	13827.026 KPa.	Int. Design Temperature	400 °C
Material	SA-387 11	Ext. Design Temperature	200 °C
Length	1000.000 mm	External Pressure Chart	CS-3
		UNS Number	K11789
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	148.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	148.0 N./mm ²
Inside Diameter Di	1100.000 mm	Actual Stress	133.7 N./mm ²
Spec. Min. Thk t or e	64.0000 mm	Spec. Nominal Thk.	64.000 mm
Surface Area	38578.8 cm ²	SG of Contents	0.650
Element Volume	950502.1 cm ³	Weight of Contents	105.6 Kg
Empty Weight	3305.5 Kg	Operating Weight	3411.1 Kg
		Comp. All. Ope. Stress	98.4 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Efficiency E	100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (13827 \cdot 553) / (148 \cdot 1 - 0.6 \cdot 13827)$$

$$= 54.7370 + 3.0000 = 57.7370 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 7.028 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (148 \cdot 1 \cdot 61) / (553 + 0.6 \cdot 61)$$

$$= 15311 - 7.03 = 15304 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (148 \cdot 1 \cdot 64) / (550 + 0.6 \cdot 64)$$

$$= 16097 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (13827 \cdot (553 + 0.6 \cdot 61)) / (1 \cdot 61)$$

$$= 133.654 \text{ N./mm}^2$$



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Element: Channel shell

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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 5.498 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 64$, $t_r = 57.8$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.95$, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 41 °C

Min Metal Temp. at Required thickness (UCS 66.1) 38 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	14545.192 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	64.000 mm	Basic MDMT	41 °C
Stress/Thk. Ratio	0.947	Temperature Difference	3 °C
UCS-66(c) red.	No	Computed Min. Temp.	38 °C
		Joint Efficiency E^*	100 %

External Pressure Results for: Channel shell

Cylindrical Shell From 30 to Tsht Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193602528 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
61.000	1228.00	999.97	20.13	0.8143	0.0201574	118.55

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 119) / (3 \cdot 20.1) = 7852 \text{ KPa.}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
3.635	1228.00	999.97	337.79	0.8143	0.0002651	25.67

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 25.7) / (3 \cdot 338) = 101 \text{ KPa.}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
61.000	1228.00	70421.23	20.13	50.0000	0.0027143	97.87

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 97.9) / (3 \cdot 20.1) = 6482 \text{ KPa.}$$

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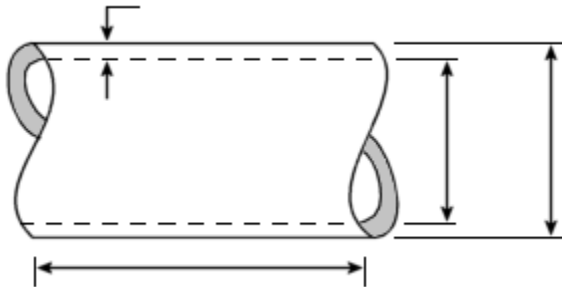
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Element: Shell

Step: 7 9:42am Apr 21,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2021



Internal Pressure Results for: Shell

Cylindrical Shell Design Information:			
Design Pressure P	4550.108 KPa.	Int. Design Temperature	454 °C
Material	SA-387 22	Ext. Design Temperature	200 °C
Length	4000.000 mm	External Pressure Chart	CS-3
		UNS Number	K21590
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	127.8 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	148.0 N./mm ²
Inside Diameter Di	1100.000 mm	Actual Stress	103.4 N./mm ²
Spec. Min. Thk t or e	28.0000 mm	Spec. Nominal Thk.	28.000 mm
Surface Area	145267.2 cm ²	SG of Contents	0.010
Element Volume	2880028.8 cm ³	Weight of Contents	105.6 Kg
Empty Weight	3305.5 Kg	Operating Weight	3411.1 Kg
		Comp. All. Ope. Stress	77.9 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Efficiency E	100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (4550 \cdot 553) / (128 \cdot 1 - 0.6 \cdot 4550)$$

$$= 20.1230 + 3.0000 = 23.1230 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.108 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (128 \cdot 1 \cdot 25) / (553 + 0.6 \cdot 25)$$

$$= 5624 - 0.11 = 5624 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (148 \cdot 1 \cdot 28) / (550 + 0.6 \cdot 28)$$

$$= 7311 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (4550 \cdot (553 + 0.6 \cdot 25)) / (1 \cdot 25)$$

$$= 103.384 \text{ N./mm}^2$$



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Element: Shell

Step: 7 9:42am Apr 21,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.482 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 28$, $t_r = 18.4$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.74$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 22 °C

Min Metal Temp. at Required thickness (UCS 66.1) 7 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	4822.231 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	28.000 mm	Basic MDMT	22 °C
Stress/Thk. Ratio	0.735	Temperature Difference	15 °C
UCS-66(c) red.	No	Computed Min. Temp.	7 °C
		Joint Efficiency E^*	100 %

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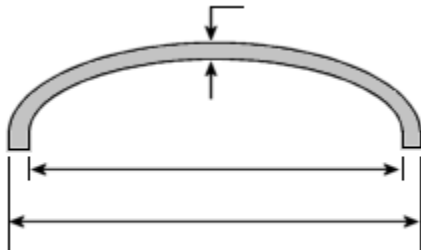
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Element: Shell head

Step: 8 9:42am Apr 21,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2021



Internal Pressure Results for: Shell head

Elliptical Head Design Information:			
Design Pressure P	4550.099 KPa.	Int. Design Temperature	454 °C
Material	SA-387 22	Ext. Design Temperature	200 °C
Length	50.000 mm	External Pressure Chart	CS-3
		UNS Number	K21590
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	127.8 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	148.0 N./mm ²
Inside Diameter Di	1100.000 mm	Actual Stress	120.6 N./mm ²
Spec. Min. Thk t or e	23.8000 mm	Spec. Nominal Thk.	28.000 mm
Surface Area	16301.5 cm ²	SG of Contents	0.010
Element Volume	221783.8 cm ³	Weight of Contents	105.6 Kg
Empty Weight	3305.5 Kg	Operating Weight	3411.1 Kg
		Comp. All. Ope. Stress	77.9 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Efficiency E	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.750 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.875 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (4550 \cdot 1106 \cdot 0.99) / (2 \cdot 128 \cdot 1 - 0.2 \cdot 4550)$$

$$= 19.6216 + 3.0000 = 22.6216 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.098 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 128 \cdot 1 \cdot 20.8) / (0.99 \cdot 1106 + 0.2 \cdot 20.8)$$

$$= 4822 - 0.098 = 4822 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 148 \cdot 1 \cdot 23.8) / (1 \cdot 1100 + 0.2 \cdot 23.8)$$

$$= 6376 \text{ KPa.}$$



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Element: Shell head

Step: 8 9:42am Apr 21,2026

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (4550 \cdot (0.99 \cdot 1106 + 0.2 \cdot 20.8)) / (2 \cdot 1 \cdot 20.8)$$

$$= 120.565 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (4550 \cdot 553) / (128 \cdot 1 - 0.6 \cdot 4550) + 3$$

$$= 23.123 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.098 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (128 \cdot 1 \cdot 25) / (553 + 0.6 \cdot 25)$$

$$= 5624 - 0.098 = 5624 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (1106 / (2 \cdot 278))^2) / 6$$

$$= 0.992825$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 10.448 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 23.8$, $t_r = 17.9$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.86$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	17 °C
Min Metal Temp. at Required thickness (UCS 66.1)	10 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 28$, $t_r = 18.4$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.74$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	7 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	4822.231 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	28.000 mm	Basic MDMT	22 °C
Stress/Thk. Ratio	0.735	Temperature Difference	15 °C
UCS-66(c) red.	No	Computed Min. Temp.	7 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 13.750 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.875 mm

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Pressure Test Results: Shop Test Step: 9 9:42am Apr 21,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

Pressure per UG99b = 1.30 * M.A.W.P. * Sa/S 6268.901 KPa.
 Pressure per UG99b[35] = 1.30 * Design Pres * Sa/S 5915.000 KPa.
 Pressure per UG99c = 1.30 * M.A.P. - Head(Hyd) 8272.948 KPa.
 Pressure per UG100 = 1.10 * M.A.W.P. * Sa/S 5304.455 KPa.
 Pressure per PED = max(1.43*DP, 1.25*DP*ratio) 6506.500 KPa.
 Pressure per PED&UG99b = max(1.43*DP, 1.3*DP*ratio) 6506.500 KPa.
 Pressure per App 27-4 = M.A.W.P. 4822.231 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

Pressure per UG99b = 1.30 * M.A.W.P. * Sa/S 18908.748 KPa.
 Pressure per UG99b[35] = 1.30 * Design Pres * Sa/S 17965.998 KPa.
 Pressure per UG99c = 1.30 * M.A.P. - Head(Hyd) 19319.895 KPa.
 Pressure per UG100 = 1.10 * M.A.W.P. * Sa/S 15999.712 KPa.
 Pressure per PED = max(1.43*DP, 1.25*DP*ratio) 19762.598 KPa.
 Pressure per PED&UG99b = max(1.43*DP, 1.3*DP*ratio) 19762.598 KPa.
 Pressure per App 27-4 = M.A.W.P. 14545.192 KPa.

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Channel shell	174.0	279.0	0.623	18919.53
Shell	127.1	279.0	0.456	6279.68
Shell head	145.8	279.0	0.522	6279.68

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3- Tube inlet	Nozzle	148.00	148.00	1.000
N4- Tube outlet	Nozzle	148.00	148.00	1.000
D1	Nozzle	148.00	148.00	1.000
Shell inlet-N1	Nozzle	148.00	127.78	1.158
Shell outlet- N	Nozzle	148.00	127.78	1.158
V1	Nozzle	148.00	127.78	1.158
D2	Nozzle	148.00	127.78	1.158

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
CH COVER	148.00	148.00	1.000
CH Flange	148.00	148.00	1.000
Channel shell	148.00	148.00	1.000
Shell	148.00	127.78	1.158
Shell head	148.00	127.78	1.158

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):



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Pressure Test Results: Shop Test

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Description	Ambient	Operating	Ratio
Tube Material	118.00	111.78	1.056
Tubesheet Material	148.00	148.00	1.000

Minimum 1.000**Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):**

Description	Ambient	Operating	Ratio
N3- Tube inlet	68.03	192.40	0.354
N4- Tube outlet	68.03	192.40	0.354
D1	57.75	192.40	0.300
Shell inlet-N1	78.54	192.40	0.408
Shell outlet- N	99.48	192.40	0.517
V1	14.59	192.40	0.076
D2	14.59	192.40	0.076

Elements Suitable for Test Pressure.

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Pressure Test Results: Field Test Step: 10 9:42am Apr 21,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

Pressure per UG99b = $1.30 * \text{M.A.W.P.} * \text{Sa/S}$ 6268.901 KPa.
 Pressure per UG99b[35] = $1.30 * \text{Design Pres} * \text{Sa/S}$ 5915.000 KPa.
 Pressure per UG99c = $1.30 * \text{M.A.P.} - \text{Head(Hyd)}$ 8272.948 KPa.
 Pressure per UG100 = $1.10 * \text{M.A.W.P.} * \text{Sa/S}$ 5304.455 KPa.
 Pressure per PED = $\max(1.43*DP, 1.25*DP*ratio)$ 6506.500 KPa.
 Pressure per PED&UG99b = $\max(1.43*DP, 1.3*DP*ratio)$ 6506.500 KPa.
 Pressure per App 27-4 = M.A.W.P. 4822.231 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

Pressure per UG99b = $1.30 * \text{M.A.W.P.} * \text{Sa/S}$ 18908.748 KPa.
 Pressure per UG99b[35] = $1.30 * \text{Design Pres} * \text{Sa/S}$ 17965.998 KPa.
 Pressure per UG99c = $1.30 * \text{M.A.P.} - \text{Head(Hyd)}$ 19319.895 KPa.
 Pressure per UG100 = $1.10 * \text{M.A.W.P.} * \text{Sa/S}$ 15999.712 KPa.
 Pressure per PED = $\max(1.43*DP, 1.25*DP*ratio)$ 19762.598 KPa.
 Pressure per PED&UG99b = $\max(1.43*DP, 1.3*DP*ratio)$ 19762.598 KPa.
 Pressure per App 27-4 = M.A.W.P. 14545.192 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= $1.3 * 4550 * 1$

= 5915.000 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= $1.3 * 13820 * 1$

= 17965.998 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Channel shell	173.8	279.0	0.623	17976.78
Shell	134.6	279.0	0.483	5925.78
Shell head	157.0	279.0	0.563	5925.78

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3- Tube inlet	Nozzle	148.00	148.00	1.000
N4- Tube outlet	Nozzle	148.00	148.00	1.000
D1	Nozzle	148.00	148.00	1.000
Shell inlet-N1	Nozzle	148.00	127.78	1.158
Shell outlet- N	Nozzle	148.00	127.78	1.158
V1	Nozzle	148.00	127.78	1.158
D2	Nozzle	148.00	127.78	1.158

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
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Pressure Test Results: Field Test

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CH COVER	148.00	148.00	1.000
CH Flange	148.00	148.00	1.000
Channel shell	148.00	148.00	1.000
Shell	148.00	127.78	1.158
Shell head	148.00	127.78	1.158

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	118.00	111.78	1.056
Tubesheet Material	148.00	148.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N3- Tube inlet	64.64	192.40	0.336
N4- Tube outlet	64.64	192.40	0.336
D1	54.87	192.40	0.285
Shell inlet-N1	74.11	192.40	0.385
Shell outlet- N	93.87	192.40	0.488
V1	13.77	192.40	0.072
D2	13.77	192.40	0.072

Elements Suitable for Test Pressure.

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External Pressure Calculations: Step: 11 9:42am Apr 21,2026

External Pressure Results Summary:

External Pressure Calculation Results:

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External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	...	267	No Calc	No Calc
20	30	No Calc	...	262	No Calc	No Calc
30	Tsht	999.97	1228	61	0.020157	118.555
Tsht	40	No Calc	...	61	No Calc	No Calc
40	50	4141.7	1156	25	0.0011757	81.3849
50	60	No Calc	1147.6	20.8	0.0025173	96.5739

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	270	201.828	101.3	No Calc
20	30	265	172.872	101.3	No Calc
30	Tsht	64	6.63544	101.3	7852
Tsht	40	64	No Calc	101.3	No Calc
40	50	28	No Calc	...	2347
50	60	23.8	No Calc	...	1945

Minimum

1945

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	999.97	70421.2	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	50	4141.7	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Flg Calc [Ext P]: New Flange

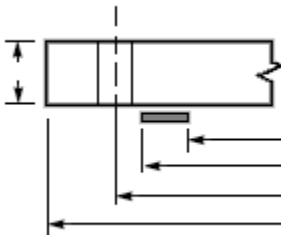
Step: 12 9:42am Apr 21,2026

Flange Input Data Values

Description: New Flange :

CH COVER

Flange Type	TEMA Channel Cover		
Design Pressure	P	-101.30	KPa.
Design Temperature		200	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	1655.000	mm
Flange Thickness	t	270.0000	mm
Flange Material	SA-336 F11		
Flange Material UNS number	K11572		
Flange Allowable Stress At Temperature	Sfo	148.01	N./mm ²
Flange Allowable Stress At Ambient	Sfa	148.00	N./mm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	159.01	N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00	N./mm ²
Perimeter along the Center of the Bolts	L	4633.849	mm
Youngs Modulus for Blind Flange Matl.	Sy	181790.67	N./mm ²
Allowed Channel Cover Deflection	dfAllow	1.3750	mm
Diameter of Bolt Circle	C	1475.000	mm
Nominal Bolt Diameter	a	88.9000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	24		
Flange Face Outside Diameter	Fod	1193.000	mm
Flange Face Inside Diameter	Fid	1100.000	mm
Flange Facing Sketch	1, Code Sketch 1a		
Gasket Outside Diameter	Go	1180.000	mm
Gasket Inside Diameter	Gi	1132.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Column for Gasket Seating	2, Code Column II		
Gasket Thickness	tg	4.5000	mm



Bolted Flange Calculation Results:

Gasket Contact Width,	$N = (Go - Gi)/2$	24.000 mm
Basic Gasket Width,	$bo = N/2$	12.000 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	8.729 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	1162.541 mm



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Flg Calc [Ext P]: New Flange

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Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H]:

$$= \pi/4 \cdot G^2 \cdot P_{eq}$$

$$= \pi/4 \cdot 1163^2 \cdot -101$$

$$= -10965.104 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.73 \cdot \pi \cdot 1163 \cdot 3 \cdot -101$$

$$= -1976.027 \text{ Kg}$$

Operating Bolt Load [Wm1]:

$$= \max(H + H_p + H'p, 0)$$

$$= \max(-10965 + -1976 + 0, 0)$$

$$= 0.000 \text{ Kg}$$

Gasket Seating Bolt Load [Wm2]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 8.73 \cdot \pi \cdot 1163 + 0 \cdot 0 \cdot 0$$

$$= 224151.312 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max(W_{m1}/S_b, W_{m2}/S_a)$$

$$= \max(0/159, 224151/159)$$

$$= 138.255 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts per App. 2 eq. (3) [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 88.9 + 6 \cdot 267/(3 + 0.5)$$

$$= 635.514 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1475 \cdot \sin(\pi / 24)$$

$$= 192.526 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per App. 2 eq. (7) [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 193 / (2 \cdot 88.9 + 267) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	138.255	1354.681	
Radial Distance between Bolts and Edge:	82.550	90.000	
Circ. Spacing between the Bolts:	180.975	192.526	635.514

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 1355 \cdot 159 / (68.9 \cdot \pi (1180 + 1132))$$

$$= 43.009 \text{ mm} [\text{Note: Exceeds actual gasket width, 24.000 }]$$

Flange Design Bolt Load, Gasket Seating [W]:

$$= S_a (A_m + A_b) / 2$$

$$= 159 (138 + 1355) / 2$$

$$= 1210240.88 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$



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Flg Calc [Ext P]: New Flange

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$$= (1475 - 1163) / 2$$

$$= 156.2293 \text{ mm}$$

Tangential Flange Stress, Flat Head (UG-34), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 0 \cdot 156 \cdot 1 / (267^2 \cdot 1163) + 0.3 \cdot 1 \cdot 101 \cdot 1163^2 / 267^2$$

$$= 0.58 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (UG-34), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 1210240 \cdot 156 \cdot 1 / (267^2 \cdot 1163)$$

$$= 42.51 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 0 / 1355$$

$$= 0.00 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (224151 / 1355)$$

$$= 16.23 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	0.58	148.01	42.51	148.00
Bolting	0.00	159.01	16.23	159.00

TEMA Channel Cover Deflection Calculations:

Actual deflection at center per TEMA formula 0.5337 mm

Allowed deflection at center of Cover 1.3750 mm

Reqd. Blind Flange Thickness at Center 146.096 mm

Reqd. Blind Flange Thickness at Gasket 146.096 mm

Min. Required Flange Thickness per TEMA 194.768 mm

Stress Check: OK for External Pressure -101.3 KPa.

Estimated Finished Weight of Flange at given Thk. 4501.7 kg.

Estimated Unfinished Weight of Forging at given Thk 4501.7 kg.

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Flg Calc [Ext P]: CH Flange

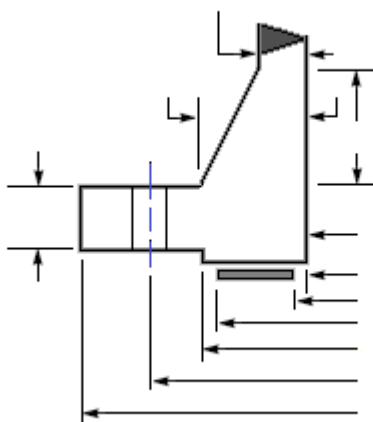
Step: 13 9:42am Apr 21,2026

Flange Input Data Values

Description: CH Flange :

CH Flange

Flange Type	Integral Weld Neck		
Design Pressure	P	-101.30	KPa.
Design Temperature		200	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	1100.000	mm
Flange Outside Diameter	A	1655.000	mm
Flange Thickness	t	265.0000	mm
Thickness of Hub at Small End	go	64.0000	mm
Thickness of Hub at Large End	gl	80.0000	mm
Length of Hub	h	96.0000	mm
Flange Material	SA-336 F11		
Flange Material UNS number	K11572		
Flange Allowable Stress At Temperature	Sfo	148.01	N./mm ²
Flange Allowable Stress At Ambient	Sfa	148.00	N./mm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	159.01	N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00	N./mm ²
Diameter of Bolt Circle	C	1475.000	mm
Nominal Bolt Diameter	a	88.9000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	24		
Flange Face Outside Diameter	Fod	1193.000	mm
Flange Face Inside Diameter	Fid	1100.000	mm
Flange Facing Sketch	1, Code Sketch 1a		
Gasket Outside Diameter	Go	1180.000	mm
Gasket Inside Diameter	Gi	1132.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Column for Gasket Seating	2, Code Column II		
Gasket Thickness	tg	4.5000	mm





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Flg Calc [Ext P]: CH Flange

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Bolted Flange Calculation Results:

Corroded Flange ID,	Bcor = B+2*ci	1106.000	mm
Corroded Large Hub,	g1Cor = g1-ci	77.000	mm
Corroded Small Hub,	g0Cor = go-ci	61.000	mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	107.500	mm
Gasket Contact Width,	N = (Go - Gi)/2	24.000	mm
Basic Gasket Width,	bo = N/2	12.000	mm
Effective Gasket Width,	b = Cb*sqrt(bo)	8.729	mm
Gasket Reaction Diameter,	G = Go - 2 * b	1162.541	mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H]:

$$= \pi/4 \cdot G^2 \cdot Peq$$

$$= \pi/4 \cdot 1163^2 \cdot -101$$

$$= -10965.104 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.73 \cdot \pi \cdot 1163 \cdot 3 \cdot -101$$

$$= -1976.027 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 1106^2 \cdot -101$$

$$= -9924.443 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= -10965 - -9924$$

$$= -1040.661 \text{ Kgf}$$

Operating Bolt Load [Wm1]:

$$= \max(H + Hp + H'p, 0)$$

$$= \max(-10965 + -1976 + 0, 0)$$

$$= 0.000 \text{ Kgf}$$

Gasket Seating Bolt Load [Wm2]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.9 \cdot 8.73 \cdot \pi \cdot 1163 + 0 \cdot 0$$

$$= 224151.312 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max(Wm1/Sb, Wm2/Sa)$$

$$= \max(0/159, 224151/159)$$

$$= 138.255 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts per App. 2 eq. (3) [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 88.9 + 6 \cdot 265/(3 + 0.5)$$

$$= 632.086 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1475 \cdot \sin(\pi / 24)$$

$$= 192.526 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per App. 2 eq. (7) [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 193 / (2 \cdot 88.9 + 265) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):



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Flg Calc [Ext P]: CH Flange

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	Minimum	Actual	Maximum
Bolt Area:	138.255	1354.681	
Radial Distance between Hub and Bolts:	104.775	107.500	
Radial Distance between Bolts and Edge:	82.550	90.000	
Circ. Spacing between the Bolts:	180.975	192.526	632.086

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 1355 \cdot 159 / (68.9 \cdot \pi (1180 + 1132))$$

$$= 43.009 \text{ mm [Note: Exceeds actual gasket width, 24.000]}$$

Note: Recommended Min. Width for Sheet and Composite Gaskets per table 2-4 :

$$= 32.000 \text{ mm [Note: Exceeds actual gasket width, 24.000]}$$

Flange Design Bolt Load, Gasket Seating [W]:

$$= Sa (Am + Ab) / 2$$

$$= 159 (138 + 1355) / 2$$

$$= 1210240.88 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1475 - 1163) / 2$$

$$= 156.2293 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= (R + g1 + hg) / 2$$

$$= (107 + 77 + 156) / 2$$

$$= 170.3646 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= R + (g1 / 2)$$

$$= 107 + (77 / 2.0)$$

$$= 146.0000 \text{ mm}$$

Summary of Moments for External Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	-9924.	-10.2293	1.0000	102.
Face Pressure, Mt	-1041.	14.1354	1.0000	-15.
Gasket Seating, Matm	1210241.	156.2293	1.0000	189079.

Total Moment for Operation, Mop 87. Kg-m.

Total Moment for Gasket seating, Matm 189079. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 259.742 mm

Hub Ratio, h/h0 = HL / H0 0.370

Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.262

Flange Factors for Integral Flange:

Factor F 0.882

Factor V 0.415

Factor f 1.000

Factors from Figure 2-7.1 K = 1.496

T = 1.712 U = 5.483

Y = 4.990 Z = 2.614

e = 0.0034 1/mm d = 0.12774E+08 mm³

Stress Factors Alpha = 1.900

Beta = 2.200 Gamma = 1.110

Delta = 1.457 Lambda = 2.566



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Flg Calc [Ext P]: CH Flange

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Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g^{1^2})$$

$$= (1 \cdot 86.8 / 1106) / (2.57 \cdot 77^2)$$

$$= 0.05 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g^{1^2})$$

$$= (1 \cdot 189079 / 1106) / (2.57 \cdot 77^2)$$

$$= 110.18 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.2 \cdot 86.8 / 1106) / (2.57 \cdot 265^2)$$

$$= 0.01 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.2 \cdot 189079 / 1106) / (2.57 \cdot 265^2)$$

$$= 20.47 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (4.99 \cdot 86.8 / (265^2 \cdot 1106)) - 2.61 \cdot 0.0094$$

$$= 0.03 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (4.99 \cdot 189079 / (265^2 \cdot 1106)) - 2.61 \cdot 20.5$$

$$= 65.62 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (0.051 + \max(0.0094, 0.03)) / 2$$

$$= 0.04 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (110 + \max(20.5, 65.6)) / 2$$

$$= 87.90 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 0 / 1355$$

$$= 0.00 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (224151 / 1355)$$

$$= 16.23 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	0.05	222.01	110.18	222.00
Radial Flange	0.01	148.01	20.47	148.00
Tangential Flange	0.03	148.01	65.62	148.00
Maximum Average	0.04	148.01	87.90	148.00
Bolting	0.00	159.01	16.23	159.00

Minimum Required Flange Thickness

172.872 mm

Stress Check: OK for External Pressure

-101.3 KPa.

Estimated Finished Weight of Flange at given Thk.

2663.7 kg.



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Flg Calc [Ext P]: CH Flange

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Estimated Unfinished Weight of Forging at given Thk 3360.0 kg.

Flange Rigidity Based on Required Thickness [ASME]:

Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:
= $52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
= $52.14 \cdot 189079 / 1 \cdot 9807 \cdot 0.41 / (1.33 \cdot 204092 \cdot 61^2 \cdot 260 \cdot 0.3)$
= 0.509 (should be ≤ 1)

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:
= $52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
= $52.14 \cdot 86.8 / 1 \cdot 9807 \cdot 0.41 / (1.33 \cdot 193337 \cdot 61^2 \cdot 260 \cdot 0.3)$
= 0.000 (should be ≤ 1)

Flange Rigidity Based on Given Thickness [ASME]:

Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:
= $52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
= $52.14 \cdot 189079 / 1 \cdot 9807 \cdot 0.41 / (2.57 \cdot 204092 \cdot 61^2 \cdot 260 \cdot 0.3)$
= 0.264 (should be ≤ 1)

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:
= $52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
= $52.14 \cdot 86.8 / 1 \cdot 9807 \cdot 0.41 / (2.57 \cdot 193337 \cdot 61^2 \cdot 260 \cdot 0.3)$
= 0.000 (should be ≤ 1)

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Element and Detail Weights:

Step: 14

9:42am Apr 21,2026

Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	4635.12	...	4635.12	...	463.512
20	30	2738.2	353574	2708.47	354572	273.82
30	40	1813.9	950502	1733.33	960899	181.39
40	50	3076.15	2880028	2753.86	2921618	307.615
50	60	368.933	221784	329.405	225171	36.8933
Total		12632	4405888.50	12160	4462260.50	1263



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset, Dtl. Cent. mm	Description
10	Ins1	156.897	137.5	...	...	Ins: 10
20	Liqd	229.683	180.5	...	...	Liquid: 20
20	Ins1	101.091	175	...	...	Ins: 20
30	Liqd	617.449	500	...	...	Liquid: 20
30	Ins1	228.845	500	...	...	Ins: 40
30	Noz1	62.2858	300	583.325	...	N3- Tube inlet
30	Noz1	62.2858	300	-583.325	...	N4- Tube outlet
30	Noz1	33.1948	700	-571.425	...	D1
30	Wght	250	500	...	...	Pass Partition
40	Sad1	226.202	300	788.5	...	Left Saddle
40	Sad1	240.151	3550	838.5	...	Right saddle
40	Liqd	28.7827	2000	...	...	Liquid: 40
40	Ins1	867.552	2000	...	...	Ins: 40
40	Noz1	481.785	1900	-931	...	Shell inlet-N1
40	Noz1	470.689	1800	931	...	Shell outlet- N
40	Noz1	22.3748	500	569.088	...	V1
40	Noz1	21.8271	3200	-569.088	...	D2
40	Wght	250	300	...	-1000	Fire proofing
40	Wght	250	3550	...	-1000	Fire proofing
50	Liqd	2.16486	141.667	50	...	Liquid: 50
50	Ins1	122.077	150	...	...	Ins: 40
30	FTsh	1912.86	1103	...	...	Tubesheet
30	Tube	3080.34	2825	...	...	Tubes
40	Bafl	402.49	1828	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	466.4
Liquid	878.1
Insulation	1476.5
Nozzles	1154.4
Weights	750.0
Exchanger Components	5395.7
Liquid in Tubes	380.3

Sum of the Detail Weights 10501.3 kg.

Weight Summation Results: (kg.)



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Element and Detail Weights:

Step: 14 9:42am Apr 21,2026

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	13895.5	13895.5	13895.5	13895.5	13895.5	13895.5
Saddles	466.4	466.4	466.4	466.4	466.4	466.4
Nozzles	1154.4	1154.4	1154.4	1154.4	1154.4	1154.4
Wld Weights	250.0	250.0	250.0	250.0	250.0	250.0
Exchanger	5395.7	5395.7	5395.7	5395.7	5395.7	5395.7
Insulation	...	...	...	1476.5	1476.5	1476.5
Empty Weights	...	...	...	500.0	500.0	...
Test Weights	...	500.0	...	...	...	...
Ope Weights	...	...	...	...	...	500.0
Ope. Liquid	...	...	...	...	...	878.1
Tube Ope Lqd	...	...	...	...	...	380.3
Test Liquid	...	4403.2	...	...	...	...
Tube Tst Lqd	...	585.1	...	...	...	...
Totals	21162.0	26650.3	21162.0	23138.5	23138.5	24396.9

Field Installation Options:

Insulation installed before lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare Weight without Removable Internals	21162.0 kg.
Shop Test Wt.	- Fabricated Weight + Water (Full)	26650.3 kg.
Shipping Wt.	- Fab. Weight + removable Intls.+ Shipping App.	21162.0 kg.
Erected Wt.	- Fab. Wt + or - loose items (trays,platforms etc.)	23138.5 kg.
Ope. Wt. no Liq	- Fab. Weight + Internals. + Details + Weights	23138.5 kg.
Operating Wt.	- Empty Weight + Operating Liq. Uncorroded	24396.9 kg.
Oper. Wt. + CA	- Corr Wt. + Operating Liquid	23877.5 kg.
Field Test Wt.	- Empty Weight + Water (Full)	28126.8 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	585436.8 cm ³
Weight of Ope Liq in tubes :	380.3 kg.
Weight of Water in tubes :	585.1 kg.

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	35966.4	...
20	30	30003.2	12751.7
30	40	38578.8	34557.5
40	50	145267	138230
50	60	16301.5	14844.1

Total	266117.1	200383.4 cm ²
Total	286.4	215.7 ft ²



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Nozzle Flange MAWP:

Step: 15 9:43am Apr 21,2026

Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating		Design Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44(b)	50%	DNV
N3- Tube inlet	18310.0	25860.0	400	1500	GR 1.9	...	...	...	...
N4- Tube outle	18310.0	25860.0	400	1500	GR 1.9	...	...	...	...
D1	18310.0	25860.0	400	1500	GR 1.9	...	...	...	...
V1	6632.8	15320.0	454	900	GR 1.1	...	...	...	...
D2	6632.8	15320.0	454	900	GR 1.1	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 6632.800 KPa.

Lowest Flange Pressure Rating was (Amb)[ShellSide]: 15320.001 KPa.

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 18309.998 KPa.

Lowest Flange Pressure Rating was (Amb)[TubeSide]: 25859.998 KPa.

Pressure Ratings are per ASME B16.5 2017 Metric Edition

Large Diameter Flange ratings per ASME B16.47 2017 edition.

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.

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Wind Load Calculation:

Step: 16 9:43am Apr 21,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	16000 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$\begin{aligned}
 &= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_{zbar}))) \\
 &= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.2 \cdot 0.93)/(1 + 1.7 \cdot 3.4 \cdot 0.2))) \\
 &= \min(0.85, 0.89) \\
 &= 0.850
 \end{aligned}$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.558
Structure Height to Diameter ratio	4.493

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Z_g = 274320 \text{ mm}$

Effective Height [z]:

$$\begin{aligned}
 &= \text{Centroid Height} + \text{Vessel Base Elevation} \\
 &= 1170 + 16000 = 17170 \text{ mm} \\
 &= 56.3 \text{ ft. Imperial Units}
 \end{aligned}$$

Velocity Pressure coefficient evaluated at height z [Kz]:

$$\begin{aligned}
 &\text{Because } z (56.3 \text{ ft.}) > 15 \text{ ft.} \\
 &= 2.01 (z / Z_g)^{2/\alpha} \\
 &= 2.01 (56.3/900)^{2/9.5} \\
 &= 1.12
 \end{aligned}$$

Type of Hill: No Hill



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Wind Load Calculation:

Step: 16 9:43am Apr 21,2026

Wind Directionality Factor [Kd]:

= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

= (1 + K1 • K2 • K3)²

= (1 + 0 • 0 • 0)²

= 1

Velocity Pressure evaluated at height z, Imperial Units [qz]:

= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)

= max(16, 0.00256 • 1.12 • 1 • 1 • 1 • 60.4²)

= 16 psf [78.1] Kgs/m²

Force on the first element [F]:

= qz • G • Cf • WindArea

= 16 • 0.85 • 0.56 • 6.82

= 51.8 lbs. [23.5] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
CH COVER	17170.0	0.000	0.000	0.000	1.122	1.000	78.120
CH Flange	17170.0	0.000	0.000	0.000	1.122	1.000	78.120
Channel shell	17170.0	0.000	0.000	0.000	1.122	1.000	78.120
Shell	17170.0	0.000	0.000	0.000	1.122	1.000	78.120
Shell head	17170.0	0.000	0.000	0.000	1.122	1.000	78.120

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
Pass Partitio	0.00	17170.00	78.12	0.00
Fire proofing	0.00	17170.00	78.12	0.00
Fire proofing	0.00	17170.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	17170	2346	6334.2	78.12	25.8269
20	30	17170	1669.03	6025.2	78.12	24.567
30	40	17170	1833.6	18336	78.12	74.7627
40	50	17170	1747.2	69888	78.12	284.96
50	60	17170	1737.12	5971.16	78.12	24.3466

Note:

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation:

Step: 17 9:43am Apr 21,2026

Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\ SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot hnx$ where $Ct = 0.020$, $x = 0.75$ and hn = Structural Height (ft.)
 $= 0.020(6.550.75)$
 $= 0.082$ seconds

The Coefficient Cu from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/33)$
 $= 0.030$

Check the Value of T which is the smaller of $Cu \cdot Ta$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.082, 0.03) \text{ per 12.8.2}$
 $= 0.030$
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 $= 0.3 \cdot Sds \cdot W \cdot I$
 $= 0.3 \cdot 0.16 \cdot 23878 \cdot 1.25$
 $= 1403.999$ Kgf

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.16 \cdot 23878$
 $= 748.8$ Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	827.5	3411.08	281.48



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Earthquake Load Calculation:

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20	30	550	3411.08	187.086
30	40	550	3411.08	187.086
40	Sad1	550	3411.08	187.086
Sad1	50	550	3411.08	187.086
40	50	550	3411.08	187.086
50	60	550	3411.08	187.086

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Center of Gravity Calculation: Step: 18 9:43am Apr 21,2026

Shop/Field Installation Options :

Insulation is installed in the Field before being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	3821.454 mm
Center of Gravity of the Liquid	1064.501 mm
Center of Gravity of the Insulation	2979.673 mm
Center of Gravity of the Nozzles	3332.556 mm
Center of Gravity of the Added Weights (Operating)	2895.850 mm
Center of Gravity of the Added Weights (Empty)	2895.850 mm
Center of Gravity of the Tubesheet(s)	1744.850 mm
Center of Gravity of the Tubes	3466.850 mm
Center of Gravity of the Baffles	3675.850 mm
Center of Gravity of Bare Shell New and Cold	1428.891 mm
Center of Gravity of Bare Shell Corroded	1354.152 mm
Vessel CG in the Operating Condition	2012.916 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	1992.091 mm
Vessel CG in the Test Condition	2207.720 mm

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Saddle Calcs: Operating Case: Step: 19 9:43am Apr 21,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (250.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$$

$$= \min(203 + 1.56 \cdot \sqrt{ 566 \cdot 25 }, 2 \cdot 300)$$

$$= 388.4859 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	565.50	mm
Shell Thickness used in this Case	t	28.000	mm
Stiffened Vessel Length per 4.15.6	L	4050.00	mm
Distance from Saddle to Vessel tangent	a1 or a	300.00	mm
Saddle Width	b1 or b	203.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		127.78	N./mm ²
Head Allowable Stress used in Calculation		148.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1070.00	mm
Coefficient of Friction	mu	0.40	
Saddle Force Q, Operating Case		27204.05	Kgf
Pressure used in Analysis	P	4550.054	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	43.48	127.78
Long. Stress at Bottom of Midspan	59.45	127.78
Long. Stress at Top of Saddles	49.74	127.78
Long. Stress at Bottom of Saddles	52.42	127.78
Tangential Shear in Shell	18.82	102.22
Circ. Stress at Horn of Saddle	18.05	159.72
Circ. Compressive Stress in Shell	2.09	127.78

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (434/2 + 0) \cdot 1070/1027$$

$$= 678.9 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl, Sum of X Forces}) \cdot B / Ls$$

$$= \max(206, 0) \cdot 1070/3250$$

$$= 67.8 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl, Sum of X Forces}) \cdot B / Ls$$

$$= \max(1404, 0) \cdot 1070/3250$$

$$= 462.2 \text{ Kgf}$$



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Saddle Calcs: Operating Case: Step: 19 9:43am Apr 21,2026

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (1404/2 + 0) \cdot 1070/1027$$

$$= 2193.9 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max (\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 25010 + \max (67.8, 679, 462, 2194)$$

$$= 27204.1 \text{ Kg}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (1.46 \cdot 1.2)^2)$$

$$= 2020.437 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 27430.25 Kg

Transverse Shear Load 702.00 Kg

Longitudinal Shear Load 1404.00 Kg

Formulas and Substitutions for Horizontal Vessel Analysis:

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min (2 (\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min (2 (1100/2 + 28 + 15) \sin(120/2), 2 \cdot 566)$$

$$= 1027.106 \text{ mm}$$

The Computed K values from Table 4.15.1:

$$K1 = 0.1066 \quad K2 = 1.1707 \quad K3 = 0.8799 \quad K4 = 0.4011$$

$$K5 = 0.7603 \quad K6 = 0.0529 \quad K7 = 0.0156 \quad K8 = 0.3405$$

$$K9 = 0.2711 \quad K10 = 0.0581 \quad K1* = 0.1923$$

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m^2)/3L)]$$

$$= -27204 \cdot 300 [1 - (1 - 300/4050 + (565^2 - 0^2)/(2 \cdot 300 \cdot 4050)) / (1 + (4 \cdot 0)/(3 \cdot 4050))]$$

$$= 469.5 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 (1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/3L) - 4a/L$$

$$= 27204 \cdot 4050/4 (1 + 2(565^2 - 0^2)/(4050^2)) / (1 + (4 \cdot 0)/(3 \cdot 4050)) - 4 \cdot 300/4050$$

$$= 20457.3 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 4550 \cdot 566/(2 \cdot 25) - 20457/(\pi \cdot 566^2 \cdot 25)$$

$$= 43.48 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 4550 \cdot 566/(2 \cdot 25) + 20457/(\pi \cdot 566^2 \cdot 25)$$

$$= 59.45 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= 4550 \cdot 566/(2 \cdot 25) - 469/(0.11 \cdot \pi \cdot 566^2 \cdot 25)$$

$$= 49.74 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:



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Saddle Calcs: Operating Case: Step: 19 9:43am Apr 21,2026

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 4550 \cdot 566 / (2 \cdot 25) + 469 / (0.19 \cdot \pi \cdot 566^2 \cdot 25)$$

$$= 52.42 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 27204 (4050 - 2 \cdot 300) / (4050 + (4 \cdot 0 / 3))$$

$$= 23173.8 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ 2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 23174 / (566 \cdot 25)$$

$$= 18.82 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(566 \cdot 25)}$$

$$= 92.743 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ 6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 27204 \cdot 0.1 / (25(203 + 92.7 + 92.7))$$

$$= -2.09 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.23) [σ 7*]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -27204 / (4 \cdot 25(203 + 92.7 + 92.7)) - 12 \cdot 0.016 \cdot 27204 \cdot 566 / (4050 \cdot 25^2)$$

$$= -18.05 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(203 + 1.56 \cdot \sqrt{(566 \cdot 25)}, 2 \cdot 300)$$

$$= 388.49 \text{ mm}$$

Distance between Saddle Supports [Ls]:

$$= 3250.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.14216\text{E-}04 \cdot 3250 (454 - 21.1)$$

$$= 20.000 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1200.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	430.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	310.0	N./
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(553 \cdot 25)} + 250)^3) / 12$$

$$= 56.435 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 56.4 + 108 (211 - 12.5)^2$$



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= 42567.391 cm⁴

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	433.4	25.0	12.5	108.4	135445.1	0.426E+05
Wearplate	250.0	15.0	32.5	37.5	121875.0	0.119E+05
Web	10.0	457.0	268.5	45.7	1227045.0	0.949E+04
BasePlate	430.0	20.0	507.0	86.0	4360200.0	0.756E+05
Totals	...	...	...	277.6	5844565.5	0.140E+06

Distance to Centroid [C1]:

= AY / A

= 5844565/278

= 210.572 mm

Angle [β]:

= 180 - Saddle Angle/2

= 180 - 120/2

= 120.0

Saddle Splitting Coefficient [K1]:

= (1 + cos(β) - 0.5•sin(β)²)/(π - β + sin(β)cos(β))

= (1 + cos(120) - 0.5•sin(120)²)/(π - 2.09 + sin(120)cos(120))

= 0.2035

Saddle Splitting Force [Fh]:

= K1 • Q

= 0.2 • 27204

= 5536.6143 Kgf

Tension Stress, St = (Fh/As) = 3.2090 N./mm²

Allowed Stress, Sa = 0.6 • Yield Str = 185.9966 N./mm²

Saddle Splitting Dimension [d]:

= B - R • sin(theta/2)/(theta/2 in radians)

= 1070 - 553 • sin(120/2)/1.05

= 612.673 mm

Bending Moment, M = Fh • d = 3392.1992 Kg-m.

Bending Stress, Sb = (M • C1 / I) = 5.0197 N./mm²

Allowed Stress, Sa = 2/3 • Yield Str = 206.6629 N./mm²

Minimum Thickness of Baseplate per Moss:

= √(3(Q + SaddleWt)BasePlateWidth / (4 • BasePlateLength • AllStress))

= √(3(27204 + 226)430/(4 • 1200 • 207))

= 18.703 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

= 2 • cos(90 - Saddle Angle/2) (Inside Radius + Shell Thk + Wear Plate Thk)

= 2 • cos(90 - 120/2) (550 + 28 + 15)

= 1027.106 mm

Distance between Ribs [e]:

= Web Length / (Nribs - 1)

= 1027/(3 - 1)

= 513.553 mm

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2

= 514 • 430/2

= 1104.139 cm²



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Saddle Calcs: Operating Case: Step: 19 9:43am Apr 21,2026

Bearing Pressure [Bp]:

$$= Q / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 27204 / (1200 \cdot 430)$$

$$= 5.272 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 1104 \cdot 5.27$$

$$= 5821.135 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 19.3 + 25.7$$

$$= 44.978 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 5821 / 45$$

$$= 12.692 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	203.0	...	20.3	...	697.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 203 - 10$$

$$= 193.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 45$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 203 / 2$$

$$= 101.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (697 / 45) }$$

$$= 39.369 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1070 - \cos(120 / 2) (550 + 28 + 15) - 20$$

$$= 753.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 199943392 / 310) }$$

$$= 112.838$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 754 / 39.4$$

$$= 19.139$$

Bending Moment [Rm]:



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Saddle Calcs: Operating Case: Step: 19 9:43am Apr 21,2026

$$\begin{aligned}
 &= F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2 \\
 &= 1404 / (2 \cdot 1200) \cdot 514 \cdot 754 / 2 \\
 &= 113.189 \text{ Kg-m.}
 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ (19.1 < 113) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3)) \\
 &= (1 - (19.1)^2 / (2 \cdot 113^2)) 310 / \\
 &\quad (5/3 + 3 \cdot (19.1) / (8 \cdot 113) - (19.1^3) / (8 \cdot 113^3)) \\
 &= 176.6 \text{ N./mm}^2
 \end{aligned}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba} \\
 &= 12.7 / 177 + (113 \cdot 102 / 6971188) / 207 \\
 &= 0.150
 \end{aligned}$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	193.0	0.0	19.3	0.0	697.
Web	513.6	10.0	0.0	51.4	0.0	4.28
Totals	...	...	...	70.7	...	701.

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 0 / 70.7 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 203 / 2 \\
 &= 101.500 \text{ mm}
 \end{aligned}$$

Length of Inner Rib [L]:

$$\begin{aligned}
 &= \text{Saddle Height} - \text{Outside Radius} - B_{pthk} \\
 &= 1070 - \cos(593/2) (20 + 0 + 0) - 0 \\
 &= 457.000 \text{ mm}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{ (\text{Total Inertia} / \text{Total Area}) } \\
 &= \sqrt{ (701 / 70.7) } \\
 &= 31.505 \text{ mm}
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL / r \\
 &= 1 \cdot 457 / 31.5 \\
 &= 14.505
 \end{aligned}$$

Unit Force [Force,u]:

$$\begin{aligned}
 &= F_l / (2 \cdot \text{Baseplate Length}) \\
 &= 1404 / (2 \cdot 1200) \\
 &= 0.585 \text{ Kg/mm}
 \end{aligned}$$

Moment at base of inner Rib [Mbase,c]:

$$\begin{aligned}
 &= \text{Unit Force} \cdot e \cdot L \\
 &= 0.58 \cdot 514 \cdot 457 \\
 &= 137.298 \text{ Kg-m.}
 \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$\begin{aligned}
 &= \text{Bending Moment} / \text{Section Modulus} \\
 &= 137 / 69095 \\
 &= 19.487 \text{ N./mm}^2
 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ (14.5 < 113) per AISC E2-1 [Sca]:



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$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (14.5)^2 / (2 \cdot 113^2)) 310 / (5/3 + 3 \cdot (14.5) / (8 \cdot 113) - (14.5^3) / (8 \cdot 113^3))$$

$$= 179.3 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 12.6/179 + (137 \cdot 102/701) / 207$$

$$= 0.164$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4	
Total Number of Bolts in Tension/Baseplate	Nbt	2	
Bolt Material Specification		SA-307 A	
Bolt Allowable Stress	Stba	138.00	N./mm ²
Bolt Corrosion Allowance	Bca	1.5	mm
Distance from Bolts to Edge	Edgedis	50.0	mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400	cm ²
BasePlate Allowable Stress	S	148.00	N./mm ²
Area Available in a Single Bolt	BltArea	3.0428	cm ²
Saddle Load QO (Weight)	QO	25236.3	Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	462.2	Kgf
Maximum Transverse Force	Ft	702.0	Kgf
Maximum Longitudinal Force	Fl	1404.0	Kgf
Saddle Bolted to Steel Foundation		No	

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1404 / (3.04 \cdot 4)$$

$$= 11.3 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 702 / (3.04 \cdot 4)$$

$$= 5.7 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 1404 / (82.8 \cdot 4)$$

$$= 0.4157 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1070 \cdot 702 + 0$$

$$= 751.15 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 751/25236$$

$$= 29.76 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.42, 0]$$



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= 0.4157 cm²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: A

Govrn. thk, tg = 15, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.74$, Temp. Reduction = 15 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A 4 °C

Min Metal Temp. at Required thickness (UCS 66.1) -11 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (250.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

= $\min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$

= $\min(203 + 1.56 \cdot \sqrt{566 \cdot 25}, 2 \cdot 300)$

= 388.4859 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	565.50	mm
Shell Thickness used in this Case	t	28.000	mm
Stiffened Vessel Length per 4.15.6	L	4050.00	mm
Distance from Saddle to Vessel tangent	a1 or a	300.00	mm
Saddle Width	b1 or b	203.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	288.40	mm
Shell Allowable Stress used in Calculation		127.78	N./mm ²
Head Allowable Stress used in Calculation		127.78	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1170.00	mm
Coefficient of Friction	mu	0.40	
Saddle Force Q, Operating Case		3531.57	Kgf
Pressure used in Analysis	P	4550.054	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	50.57	127.78
Long. Stress at Bottom of Midspan	52.36	127.78
Long. Stress at Top of Saddles	51.72	127.78
Long. Stress at Bottom of Saddles	51.32	127.78
Tangential Shear in Shell	2.23	102.22
Circ. Stress at Horn of Saddle	2.34	159.72
Circ. Compressive Stress in Shell	0.27	127.78

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

= $Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$

= $3 (434/2 + 0) \cdot 1170/1027$



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= 742.4 Kgf

Longitudinal Saddle Reaction Force [Fwl]:

= max(Fl, Sum of X Forces) • B / Ls

= max(206, 0) • 1170/3250

= 74.2 Kgf

Saddle Reaction Force due to Earthquake Fl [Fsl]:

= max(Fl, Sum of X Forces) • B / Ls

= max(1404, 0) • 1170/3250

= 505.4 Kgf

Saddle Reaction Force due to Earthquake Ft [Fst]:

= Ftr(Ft/Num of Saddles + Z Force Load) • B / E

= 3 (1404/2 + 0) • 1170/1027

= 2399.0 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)

= 1133 + max(74.2, 742, 505, 2399)

= 3531.6 Kgf

Longitudinal Wind Force [Fl]:

= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))

= 1.1 • 766 (0 + $\pi/4$ (1.46 • 1.2)²)

= 2020.437 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 3771.72 Kgf

Transverse Shear Load 702.00 Kgf

Longitudinal Shear Load 1404.00 Kgf

Formulas and Substitutions for Horizontal Vessel Analysis:

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)

= min(2(1100/2 + 28 + 15)sin(120/2), 2•566)

= 1027.106 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0156 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1- a/L + (Rm²-hm²)/(2a•L))/(1+(4hm²)/3L)]

= -3532•300[1-(1-300/4050+(565²-288²)/

(2•300•4050))/(1+(4•288)/(3•4050))]

= -69.3 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4(1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L

= 3532•4050/4(1+2(565²-288²)/(4050²))/(1+(4•288)/

(3•4050))-4•300/4050

= 2300.4 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= P • Rm/(2t) - M2/(π •Rm²t)

= 4550 • 566/(2•25) - 2300/(π •566²•25)

= 50.57 N./mm²



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Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 4550 \cdot 566 / (2 \cdot 25) + 2300 / (\pi \cdot 566^2 \cdot 25)$$

$$= 52.36 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 4550 \cdot 566 / (2 \cdot 25) - 69.3 / (0.11 \cdot \pi \cdot 566^2 \cdot 25)$$

$$= 51.72 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 4550 \cdot 566 / (2 \cdot 25) + 69.3 / (0.19 \cdot \pi \cdot 566^2 \cdot 25)$$

$$= 51.32 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 3532 (4050 - 2 \cdot 300) / (4050 + (4 \cdot 288 / 3))$$

$$= 2747.5 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 2748 / (566 \cdot 25)$$

$$= 2.23 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{566 \cdot 25}$$

$$= 92.743 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.76 \cdot 3532 \cdot 0.1 / (25(203 + 92.7 + 92.7))$$

$$= -0.27 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_{*7}]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -3532 / (4 \cdot 25(203 + 92.7 + 92.7)) - 12 \cdot 0.016 \cdot 3532 \cdot 566 / (4050 \cdot 25^2)$$

$$= -2.34 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(203 + 1.56 \cdot \sqrt{566 \cdot 25}, 2 \cdot 300)$$

$$= 388.49 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1200.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	430.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	310.0	N./
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t}) + \text{WearPlateWidth})^3 / 12$$

$$= ((1.56 \cdot \sqrt{553 \cdot 25}) + 250)^3 / 12$$

$$= 56.435 \text{ cm}^4$$



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Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 56.4 + 108 (252 - 12.5)^2$$

$$= 62302.707 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	433.4	25.0	12.5	108.4	135445.1	0.623E+05
Wearplate	250.0	15.0	32.5	37.5	121875.0	0.181E+05
Web	10.0	557.0	318.5	55.7	1774044.9	0.169E+05
BasePlate	430.0	20.0	607.0	86.0	5220199.5	0.108E+06
Totals	...	...	...	287.6	7251564.5	0.206E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 7251564/288$$

$$= 252.179 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 3532$$

$$= 718.7504 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.3933 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 185.9966 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1170 - 553 \cdot \sin(120/2) / 1.05$$

$$= 712.673 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 512.2439 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.6163 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 206.6629 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(3532 + 240)430 / (4 \cdot 1200 \cdot 207))}$$

$$= 6.935 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (550 + 28 + 15)$$

$$= 1027.106 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 1027 / (3 - 1)$$

$$= 513.553 \text{ mm}$$



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Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 514 \cdot 430 / 2$$

$$= 1104.139 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= Q / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 3532 / (1200 \cdot 430)$$

$$= 0.684 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1104 \cdot 0.68$$

$$= 755.686 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 19.3 + 25.7$$

$$= 44.978 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 756 / 45$$

$$= 1.648 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	203.0	...	20.3	...	697.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 203 - 10$$

$$= 193.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 45$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 203 / 2$$

$$= 101.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (697 / 45) }$$

$$= 39.369 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1170 - \cos(120 / 2) (550 + 28 + 15) - 20$$

$$= 853.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 199943392 / 310) }$$

$$= 112.838$$

Slenderness ratio [KL/r]:

$$= KL / r$$



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Saddle Calcs: Operating Case: Step: 19 9:43am Apr 21,2026

$$= 1 \cdot 853/39.4$$

$$= 21.679$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 1404 / (2 \cdot 1200) \cdot 514 \cdot 853/2$$

$$= 128.210 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($21.7 < 113$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (21.7)^2 / (2 \cdot 113^2)) 310 / (5/3 + 3 \cdot (21.7) / (8 \cdot 113) - (21.7^3) / (8 \cdot 113^3))$$

$$= 175.1 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 1.65/175 + (128 \cdot 102/6971188) / 207$$

$$= 0.098$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	193.0	0.0	19.3	0.0	697.
Web	513.6	10.0	0.0	51.4	0.0	4.28
Totals	...	...	...	70.7	...	701.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/70.7$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 203/2$$

$$= 101.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1170 - \cos(593/2) (20 + 0 + 0) - 0$$

$$= 557.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (701/70.7) }$$

$$= 31.505 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 557/31.5$$

$$= 17.680$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 1404 / (2 \cdot 1200)$$

$$= 0.585 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.58 \cdot 514 \cdot 557$$

$$= 167.342 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma B,base,c$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$



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= 167/69095
= 23.751 N./mm²

Compressive Allowable, $KL/r < C_c$ ($17.7 < 113$) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
= $(1 - (17.7)^2 / (2 \cdot 113^2)) 310 /$
 $(5/3 + 3 \cdot (17.7) / (8 \cdot 113) - (17.7)^3 / (8 \cdot 113^3))$
= 177.5 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$
= $1.63 / 178 + (167 \cdot 102 / 701) / 207$
= 0.124

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4	
Total Number of Bolts in Tension/Baseplate	Nbt	2	
Bolt Material Specification		SA-307 A	
Bolt Allowable Stress	Stba	138.00	N./mm ²
Bolt Corrosion Allowance	Bca	1.5	mm
Distance from Bolts to Edge	Edgedis	50.0	mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400	cm ²
BasePlate Allowable Stress	S	148.00	N./mm ²
Area Available in a Single Bolt	BltArea	3.0428	cm ²
Saddle Load QO (Weight)	QO	1372.7	Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	505.4	Kgf
Maximum Transverse Force	Ft	702.0	Kgf
Maximum Longitudinal Force	F1	1404.0	Kgf
Saddle Bolted to Steel Foundation	No		

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

= $F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $1404 / (3.04 \cdot 4)$
= 11.3 N./mm². Must be less than 82.8 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

= $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $702 / (3.04 \cdot 4)$
= 5.7 N./mm². Must be less than 82.8 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
= $1404 / (82.8 \cdot 4)$
= 0.4157 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot Ft + \text{Sum of X Moments}$
= $1170 \cdot 702 + 0$
= 821.36 Kg-m.

Eccentricity (e):

= $Rmom / QO$
= $821 / 1373$
= 598.33 mm > Bplen/6 --> Uplift in Transverse direction

f = Bplen / 2 - Edgedis

= 1200/2 - 50



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$$= 549.99 \text{ mm}$$

Modular Ratio Of Steel/Concrete (n1):

$$= ES / EC$$

$$= 203402/21526$$

$$= 9.45$$

$$K1 = 3 (e - 0.5 \cdot B_{plen})$$

$$= 3 (598 - 0.5 \cdot 1200)$$

$$= -5.02 \text{ mm}$$

$$K2 = 6 \cdot n1 \cdot A_t / B_{pwid} \cdot (f + e)$$

$$= 6 \cdot 9.45 \cdot 6.09/430 \cdot (550 + 598)$$

$$= 92136.10 \text{ mm}^2$$

$$K3 = -K2 \cdot (0.5 \cdot B_{plen} + f)$$

$$= -92136 \cdot (0.5 \cdot 1200 + 550)$$

$$= -105955347.14 \text{ mm}^3$$

Iteratively Solving for the Effective Bearing Length:

$$Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$$

$$Y^3 + -5.02 \cdot Y^2 + 92136 \cdot Y + -105955344 = 0$$

$$Y = 410.17 \text{ mm}$$

$$\text{Num} = (B_{plen} / 2 - Y / 3 - e)$$

$$= (1200/2 - 410/3 - 598)$$

$$= -135.05$$

$$\text{Denom} = (B_{plen} / 2 - Y / 3 + f)$$

$$= (1200/2 - 410/3 + 550)$$

$$= 1013.26$$

Total Bolt Tension Force [Tforce]:

$$= -QO \cdot \text{Num} / \text{Denom}$$

$$= -1373 \cdot -135/1013$$

$$= 182.96 \text{ Kg}$$

Bolt Area Required due to Transverse Load [Bltareart]:

$$= T_{force} / (S_{tba} \cdot N_{bt})$$

$$= 183 / (138 \cdot 2)$$

$$= 0.0650 \text{ cm}^2$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[B_{ltarearl}, B_{ltarears}, B_{ltareart}]$$

$$= \max[0, 0.42, 0.065]$$

$$= 0.4157 \text{ cm}^2$$

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)

$$= 2(QO + T_{force}) / (Y \cdot B_{pwid})$$

$$= 2(1373 + 183) / (410 \cdot 430)$$

$$= 172.99 \text{ KPa.}$$

Distance from Baseplate Edge to the Web [ADIST]:

$$= (B_{plen} - W_{blength}) / 2$$

$$= (1200 - 1149) / 2$$

$$= 25.4000 \text{ mm}$$

Overturning Moment due To Bolt Tension [Mt]:

$$= T_{force} \cdot A_{dist}$$

$$= 183 \cdot 25.4$$

$$= 4.65 \text{ Kg-m.}$$



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Saddle Calcs: Operating Case: Step: 19 9:43am Apr 21,2026

Equivalent Bearing Pressure (f1):

$$\begin{aligned} &= f_c \cdot (Y - \text{Adist}) / Y \\ &= 173 \cdot (410 - 25.4) / 410 \\ &= 162.28 \text{ KPa.} \end{aligned}$$

Overturning Moment due to Bearing Pressure [Mc]:

$$\begin{aligned} &= (\text{Adist}^2 \cdot \text{Bpwid} / 6) \cdot (f1 + 2 \cdot f_c) \\ &= (25.4^2 \cdot 430 / 6) \cdot (162 + 2 \cdot 173) \\ &= 2.40 \text{ Kg-m.} \end{aligned}$$

Baseplate Required Thickness [Treq]:

$$\begin{aligned} &= \sqrt{(6 \cdot \max(\text{Mt}, \text{Mc}) / (\text{Bpwid} \cdot \text{Sba}))} \\ &= \sqrt{(6 \cdot \max(4.65, 2.4) / (430 \cdot 222))} \\ &= 1.6925 \text{ mm} \end{aligned}$$

MDMT Calculations for: Right saddle, Saddle to Wearplate Junction, Curve: A

Govrn. thk, tg = 15, c = 3 mm, E* = 1

Thickness Ratio = tr * E* / (tg - c) = 0.74, Temp. Reduction = 15 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve A 4 °C

Min Metal Temp. at Required thickness (UCS 66.1) -11 °C

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Test Case:

Step: 20 9:43am Apr 21,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (250.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$$

$$= \min(203 + 1.56 \cdot \sqrt{ 564 \cdot 28 }, 2 \cdot 300)$$

$$= 399.0393 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	564.00	mm
Shell Thickness used in this Case	t	28.000	mm
Stiffened Vessel Length per 4.15.6	L	4050.00	mm
Distance from Saddle to Vessel tangent	a1 or a	300.00	mm
Saddle Width	b1 or b	203.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		279.01	N./mm ²
Head Allowable Stress used in Calculation		279.01	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1070.00	mm
Coefficient of Friction	mu	0.40	
Saddle Force Q, Test Case, no Ext. Forces		26265.63	Kgf
Pressure used in Analysis	P	5920.528	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	52.71	279.01
Long. Stress at Bottom of Midspan	66.55	279.01
Long. Stress at Top of Saddles	58.16	279.01
Long. Stress at Bottom of Saddles	60.45	279.01
Tangential Shear in Shell	16.27	165.24
Circ. Stress at Horn of Saddle	14.41	418.52
Circ. Compressive Stress in Shell	1.75	279.01

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (217/2 + 0) \cdot 1070/1027$$

$$= 339.5 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_l, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(103, 0) \cdot 1070/3250$$

$$= 33.9 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 25926 + \max(33.9, 339, 0, 0)$$

$$= 26265.6 \text{ Kgf}$$



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Saddle Calcs: Test Case:

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Longitudinal Wind Force [F_L]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (1.46 \cdot 1.2)^2)$$

$$= 2020.437 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 26491.83 Kg_f

Transverse Shear Load 108.62 Kg_f

Longitudinal Shear Load 103.02 Kg_f

Formulas and Substitutions for Horizontal Vessel Analysis:

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1100/2 + 28 + 15) \sin(120/2), 2 \cdot 564)$$

$$= 1027.106 \text{ mm}$$

The Computed K values from Table 4.15.1:

$$K1 = 0.1066 \quad K2 = 1.1707 \quad K3 = 0.8799 \quad K4 = 0.4011$$

$$K5 = 0.7603 \quad K6 = 0.0529 \quad K7 = 0.0157 \quad K8 = 0.3405$$

$$K9 = 0.2711 \quad K10 = 0.0581 \quad K1^* = 0.1923$$

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M₁]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h m^2)/(2a \cdot L)) / (1 + (4h m^2)/3L)]$$

$$= -26266 \cdot 300 [1 - (1 - 300/4050 + (564^2 - 0^2)/$$

$$(2 \cdot 300 \cdot 4050) / (1 + (4 \cdot 0) / (3 \cdot 4050))]$$

$$= 447.8 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M₂]:

$$= Q \cdot L / 4 (1 + 2(R_m^2 - h m^2)/(L^2)) / (1 + (4h m^2)/(3L)) - 4a/L$$

$$= 26266 \cdot 4050 / 4 (1 + 2(564^2 - 0^2)/(4050^2)) / (1 + (4 \cdot 0) /$$

$$(3 \cdot 4050)) - 4 \cdot 300 / 4050$$

$$= 19746.1 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= 5921 \cdot 564 / (2 \cdot 28) - 19746 / (\pi \cdot 564^2 \cdot 28)$$

$$= 52.71 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 5921 \cdot 564 / (2 \cdot 28) + 19746 / (\pi \cdot 564^2 \cdot 28)$$

$$= 66.55 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M_1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 5921 \cdot 564 / (2 \cdot 28) - 448 / (0.11 \cdot \pi \cdot 564^2 \cdot 28)$$

$$= 58.16 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M_1 / (K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 5921 \cdot 564 / (2 \cdot 28) + 448 / (0.19 \cdot \pi \cdot 564^2 \cdot 28)$$

$$= 60.45 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h m / 3))$$

$$= 26266 (4050 - 2 \cdot 300) / (4050 + (4 \cdot 0 / 3))$$

$$= 22374.4 \text{ Kg_f}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:



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$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 22374 / (564 \cdot 28)$$

$$= 16.27 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{ (Rm \cdot t) }$$

$$= 0.78 \cdot \sqrt{ (564 \cdot 28) }$$

$$= 98.020 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t (b + X1 + X2))$$

$$= -0.76 \cdot 26266 \cdot 0.1 / (28 (203 + 98 + 98))$$

$$= -1.75 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.23) [σ_7^*]:

$$= -Q / (4 \cdot t (b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot Rm / (L \cdot t^2)$$

$$= -26266 / (4 \cdot 28 (203 + 98 + 98)) -$$

$$12 \cdot 0.016 \cdot 26266 \cdot 564 / (4050 \cdot 28^2)$$

$$= -14.41 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{ (Rm \cdot t) }, 2a)$$

$$= \min(203 + 1.56 \cdot \sqrt{ (564 \cdot 28) }, 2 \cdot 300)$$

$$= 399.04 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1200.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	430.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	310.0	N./
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{ (R \cdot t) } + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{ (550 \cdot 28) } + 250) 28^3) / 12$$

$$= 81.147 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 81.1 + 124 (202 - 14)^2$$

$$= 44091.402 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	443.6	28.0	14.0	124.2	173887.6	0.441E+05
Wearplate	250.0	15.0	35.5	37.5	133125.0	0.104E+05
Web	10.0	457.0	271.5	45.7	1240754.9	0.101E+05
BasePlate	430.0	20.0	510.0	86.0	4385999.5	0.815E+05
Totals	...	...	...	293.4	5933767.0	0.146E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 5933767 / 293$$

$$= 202.238 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$



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Saddle Calcs: Test Case:

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$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 26266$$

$$= 5345.6255 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 3.0983 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 185.9966 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1070 - 550 \cdot \sin(120/2) / 1.05$$

$$= 615.154 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 3288.4460 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 4.4623 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 206.6629 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(26266 + 226)430 / (4 \cdot 1200 \cdot 207)}$$

$$= 18.381 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (550 + 28 + 15)$$

$$= 1027.106 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 1027 / (3 - 1)$$

$$= 513.553 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot B_{\text{pwid}} / 2$$

$$= 514 \cdot 430 / 2$$

$$= 1104.139 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= Q / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 26266 / (1200 \cdot 430)$$

$$= 5.090 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1104 \cdot 5.09$$

$$= 5620.332 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 19.3 + 25.7$$

$$= 44.978 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$



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Saddle Calcs: Test Case:

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$$= 5620/45$$

$$= 12.254 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	203.0	...	20.3	...	697.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 203 - 10$$

$$= 193.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/45$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 203/2$$

$$= 101.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(697/45)}$$

$$= 39.369 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1070 - \cos(120/2) (550 + 28 + 15) - 20$$

$$= 753.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943392/310)}$$

$$= 112.838$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 754/39.4$$

$$= 19.139$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 103 / (2 \cdot 1200) \cdot 514 \cdot 754/2$$

$$= 8.305 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($19.1 < 113$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (19.1)^2 / (2 \cdot 113^2)) 310 /$$

$$(5/3 + 3 \cdot (19.1) / (8 \cdot 113) - (19.1^3) / (8 \cdot 113^3))$$

$$= 176.6 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 12.3/177 + (8.3 \cdot 102/6971188) / 207$$

$$= 0.075$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction



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Saddle Calcs: Test Case:

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	B	D	Y	A	AY	Io
Rib	10.0	193.0	0.0	19.3	0.0	697.
Web	513.6	10.0	0.0	51.4	0.0	4.28
Totals	...	...	...	70.7	...	701.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/70.7$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 203/2$$

$$= 101.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1070 - \cos(593/2)(20 + 0 + 0) - 0$$

$$= 457.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(701/70.7)}$$

$$= 31.505 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 457/31.5$$

$$= 14.505$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 103/(2 \cdot 1200)$$

$$= 0.043 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.043 \cdot 514 \cdot 457$$

$$= 10.074 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 10.1/69095$$

$$= 1.430 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($14.5 < 113$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (14.5)^2 / (2 \cdot 113^2)) 310 / (5/3 + 3 \cdot (14.5) / (8 \cdot 113) - (14.5)^3 / (8 \cdot 113^3))$$

$$= 179.3 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$$

$$= 12.1/179 + (10.1 \cdot 102/701) / 207$$

$$= 0.075$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4	
Total Number of Bolts in Tension/Baseplate	Nbt	2	
Bolt Material Specification		SA-307 A	
Bolt Allowable Stress	Stba	138.00	N./mm ²
Bolt Corrosion Allowance	Bca	1.5	mm
Distance from Bolts to Edge	Edgedis	50.0	mm



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Saddle Calcs: Test Case:

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User Defined Area in a Single Bolt	UserBoltArea	4.0400	cm ²
BasePlate Allowable Stress	S	148.00	N./mm ²
Area Available in a Single Bolt	BltArea	3.0428	cm ²
Saddle Load Q0 (Weight)	Q0	26152.4	Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	33.9	Kgf
Maximum Transverse Force	Ft	108.6	Kgf
Maximum Longitudinal Force	F1	103.0	Kgf
Saddle Bolted to Steel Foundation		No	

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 103 / (3.04 \cdot 4)$$

$$= 0.8 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 109 / (3.04 \cdot 4)$$

$$= 0.9 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (Q0 > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbols})$$

$$= 103 / (82.8 \cdot 4)$$

$$= 0.0305 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1070 \cdot 109 + 0$$

$$= 116.22 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 116 / 26152$$

$$= 4.44 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.031, 0]$$

$$= 0.0305 \text{ cm}^2$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (250.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$$

$$= \min(203 + 1.56 \cdot \sqrt{564 \cdot 28}, 2 \cdot 300)$$

$$= 399.0393 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	564.00	mm
Shell Thickness used in this Case	t	28.000	mm



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Saddle Calcs: Test Case: Step: 20 9:43am Apr 21,2026

Stiffened Vessel Length per 4.15.6	L	4050.00	mm
Distance from Saddle to Vessel tangent	a1 or a	300.00	mm
Saddle Width	b1 or b	203.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	286.90	mm
Shell Allowable Stress used in Calculation		279.01	N./mm ²
Head Allowable Stress used in Calculation		279.01	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1170.00	mm
Coefficient of Friction	mu	0.40	
Saddle Force Q, Test Case, no Ext. Forces		628.94	Kgf
Pressure used in Analysis	P	5920.528	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	59.49	279.01
Long. Stress at Bottom of Midspan	59.78	279.01
Long. Stress at Top of Saddles	59.67	279.01
Long. Stress at Bottom of Saddles	59.61	279.01
Tangential Shear in Shell	0.36	165.24
Circ. Stress at Horn of Saddle	0.34	418.52
Circ. Compressive Stress in Shell	0.04	279.01

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (217/2 + 0) \cdot 1170/1027$$

$$= 371.2 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_l, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(103, 0) \cdot 1170/3250$$

$$= 37.1 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 258 + \max(37.1, 371, 0, 0)$$

$$= 628.9 \text{ Kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (1.46 \cdot 1.2)^2)$$

$$= 2020.437 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	869.09	Kgf
Transverse Shear Load	108.62	Kgf
Longitudinal Shear Load	103.02	Kgf

Formulas and Substitutions for Horizontal Vessel Analysis:

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:



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Saddle Calcs: Test Case:

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$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1100/2 + 28 + 15) \sin(120/2), 2 \cdot 564)$$

$$= 1027.106 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0157	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m^2)/3L)]$$

$$= -629 \cdot 300 [1 - (1 - 300/4050 + (564^2 - 287^2)/(2 \cdot 300 \cdot 4050))/(1 + (4 \cdot 287^2)/(3 \cdot 4050))]$$

$$= -12.3 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 (1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/3L) - 4a/L$$

$$= 629 \cdot 4050/4 (1 + 2(564^2 - 287^2)/(4050^2))/(1 + (4 \cdot 287^2)/(3 \cdot 4050)) - 4 \cdot 300/4050$$

$$= 409.9 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 5921 \cdot 564/(2 \cdot 28) - 410/(\pi \cdot 564^2 \cdot 28)$$

$$= 59.49 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 5921 \cdot 564/(2 \cdot 28) + 410/(\pi \cdot 564^2 \cdot 28)$$

$$= 59.78 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= 5921 \cdot 564/(2 \cdot 28) - 12.3/(0.11 \cdot \pi \cdot 564^2 \cdot 28)$$

$$= 59.67 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m/(2t) + M1/(K1* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 5921 \cdot 564/(2 \cdot 28) + 12.3/(0.19 \cdot \pi \cdot 564^2 \cdot 28)$$

$$= 59.61 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 629 (4050 - 2 \cdot 300)/(4050 + (4 \cdot 287/3))$$

$$= 489.5 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 490/(564 \cdot 28)$$

$$= 0.36 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(564 \cdot 28)}$$

$$= 98.020 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.76 \cdot 629 \cdot 0.1/(28(203 + 98 + 98))$$

$$= -0.04 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_{7*}]:

$$= -Q/(4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot R_m/(L \cdot t^2)$$



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$$= -629 / (4 \cdot 28 (203 + 98 + 98)) - 12 \cdot 0.016 \cdot 629 \cdot 564 / (4050 \cdot 28^2) = -0.34 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a) = \min(203 + 1.56 \cdot \sqrt{(564 \cdot 28)}, 2 \cdot 300) = 399.04 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1200.0000	mm
Baseplate Thickness	Bpthk	20.0000	mm
Baseplate Width	Bpwid	430.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	310.0	N./
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.400	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12 = ((1.56 \cdot \sqrt{(550 \cdot 28)} + 250)28^3) / 12 = 81.147 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2 = 81.1 + 124 (242 - 14)^2 = 64673.371 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	443.6	28.0	14.0	124.2	173887.6	0.647E+05
Wearplate	250.0	15.0	35.5	37.5	133125.0	0.160E+05
Web	10.0	557.0	321.5	55.7	1790754.8	0.179E+05
BasePlate	430.0	20.0	610.0	86.0	5246000.0	0.116E+06
Totals	...	...	...	303.4	7343767.5	0.215E+06

Distance to Centroid [C1]:

$$= AY / A = 7343767 / 303 = 242.045 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2 = 180 - 120 / 2 = 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) = (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) = 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q = 0.2 \cdot 629 = 128.0031 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.0701 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 185.9966 \text{ N./mm}^2$$



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Saddle Calcs: Test Case:

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Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1170 - 550 \cdot \sin(120/2) / 1.05$$

$$= 715.154 \text{ mm}$$

$$\text{Bending Moment, } M = F_h \cdot d = 91.5437 \text{ Kg-m.}$$

$$\text{Bending Stress, } S_b = (M \cdot C_1 / I) = 0.1010 \text{ N./mm}^2$$

$$\text{Allowed Stress, } S_a = 2/3 \cdot \text{Yield Str} = 206.6629 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(629 + 240) \cdot 430) / (4 \cdot 1200 \cdot 207)}$$

$$= 3.329 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (550 + 28 + 15)$$

$$= 1027.106 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1027 / (3 - 1)$$

$$= 513.553 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 514 \cdot 430 / 2$$

$$= 1104.139 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= Q / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 629 / (1200 \cdot 430)$$

$$= 0.122 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 1104 \cdot 0.12$$

$$= 134.581 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 19.3 + 25.7$$

$$= 44.978 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 135 / 45$$

$$= 0.293 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	203.0	...	20.3	...	697.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 203 - 10$$

$$= 193.000 \text{ mm}$$



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Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/45$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 203/2$$

$$= 101.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(697/45)}$$

$$= 39.369 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1170 - \cos(120/2) (550 + 28 + 15) - 20$$

$$= 853.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943392/310)}$$

$$= 112.838$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 853/39.4$$

$$= 21.679$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 103 / (2 \cdot 1200) \cdot 514 \cdot 853/2$$

$$= 9.407 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($21.7 < 113$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (21.7)^2 / (2 \cdot 113^2)) 310 / (5/3 + 3 \cdot (21.7) / (8 \cdot 113) - (21.7^3) / (8 \cdot 113^3))$$

$$= 175.1 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 0.29/175 + (9.41 \cdot 102/6971188) / 207$$

$$= 0.008$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	193.0	0.0	19.3	0.0	697.
Web	513.6	10.0	0.0	51.4	0.0	4.28
Totals	...	...	...	70.7	...	701.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/70.7$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 203/2$$

$$= 101.500 \text{ mm}$$



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Saddle Calcs: Test Case:

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Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1170 - \cos(593/2)(20 + 0 + 0) - 0$$

$$= 557.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{701/70.7}$$

$$= 31.505 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 557/31.5$$

$$= 17.680$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 103/(2 \cdot 1200)$$

$$= 0.043 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.043 \cdot 514 \cdot 557$$

$$= 12.278 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 12.3/69095$$

$$= 1.743 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($17.7 < 113$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (17.7)^2 / (2 \cdot 113^2)) 310 / (5/3 + 3 \cdot (17.7) / (8 \cdot 113) - (17.7)^3 / (8 \cdot 113^3))$$

$$= 177.5 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 0.29/178 + (12.3 \cdot 102/701) / 207$$

$$= 0.010$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4	
Total Number of Bolts in Tension/Baseplate	Nbt	2	
Bolt Material Specification		SA-307 A	
Bolt Allowable Stress	Stba	138.00	N./mm ²
Bolt Corrosion Allowance	Bca	1.5	mm
Distance from Bolts to Edge	Edgedis	50.0	mm
User Defined Area in a Single Bolt	UserBoltArea	4.0400	cm ²
BasePlate Allowable Stress	S	148.00	N./mm ²
Area Available in a Single Bolt	BltArea	3.0428	cm ²
Saddle Load Q0 (Weight)	Q0	497.9	Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	37.1	Kgf
Maximum Transverse Force	Ft	108.6	Kgf
Maximum Longitudinal Force	F1	103.0	Kgf
Saddle Bolted to Steel Foundation		No	

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 103/(3.04 \cdot 4)$$

$$= 0.8 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:



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Saddle Calcs: Test Case: Step: 20 9:43am Apr 21, 2026

$$\begin{aligned}
 &= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\
 &= 109 / (3.04 \cdot 4) \\
 &= 0.9 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.
 \end{aligned}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_t / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 = $103 / (82.8 \cdot 4)$
 = 0.0305 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = $1170 \cdot 109 + 0$
 = 127.08 Kg-m.

Eccentricity (e):
 = R_{mom} / QO
 = 127/498
 = 255.23 mm > Bplen/6 --> Uplift in Transverse direction

$f = B_{plen} / 2 - \text{Edgedis}$
 = $1200/2 - 50$
 = 549.99 mm

Modular Ratio Of Steel/Concrete (n1):
 = E_S / E_C
 = 203402/21526
 = 9.45

$K1 = 3 (e - 0.5 \cdot B_{plen})$
 = $3 (255 - 0.5 \cdot 1200)$
 = -1034.32 mm

$K2 = 6 \cdot n1 \cdot A_t / B_{pwid} \cdot (f + e)$
 = $6 \cdot 9.45 \cdot 6.09/430 \cdot (550 + 255)$
 = 64607.15 mm²

$K3 = -K2 \cdot (0.5 \cdot B_{plen} + f)$
 = $-64607 \cdot (0.5 \cdot 1200 + 550)$
 = -74297403.77 mm³

Iteratively Solving for the Effective Bearing Length:
 $Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$
 $Y^3 + -1034 \cdot Y^2 + 64607 \cdot Y + -74297408 = 0$
 $Y = 1040.83 \text{ mm}$

$\text{Num} = (B_{plen} / 2 - Y / 3 - e)$
 = $(1200/2 - 1041/3 - 255)$
 = -2.17

$\text{Denom} = (B_{plen} / 2 - Y / 3 + f)$
 = $(1200/2 - 1041/3 + 550)$
 = 803.04

Total Bolt Tension Force [Tforce]:
 = $-QO \cdot \text{Num} / \text{Denom}$
 = $-498 \cdot -2.17/803$
 = 1.35 Kg



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Bolt Area Required due to Transverse Load [Bltareart]:

$$= T_{force} / (S_{tba} \cdot N_{bt})$$

$$= 1.35 / (138 \cdot 2)$$

$$= 0.0005 \text{ cm}^2$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.031, 0.00048]$$

$$= 0.0305 \text{ cm}^2$$

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)

$$= 2(QO + T_{force}) / (Y \cdot B_{pwid})$$

$$= 2(498 + 1.35) / (1041 \cdot 430)$$

$$= 21.88 \text{ KPa.}$$

Distance from Baseplate Edge to the Web [ADIST]:

$$= (B_{plen} - WebLength) / 2$$

$$= (1200 - 1149) / 2$$

$$= 25.4000 \text{ mm}$$

Overturning Moment due To Bolt Tension [Mt]:

$$= T_{force} \cdot A_{dist}$$

$$= 1.35 \cdot 25.4$$

$$= 0.03 \text{ Kg-m.}$$

Equivalent Bearing Pressure (f1):

$$= f_c \cdot (Y - A_{dist}) / Y$$

$$= 21.9 \cdot (1041 - 25.4) / 1041$$

$$= 21.34 \text{ KPa.}$$

Overturning Moment due to Bearing Pressure [Mc]:

$$= (A_{dist}^2 \cdot B_{pwid} / 6) \cdot (f1 + 2 \cdot f_c)$$

$$= (25.4^2 \cdot 430 / 6) \cdot (21.3 + 2 \cdot 21.9)$$

$$= 0.31 \text{ Kg-m.}$$

Baseplate Required Thickness [Treq]:

$$= \sqrt{(6 \cdot \max(Mt, Mc) / (B_{pwid} \cdot S_{ba}))}$$

$$= \sqrt{(6 \cdot \max(0.034, 0.31) / (430 \cdot 222))}$$

$$= 0.4350 \text{ mm}$$

Review notes about nozzle loadings and supports in the Warnings report.

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Nozzle Calcs.: N3- Tube inlet

Nozl: 15 9:43am Apr 21,2026

Input, Nozzle Desc: N3- Tube inlet

From: 30

Pressure for Reinforcement Calculations	P	13819.999	KPa.
Temperature for Internal Pressure	Temp	400	°C
Design External Pressure	Pext	101.30	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	148.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1100.00	mm
Design Length of Section	L	999.9695	mm
Shell Finished (Minimum) Thickness	t	64.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		941.85	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	148.00	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	20.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		40.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	60.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		1500	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



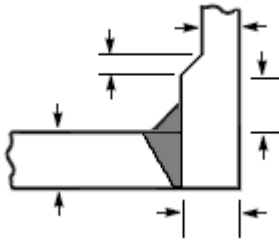
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Nozzle Calcs.: N3- Tube inlet

Nozl: 15 9:43am Apr 21,2026



Hub Nozzle (Set-in)

Reinforcement CALCULATION, Description: N3- Tube inlet

ASME Code, Section VIII, Div. 1, 2021, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.624 in.
Actual Thickness Used in Calculation 0.438 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $13820 \cdot 553 / (148 \cdot 1 - 0.6 \cdot 13820)$
= 54.7075 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $13820 \cdot 36.3 / (148 \cdot 1 - 0.6 \cdot 13820)$
= 3.5936 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4354 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 28.9/0.5773
= 50.0126 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	210.9000	mm
Parallel to Vessel Wall	Rn+tn+t	105.4500	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	70.3256	mm

Weld Strength Reduction Factor [fr1]:

= min(1, Sn/Sv)
= min(1, 148/148)
= 1.000

Weld Strength Reduction Factor [fr2]:

= min(1, Sn/Sv)
= min(1, 148/148)
= 1.000



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Nozzle Calcs.: N3- Tube inlet

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Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)

= min(1, 1)

= 1.000

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	39.745	1.321	NA
Area in Shell	A1	8.699	79.307	NA
Area in Nozzle Wall	A2	6.374	10.816	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.250	2.250	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	34.650	34.650	NA
TOTAL AREA AVAILABLE	Atot	51.973	127.022	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

= (d • tr•F + 2 • tn • tr•F • (1-fr1)) UG-37(c)

= (72.6•54.7•1+2•37•54.7•1•(1-1))

= 39.745 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

= d(E1•t - F•tr) - 2 • tn(E1•t - F•tr) • (1 - fr1)

= 138 (1 • 61 - 1 • 54.7) - 2 • 37

(1 • 61 - 1 • 54.7) • (1 - 1)

= 8.699 cm²

Area Available in Nozzle Projecting Outward [A2]:

= (2 • tlnp)(tn - trn)fr2

= (2 • 70.3)(8.13 - 3.59)1

= 6.374 cm²

Area Available in Inward Weld + Outward Weld [A41 + A43]:

= Wo² • fr2 + (Wi-can/0.707)² • fr2

= 15² • 1 + (0)² • 1

= 2.250 cm²

Area Available in the Hub Section [A6]:

= (2 • min(Tlnp,ho,Hubht)) • (Hubtk - tn) • fr2

= (2 • min(70.3,250,50)) • (40 - 11.1) • 1)

= 34.650 cm² Includes Bevel Area of 5.775 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 6.5936 mm

Wall Thickness per UG16(b), tr16b = 5.5000 mm

Wall Thickness, shell/head, internal pressure trb1 = 57.7075 mm

Wall Thickness tb1 = max(trb1, tr16b) = 57.7075 mm

Wall Thickness, shell/head, external pressure trb2 = 3.3787 mm

Wall Thickness tb2 = max(trb2, tr16b) = 5.5000 mm

Wall Thickness per table UG-45 tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]

= min[7.8, max(57.7, 5.5)]



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Nozzle Calcs.: N3- Tube inlet

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= 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)

= max(6.59, 7.8)

= 7.8000 mm

Available Nozzle Neck Thickness = $0.875 \times 11.125 = 9.735$ mm→OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	112.2,	Allowable	:	148.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	257.8 N./mm ²	Passed
Occasional	:	27.8,	Allowable	:	196.8 N./mm ²	Passed
Shear	:	62.1,	Allowable	:	103.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 9.73, tr = 3.59, c = 3 mm, E* = 1

Thickness Ratio = $tr \times E^*/(tg - c) = 0.53$, Temp. Reduction = 28 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 40, tr = 3.59, c = 3 mm, E* = 1

Thickness Ratio = $tr \times E^*/(tg - c) = 0.097$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 31 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -46 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $13820.00/25860.00 = 0.534$

Weld Size Calculations, Description: N3- Tube inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness

Nozzle Weld 6.0000 = Min per Code 10.6050 = $0.7 \times Wo$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

= max(0, (A-A1+2*tn*fr1*(E1*t-tr))Sv)

= max(0, (39.7 - 8.7 + 2 * 37 * 1 *

(1 * 61 - 54.7))148)

= 53878.52 Kgf

For hub type nozzles, A2 includes the area of the hub.



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Nozzle Calcs.: N3- Tube inlet

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Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr}2) \cdot S_v$$

$$= (41 + 2.25 - 0 \cdot 1) \cdot 148$$

$$= 65305.22 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S_v$$

$$= (41 + 0 + 2.25 + (45.1)) \cdot 148$$

$$= 133427.12 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S$$

$$= (41 + 0 + 2.25 + 0 + (45.1)) \cdot 148$$

$$= 133427.12 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 147 \cdot 15 \cdot 0.49 \cdot 148$$

$$= 25551. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 54.8) \cdot (40 - 3) \cdot 0.7 \cdot 148$$

$$= 67321. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnv} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 147 \cdot (60 - 0) \cdot 0.74 \cdot 148$$

$$= 154351. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (25551 + 67321) = 92873 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (25551 + 0 + 154351 + 0) = 179902 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (25551 + 154351 + 0) = 179902 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 92872 Kgf, must exceed W = 53878 Kgf	or W1 = 65305 Kgf
Path 2-2 = 179902 Kgf, must exceed W = 53878 Kgf	or W2 = 133427 Kgf
Path 3-3 = 179902 Kgf, must exceed W = 53878 Kgf	or W3 = 133427 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 15137 KPa.

Nozzle is O.K. for the External Pressure: 101 KPa.

The Drop for this Nozzle is : 4.9096 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 318.9096 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center



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Nozzle Calcs.: N3- Tube inlet

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of the nozzle to the edge of the diameter limit is: 105.45 mm

Input Echo, WRC297 Item 1, Description: N3- Tube inlet:

Diameter Basis for Cylindrical Shell	ID		
Shell Corrosion Allowance	3.0000	mm	
Shell Diameter	1100.000	mm	
Shell Thickness	64.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-387 11		
Vessel Cold S.I. Allowable	Smc 148.00	N./mm ²	
Vessel Hot S.I. Allowable	Smh 148.00	N./mm ²	

Using 2 * Yield for Discontinuity Stress Allowable (Div 2, 4.1.6.3), Sps.
Make sure that material properties at this temperature are not
time-dependent for Material: SA-387 11

Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	66.650	mm	
Nozzle Thickness	40.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-336 F11		
Nozzle Cold S.I. Allowable	SNmc 148.00	N./mm ²	
Nozzle Hot S.I. Allowable	SNmh 148.00	N./mm ²	

Using 2 * Yield for Discontinuity Stress Allowable (Div 2, 4.1.6.3), Sps.
Make sure that material properties at this temperature are not
time-dependent for Material: SA-336 F11

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	13820.00	KPa.
Radial Load	P	-1183.00	Kgf
Circumferential Shear	Vc	979.00	Kgf
Longitudinal Shear	Vl	979.00	Kgf
Circumferential Moment	Mc	248.00	Kg-m.
Longitudinal Moment	Ml	248.00	Kg-m.
Torsional Moment	Mt	326.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dv/Tv (19.188) is < 20
check the limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Tv, (2.291) is < 5
check the limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Tn (3.666) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
= NozzleOD + 2 * 1.65 * $\sqrt{(R_{mean}(t - ca))}$
= 146.65 + 2 * 1.65 * $\sqrt{(583.5(64.0 - 3.0))}$
= 769.235 mm

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1167.000 mm



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Nozzle Calcs.: N3- Tube inlet

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Nozzle Outside Diameter (d) = 146.650 mm
 Vessel Thickness used (T) = 61.000 mm
 Nozzle Thickness used (t) = 37.000 mm
 T / t = 1.649
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.550$$

Nr/P = 0.184
 Mr/P = 0.210
 M0/P = 0.101
 N0/P = 0.160
 MrD/Mc = 0.248
 NrDL/Mc = 0.179
 M0d/Mc = 0.152
 N0DL/Mc = -0.096
 MrD/Ml = 0.222
 NrDL/Ml = 0.168
 M0D/Ml = 0.134
 N0DL/Ml = 0.028

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	1	-1	1	-1
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-3	3	3	-3
Normal Pressure Stress	118	132	118	132
Outplane Stress Summary	117	134	124	127

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-5	5	5	-5
Inplane Pressure Stress	59	59	59	59
Inplane Stress Summary	57	61	70	50

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top	Al Top	Bu Bottom	Bl Bottom
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Nozzle Calcs.: N3- Tube inlet

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	Outside	Inside	Outside	Inside
Two * Max Shear Stress	117	134	124	127

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	1	-1	1	-1
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-4	4	4	-4
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	59	59	59	59
Outplane Stress Summary	58	62	65	53

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-6	6	6	-6
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	118	132	118	132
Inplane Stress Summary	115	135	130	123

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	115	135	130	123

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10	27	10	27



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Nozzle Calcs.: N3- Tube inlet

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Hoop Stress Summary	11	27	11	27
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-8	-8	8	8
Axial Bending (ML)	-10	10	10	-10
Axial Pressure Stress	5	5	5	5

Axial Stress Summary	-6	1	30	-1
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2

Shear Stress Summary	3	3	1	1
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	18	28	30	29
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10	27	10	27

Hoop Stress Summary	11	28	10	27
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	-8	-8	8	8
Axial Bending (Mc)	-11	11	11	-11
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5

Axial Stress Summary	-7	3	32	-2
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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Nozzle Calcs.: N3- Tube inlet

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(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	2	2	2	2
Shear Stress Summary	1	1	3	3

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	19	28	32	31

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	118	132	118	132	118	132	118	132
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	-2	2	4	-4	-3	3	9	-9
Long. Pm (SUS)	59	59	59	59	59	59	59	59
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	-2	2	8	-8	-3	3	5	-5
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	118.0	132.0	118.0	132.0	118.0	132.0	118.0	132.0
Pm+Pl (SUS)	118.0	132.0	118.0	132.0	118.0	132.0	118.0	132.0
Pm+Pl+Q (Total)	116.0	134.0	122.0	128.0	115.0	135.0	127.0	123.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	132.00	148.00	Passed
Pm+Pl (SUS)	132.00	222.00	Passed
Pm+Pl+Q (TOTAL)	135.01	443.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	10	27	10	27	10	27	10	27
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5



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Long. P1 (SUS)	-8	-8	8	8	-8	-8	8	8
Long. Q (SUS)	-5	5	15	-15	-6	6	16	-16
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	10.0	27.0	10.0	27.0	10.0	27.0	10.0	27.0
Pm+P1 (SUS)	13.2	30.1	13.3	27.1	13.2	30.1	13.3	27.1
Pm+P1+Q (Total)	19.0	27.4	28.1	29.1	19.1	27.0	29.5	30.6

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	27.00	148.00	Passed
Pm+P1 (SUS)	30.07	222.00	Passed
Pm+P1+Q (TOTAL)	30.59	443.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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Nozzle Calcs.: N4- Tube outlet

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Input, Nozzle Desc: N4- Tube outlet

From: 30

Pressure for Reinforcement Calculations	P	13827.008	KPa.
Temperature for Internal Pressure	Temp	400	°C
Design External Pressure	Pext	101.30	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	148.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1100.00	mm
Design Length of Section	L	999.9695	mm
Shell Finished (Minimum) Thickness	t	64.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		941.85	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	148.00	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	20.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		40.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	56.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		1500	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



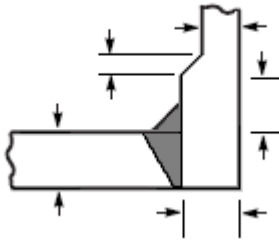
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Nozzle Calcs.: N4- Tube outlet

Nozl: 16 9:43am Apr 21,2026



Hub Nozzle (Set-in)

Reinforcement CALCULATION, Description: N4- Tube outlet

ASME Code, Section VIII, Div. 1, 2021, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.624 in.
Actual Thickness Used in Calculation	0.438 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 13827 \cdot 553 / (148 \cdot 1 - 0.6 \cdot 13827)$
 $= 54.7369 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 13827 \cdot 36.3 / (148 \cdot 1 - 0.6 \cdot 13827)$
 $= 3.5955 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4354 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 28.9 / 0.5773$
 $= 50.0126 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	210.9000	mm
Parallel to Vessel Wall	Rn+tn+t	105.4500	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	70.3256	mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 148 / 148)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 148 / 148)$
 $= 1.000$



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Nozzle Calcs.: N4- Tube outlet

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Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)

= min(1, 1)

= 1.000

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	39.766	1.321	NA
Area in Shell	A1	8.659	79.307	NA
Area in Nozzle Wall	A2	6.371	10.816	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.250	2.250	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	34.650	34.650	NA
TOTAL AREA AVAILABLE	Atot	51.930	127.022	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

= (d • tr•F + 2 • tn • tr•F • (1-fr1)) UG-37(c)

= (72.6•54.7•1+2•37•54.7•1•(1-1))

= 39.766 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

= d(E1•t - F•tr) - 2 • tn(E1•t - F•tr) • (1 - fr1)

= 138 (1 • 61 - 1 • 54.7) - 2 • 37

(1 • 61 - 1 • 54.7) • (1 - 1)

= 8.659 cm²

Area Available in Nozzle Projecting Outward [A2]:

= (2 • tlnp)(tn - trn)fr2

= (2 • 70.3)(8.13 - 3.6)1

= 6.371 cm²

Area Available in Inward Weld + Outward Weld [A41 + A43]:

= Wo² • fr2 + (Wi-can/0.707)² • fr2

= 15² • 1 + (0)² • 1

= 2.250 cm²

Area Available in the Hub Section [A6]:

= (2 • min(Tlnp,ho,Hubht)) • (Hubtk - tn) • fr2

= (2 • min(70.3,250,50)) • (40 - 11.1) • 1)

= 34.650 cm² Includes Bevel Area of 5.775 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 6.5955 mm

Wall Thickness per UG16(b), tr16b = 5.5000 mm

Wall Thickness, shell/head, internal pressure trb1 = 57.7369 mm

Wall Thickness tb1 = max(trb1, tr16b) = 57.7369 mm

Wall Thickness, shell/head, external pressure trb2 = 3.3787 mm

Wall Thickness tb2 = max(trb2, tr16b) = 5.5000 mm

Wall Thickness per table UG-45 tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]

= min[7.8, max(57.7, 5.5)]



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Nozzle Calcs.: N4- Tube outlet Nozl: 16 9:43am Apr 21, 2026

= 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)

= max(6.6, 7.8)

= 7.8000 mm

Available Nozzle Neck Thickness = $0.875 \times 11.125 = 9.735$ mm→OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	112.2,	Allowable	:	148.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	257.8 N./mm ²	Passed
Occasional	:	27.8,	Allowable	:	196.8 N./mm ²	Passed
Shear	:	61.7,	Allowable	:	103.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 9.73, tr = 3.6, c = 3 mm, E* = 1

Thickness Ratio = $tr \times E*/(tg - c) = 0.53$, Temp. Reduction = 28 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 40, tr = 3.6, c = 3 mm, E* = 1

Thickness Ratio = $tr \times E*/(tg - c) = 0.097$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 31 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -46 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $13827.01/25860.00 = 0.535$

Weld Size Calculations, Description: N4- Tube outlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	10.6050 = $0.7 \times Wo$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

= max(0, (A-A1+2*tn*fr1*(E1*t-tr))Sv)

= max(0, (39.8 - 8.66 + 2 * 37 * 1 *

(1 * 61 - 54.7))148)

= 53939.24 Kgf

For hub type nozzles, A2 includes the area of the hub.



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Nozzle Calcs.: N4- Tube outlet

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Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr}2) \cdot S_v$$

$$= (41 + 2.25 - 0 \cdot 1) \cdot 148$$

$$= 65301.12 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S_v$$

$$= (41 + 0 + 2.25 + (45.1)) \cdot 148$$

$$= 133423.02 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S$$

$$= (41 + 0 + 2.25 + 0 + (45.1)) \cdot 148$$

$$= 133423.02 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 147 \cdot 15 \cdot 0.49 \cdot 148$$

$$= 25551. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 54.8) \cdot (40 - 3) \cdot 0.7 \cdot 148$$

$$= 67321. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 147 \cdot (56 - 0) \cdot 0.74 \cdot 148$$

$$= 144061. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (25551 + 67321) = 92873 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (25551 + 0 + 144061 + 0) = 169612 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (25551 + 144061 + 0) = 169612 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 92872 Kgf, must exceed W = 53939 Kgf	or W1 = 65301 Kgf
Path 2-2 = 169612 Kgf, must exceed W = 53939 Kgf	or W2 = 133423 Kgf
Path 3-3 = 169612 Kgf, must exceed W = 53939 Kgf	or W3 = 133423 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 15137 KPa.

Nozzle is O.K. for the External Pressure: 101 KPa.

The Drop for this Nozzle is : 4.9096 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 318.9096 mm

Input Echo, WRC297 Item 1, Description: N4- Tube outle:



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Nozzle Calcs.: N4- Tube outlet Nozl: 16 9:43am Apr 21,2026

Diameter Basis for Cylindrical Shell	ID		
Shell Corrosion Allowance	3.0000	mm	
Shell Diameter	1100.000	mm	
Shell Thickness	64.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-387 11		
Vessel Cold S.I. Allowable	Smc	148.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	148.00	N./mm ²

Using 2 * Yield for Discontinuity Stress Allowable (Div 2, 4.1.6.3), Sps.
Make sure that material properties at this temperature are not
time-dependent for Material: SA-387 11

Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	66.650	mm	
Nozzle Thickness	40.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-336 F11		
Nozzle Cold S.I. Allowable	SNmc	148.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	148.00	N./mm ²

Using 2 * Yield for Discontinuity Stress Allowable (Div 2, 4.1.6.3), Sps.
Make sure that material properties at this temperature are not
time-dependent for Material: SA-336 F11

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	13827.01	KPa.
Radial Load	P	-1183.00	Kgf
Circumferential Shear	Vc	979.00	Kgf
Longitudinal Shear	Vl	979.00	Kgf
Circumferential Moment	Mc	248.00	Kg-m.
Longitudinal Moment	Ml	248.00	Kg-m.
Torsional Moment	Mt	323.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_v/T_v (19.188) is < 20
check the limitations of WRC bulletin 297.
Warning - The ratio, D_n/T_v (2.291) is < 5
check the limitations of WRC bulletin 297.
Warning - The ratio, D_n/T_n (3.666) is < 10
check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
= NozzleOD + $2 \cdot 1.65 \cdot \sqrt{R_{mean}(t - ca)}$
= 146.65 + $2 \cdot 1.65 \cdot \sqrt{583.5(64.0 - 3.0)}$
= 769.235 mm

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1167.000	mm
Nozzle Outside Diameter	(d) =	146.650	mm
Vessel Thickness used	(T) =	61.000	mm
Nozzle Thickness used	(t) =	37.000	mm
T / t	=	1.649	



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Nozzle Calcs.: N4- Tube outlet

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$$d / t = 10.000$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.550$$

Nr/P	=	0.184
Mr/P	=	0.210
MØ/P	=	0.101
NØ/P	=	0.160
MrD/Mc	=	0.248
NrDL/Mc	=	0.179
MØd/Mc	=	0.152
NØDL/Mc	=	-0.096
MrD/Ml	=	0.222
NrDL/Ml	=	0.168
MØD/Ml	=	0.134
NØDL/Ml	=	0.028

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	1	-1	1	-1
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-3	3	3	-3
Normal Pressure Stress	118	132	118	132
Outplane Stress Summary	117	134	124	127

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-5	5	5	-5
Inplane Pressure Stress	59	59	59	59
Inplane Stress Summary	57	61	70	50

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	117	134	124	127



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Nozzle Calcs.: N4- Tube outlet

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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	1	-1	1	-1
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-4	4	4	-4
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	59	59	59	59
Outplane Stress Summary	58	62	65	53

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-6	6	6	-6
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	118	132	118	132
Inplane Stress Summary	115	135	130	123

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	115	135	130	123

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10	27	10	27
Hoop Stress Summary	11	27	11	27

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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Nozzle Calcs.: N4- Tube outlet

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(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-8	-8	8	8
Axial Bending (ML)	-10	10	10	-10
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-6	1	30	-1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	3	3	1	1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	18	28	30	29

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10	27	10	27
Hoop Stress Summary	11	28	10	27

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	-8	-8	8	8
Axial Bending (Mc)	-11	11	11	-11
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-7	3	32	-2

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1



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Nozzle Calcs.: N4- Tube outlet Nozl: 16 9:43am Apr 21,2026

Shear due to Torsion	2	2	2	2
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Shear Stress Summary	1	1	3	3
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	19	28	32	31

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	118	132	118	132	118	132	118	132
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	-2	2	4	-4	-3	3	9	-9
Long. Pm (SUS)	59	59	59	59	59	59	59	59
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	-2	2	8	-8	-3	3	5	-5
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	118.0	132.0	118.0	132.0	118.0	132.0	118.0	132.0
Pm+Pl (SUS)	118.0	132.0	118.0	132.0	118.0	132.0	118.0	132.0
Pm+Pl+Q (Total)	116.0	134.0	122.0	128.0	115.0	135.0	127.0	123.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	132.00	148.00	Passed
Pm+Pl (SUS)	132.00	222.00	Passed
Pm+Pl+Q (TOTAL)	135.01	443.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	10	27	10	27	10	27	10	27
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-8	-8	8	8	-8	-8	8	8
Long. Q (SUS)	-5	5	15	-15	-6	6	16	-16
Shear Pm (SUS)	0	0	0	0	0	0	0	0



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Nozzle Calcs.: N4- Tube outlet

Nozl: 16

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Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	10.0	27.0	10.0	27.0	10.0	27.0	10.0	27.0
Pm+P1 (SUS)	13.2	30.1	13.3	27.1	13.2	30.1	13.3	27.1
Pm+P1+Q (Total)	19.0	27.4	28.1	29.1	19.1	27.0	29.5	30.6

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	27.00	148.00	Passed
Pm+P1 (SUS)	30.07	222.00	Passed
Pm+P1+Q (TOTAL)	30.59	443.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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Nozzle Calcs.: D1

Nozl: 17 9:43am Apr 21,2026

Input, Nozzle Desc: D1

From: 30

Pressure for Reinforcement Calculations	P	13827.008	KPa.
Temperature for Internal Pressure	Temp	400	°C
Design External Pressure	Pext	101.30	KPa.
Temperature for External Pressure	Tempex	200	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	148.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1100.00	mm
Design Length of Section	L	999.9695	mm
Shell Finished (Minimum) Thickness	t	64.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1341.85	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F11	
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	148.00	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	20.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		40.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	60.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		1500	
Flange Grade		GR 1.9	

The Pressure Design option was Design Pressure + static head.



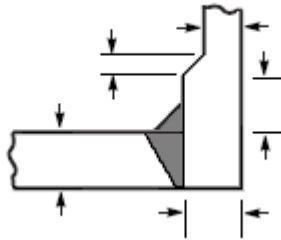
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Nozzle Calcs.: D1

Nozl: 17 9:43am Apr 21, 2026



Hub Nozzle (Set-in)

Reinforcement CALCULATION, Description: D1

ASME Code, Section VIII, Div. 1, 2021, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.687 in.
Actual Thickness Used in Calculation	0.344 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 13827 \cdot 553 / (148 \cdot 1 - 0.6 \cdot 13827)$
 $= 54.7369 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 13827 \cdot 24.4 / (148 \cdot 1 - 0.6 \cdot 13827)$
 $= 2.4176 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3455 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 31.3 / 0.5773$
 $= 54.1481 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	182.3250 mm
Parallel to Vessel Wall	Rn+tn+t	91.1625 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	68.4921 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: D1.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.4176 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm



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Nozzle Calcs.: D1

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Wall Thickness, shell/head, internal pressure	trb1 = 57.7369 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 57.7369 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.3787 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 6.4200 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1, tb2)]
 = min[6.42, max(57.7, 5.5)]
 = 6.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
 = max(5.42, 6.42)
 = 6.4200 mm

Available Nozzle Neck Thickness = $0.875 \times 8.738 = 7.645$ mm → OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 7.65, tr = 2.42, c = 3 mm, E* = 1
 Thickness Ratio = $tr \times E^* / (tg - c) = 0.52$, Temp. Reduction = 30 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 40, tr = 2.42, c = 3 mm, E* = 1
 Thickness Ratio = $tr \times E^* / (tg - c) = 0.065$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-46 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $13827.01 / 25860.00 = 0.535$

Weld Size Calculations, Description: D1

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	10.6050 = $0.7 \times Wo$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 15311 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 3.4408 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 317.4408 mm



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Nozzle Calcs.: D1

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Nozzle Calcs.: Shell inlet-N1

Nozl: 18 9:43am Apr 21,2026

Input, Nozzle Desc: Shell inlet-N1

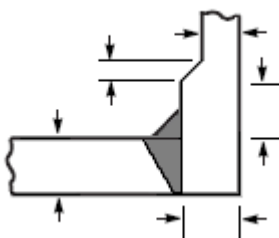
From: 40

Pressure for Reinforcement Calculations	P	4550.108	KPa.
Temperature for Internal Pressure	Temp	454	°C
Parent Material		SA-387 22	
Parent Allowable Stress at Temperature	Sv	127.78	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1100.00	mm
Shell Finished (Minimum) Thickness	t	30.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		3747.85	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F22	
Material UNS Number		K21590	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	127.78	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		30.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	32.0000	mm
Hub Height of Integral Nozzle	h	70.0000	mm
Height of Beveled Transition	L'	60.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		130.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	25.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.



Hub Nozzle (Set-in)



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Nozzle Calcs.: Shell inlet-N1

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Reinforcement CALCULATION, Description: Shell inlet-N1

ASME Code, Section VIII, Div. 1, 2021, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 30.000 in.
Actual Thickness Used in Calculation 1.260 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4550 \cdot 553 / (128 \cdot 1 - 0.6 \cdot 4550)$
= 20.1230 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4550 \cdot 384 / (128 \cdot 1 - 0.6 \cdot 4550)$
= 13.9733 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 98/0.5773
= 169.7411 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1536.0000 mm
Parallel to Vessel Wall, opening length d 768.0000 mm
Normal to Vessel Wall (Thickness Limit), no pad T1np 67.5000 mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 128/128)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 128/128)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1, 1)
= 1.000

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	154.545	NA	NA
Area in Shell	A1	52.815	NA	NA
Area in Nozzle Wall	A2	20.286	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	4.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	132.300	NA	NA
TOTAL AREA AVAILABLE	Atot	209.401	NA	NA



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Nozzle Calcs.: Shell inlet-N1

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The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (768 \cdot 20.1 \cdot 1 + 2 \cdot 127 \cdot 20.1 \cdot 1 \cdot (1 - 1))$$

$$= 154.545 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 768 (1 \cdot 27 - 1 \cdot 20.1) - 2 \cdot 127$$

$$(1 \cdot 27 - 1 \cdot 20.1) \cdot (1 - 1)$$

$$= 52.815 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp) (tn - trn) fr2$$

$$= (2 \cdot 67.5) (29 - 14) \cdot 1$$

$$= 20.286 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$$

$$= 20^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 4.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(67.5, 250, 70)) \cdot (130 - 32) \cdot 1$$

$$= 132.300 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 16.9733 \text{ mm}$

Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb1 = 23.1230 \text{ mm}$

Wall Thickness $tb1 = \max(trb1, tr16b) = 23.1230 \text{ mm}$

Wall Thickness, shell/head, external pressure $trb2 = 3.3787 \text{ mm}$

Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm}$

Wall Thickness per table UG-45 $tb3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(23.1, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(17, 11.3)$$

$$= 16.9733 \text{ mm}$$

Available Nozzle Neck Thickness = 32.0000 mm → OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained : 39.5, Allowable : 127.8 N./mm² Passed

Expansion : 0.0, Allowable : 305.2 N./mm² Passed

Occasional : 29.0, Allowable : 169.9 N./mm² Passed

Shear : 6.8, Allowable : 89.4 N./mm² Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of



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Nozzle Calcs.: Shell inlet-N1

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the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Govrn. thk, $t_g = 30$, $t_r = 17.3$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.64$, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 24 °C

Min Metal Temp. at Required thickness (UCS 66.1) 4 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 4 °C

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

= (Hub Thickness - Neck Thickness) / $\tan(30)$

= 98/0.5773

= 169.7411 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1152.0000 mm

Parallel to Vessel Wall, opening length d 576.0000 mm

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 67.5000 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure] AR		103.035	103.035
Area Available in Shell A1		26.408	26.408
Area Available in Nozzle Wall A2		20.286	20.286
Area Available in Inward Nozzle A3		0.000	0.000
Area Available in Welds A4		4.000	4.000
Area Available in Pad A5		0.000	0.000
Area Available in Hub A6		132.300	132.300
TOTAL AREA AVAILABLE Atot		182.994	182.994

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: Shell inlet-N1

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
Nozzle Weld 6.0000 = Min per Code 14.1400 = $0.7 \cdot W_o$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

= $\max(0, (A - A1 + 2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) S_v)$

= $\max(0, (155 - 52.8 + 2 \cdot 127 \cdot 1 \cdot$

$(1 \cdot 27 - 20.1)) 128)$

= 155308.80 Kgf



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Nozzle Calcs.: Shell inlet-N1

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For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$$

$$= (153 + 4 - 0 \cdot 1) \cdot 128$$

$$= 204024.89 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (153 + 0 + 4 + (68.6)) \cdot 128$$

$$= 293381.69 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$$

$$= (153 + 0 + 4 + 0 + (68.6)) \cdot 128$$

$$= 293381.69 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 1022 \cdot 20 \cdot 0.49 \cdot 128$$

$$= 204987. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 448) \cdot (130 - 3) \cdot 0.7 \cdot 128$$

$$= 1628452. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 1022 \cdot (25 - 0) \cdot 0.74 \cdot 128$$

$$= 386966. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (204987 + 1628451) = 1833439 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (204987 + 0 + 386966 + 0) = 591953 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (204987 + 386966 + 0) = 591953 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 1833439 Kgf, must exceed W = 155308 Kgf or W1 = 204024 Kgf
 Path 2-2 = 591953 Kgf, must exceed W = 155308 Kgf or W2 = 293381 Kgf
 Path 3-3 = 591953 Kgf, must exceed W = 155308 Kgf or W3 = 293381 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5292 KPa.

The Drop for this Nozzle is : 346.5817 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 626.5817 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 34.76 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 768.00 mm



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Nozzle Calcs.: Shell inlet-N1

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Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: Shell inlet-N1:

Diameter Basis for Cylindrical Shell	ID		
Shell Corrosion Allowance	3.0000	mm	
Shell Diameter	1100.000	mm	
Shell Thickness	30.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-387 22		
Vessel Cold S.I. Allowable	Smc	148.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	127.78	N./mm ²
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	762.000	mm	
Nozzle Thickness	130.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-336 F22		
Nozzle Cold S.I. Allowable	SNmc	148.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	127.78	N./mm ²

Using 2 * Yield for Discontinuity Stress Allowable (Div 2, 4.1.6.3), Sps.
Make sure that material properties at this temperature are not time-dependent for Material: SA-336 F22

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	4550.11	KPa.
Radial Load	P	-6883.56	Kgf
Circumferential Shear	Vc	4971.46	Kgf
Longitudinal Shear	Vl	4971.46	Kgf
Circumferential Moment	Mc	11468.14	Kg-m.
Longitudinal Moment	Ml	11468.14	Kg-m.
Torsional Moment	Mt	13188.37	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Dv (.881) is > 0.5
check the Limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (7.862) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
= NozzleOD + 2 * 1.65 * $\sqrt{R_{mean}(t - ca)}$
= 1022.0 + 2 * 1.65 * $\sqrt{566.5(30.0 - 3.0)}$
= 1430.127 mm

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1133.000 mm
Nozzle Outside Diameter	(d) =	1022.000 mm
Vessel Thickness used	(T) =	27.000 mm
Nozzle Thickness used	(t) =	127.000 mm
T / t	=	0.213



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Nozzle Calcs.: Shell inlet-N1

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$$d / t = 10.000$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 5.843$$

Nr/P	=	0.042
Mr/P	=	0.068
MØ/P	=	0.022
NØ/P	=	0.048
MrD/Mc	=	0.246
NrDL/Mc	=	0.085
MØd/Mc	=	0.072
NØDL/Mc	=	0.080
MrD/Ml	=	0.054
NrDL/Ml	=	0.038
MØD/Ml	=	0.017
NØDL/Ml	=	0.043

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-6	-6	6	6
Outplane Bending (ML)	-15	15	15	-15
Normal Pressure Stress	90	95	90	95
Outplane Stress Summary	85	96	129	78

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	37	-37	37	-37
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-48	48	48	-48
Inplane Pressure Stress	45	45	45	45
Inplane Stress Summary	32	54	141	-31

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	4	4	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	85	97	141	109



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Nozzle Calcs.: Shell inlet-N1

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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	-12	-12	12	12
Outplane Bending (Mc)	-65	65	65	-65
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	45	45	45	45
Outplane Stress Summary	-15	90	139	-15

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	37	-37	37	-37
Inplane Membrane (Mc)	-12	-12	12	12
Inplane Bending (Mc)	-222	222	222	-222
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	90	95	90	95
Inplane Stress Summary	-102	271	368	-147

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	102	271	368	148

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-6	-6	6	6
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	14	19	14	19
Hoop Stress Summary	12	17	25	30

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
--------------------	----	----	----	----



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Nozzle Calcs.: Shell inlet-N1

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(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	7	7	7	7
Axial Stress Summary	5	5	11	6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	12	17	25	30

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-12	-12	12	12
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	14	19	14	19
Hoop Stress Summary	6	11	30	36

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-8	8	8	-8
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	7	7	7	7
Axial Stress Summary	-1	12	18	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0



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Nozzle Calcs.: Shell inlet-N1 Nozl: 18 9:43am Apr 21,2026

Shear due to Torsion	0	0	0	0
----------------------	---	---	---	---

Shear Stress Summary	0	0	0	0
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	7	12	30	36

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	90	95	90	95	90	95	90	95
Circ. Pl (SUS)	-2	-2	10	10	-9	-9	15	15
Circ. Q (SUS)	-3	3	27	-27	-185	185	259	-259
Long. Pm (SUS)	45	45	45	45	45	45	45	45
Long. Pl (SUS)	-2	-2	8	8	-8	-8	16	16
Long. Q (SUS)	-11	11	85	-85	-53	53	77	-77
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	90.0	95.0	90.0	95.0	90.0	95.0	90.0	95.0
Pm+Pl (SUS)	88.0	93.0	100.0	105.0	81.0	86.0	105.0	110.0
Pm+Pl+Q (Total)	85.2	96.2	138.1	110.0	104.0	271.0	364.0	149.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	95.00	127.78	Passed
Pm+Pl (SUS)	110.02	191.67	Passed
Pm+Pl+Q (TOTAL)	364.04	383.34	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	14	19	14	19	14	19	14	19
Circ. Pl (SUS)	-2	-2	10	10	-8	-8	16	16
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	0	0	2	-2	-7	7	9	-9
Shear Pm (SUS)	0	0	0	0	0	0	0	0



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Nozzle Calcs.: Shell inlet-N1

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Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	14.0	19.0	14.0	19.0	14.0	19.0	14.0	19.0
Pm+P1 (SUS)	12.0	17.0	24.0	29.0	6.0	11.0	30.0	35.0
Pm+P1+Q (Total)	12.0	17.0	24.0	29.0	7.0	13.0	30.0	36.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	19.00	127.78	Passed
Pm+P1 (SUS)	35.00	191.67	Passed
Pm+P1+Q (TOTAL)	36.00	383.34	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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Nozzle Calcs.: Shell outlet- N

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Input, Nozzle Desc: Shell outlet- N

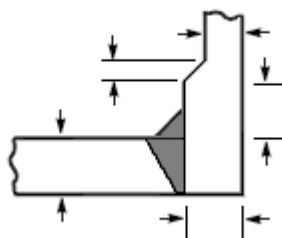
From: 40

Pressure for Reinforcement Calculations	P	4550.000	KPa.
Temperature for Internal Pressure	Temp	454	°C
Parent Material		SA-387 22	
Parent Allowable Stress at Temperature	Sv	127.78	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1100.00	mm
Shell Finished (Minimum) Thickness	t	30.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		3647.85	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F22	
Material UNS Number		K21590	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	127.78	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		30.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	25.0000	mm
Hub Height of Integral Nozzle	h	70.0000	mm
Height of Beveled Transition	L'	60.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		130.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	25.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.



Hub Nozzle (Set-in)



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Nozzle Calcs.: Shell outlet- N Nozl: 19 9:43am Apr 21,2026

Reinforcement CALCULATION, Description: Shell outlet- N

ASME Code, Section VIII, Div. 1, 2021, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 30.000 in.
Actual Thickness Used in Calculation 0.984 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4550 \cdot 553 / (128 \cdot 1 - 0.6 \cdot 4550)$
= 20.1225 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $4550 \cdot 384 / (128 \cdot 1 - 0.6 \cdot 4550)$
= 13.9730 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 105/0.5773
= 181.8654 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1536.0000 mm
Parallel to Vessel Wall, opening length d 768.0000 mm
Normal to Vessel Wall (Thickness Limit), no pad Tlnp 67.5000 mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 128/128)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 128/128)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1, 1)
= 1.000

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	154.541	NA	NA
Area in Shell	A1	52.819	NA	NA
Area in Nozzle Wall	A2	10.836	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	4.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	141.750	NA	NA
TOTAL AREA AVAILABLE	Atot	209.405	NA	NA



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Nozzle Calcs.: Shell outlet- N Nozl: 19 9:43am Apr 21,2026

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (768 \cdot 20.1 \cdot 1 + 2 \cdot 127 \cdot 20.1 \cdot 1 \cdot (1 - 1))$$

$$= 154.541 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 768 (1 \cdot 27 - 1 \cdot 20.1) - 2 \cdot 127$$

$$(1 \cdot 27 - 1 \cdot 20.1) \cdot (1 - 1)$$

$$= 52.819 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp) (tn - trn) fr2$$

$$= (2 \cdot 67.5) (22 - 14) \cdot 1$$

$$= 10.836 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$$

$$= 20^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 4.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(67.5, 250, 70)) \cdot (130 - 25) \cdot 1$$

$$= 141.750 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 16.9730 \text{ mm}$

Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb1 = 23.1225 \text{ mm}$

Wall Thickness $tb1 = \max(trb1, tr16b) = 23.1225 \text{ mm}$

Wall Thickness, shell/head, external pressure $trb2 = 3.3787 \text{ mm}$

Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm}$

Wall Thickness per table UG-45 $tb3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(23.1, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(17, 11.3)$$

$$= 16.9730 \text{ mm}$$

Available Nozzle Neck Thickness = 25.0000 mm → OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained : 52.5, Allowable : 127.8 N./mm² Passed

Expansion : 0.0, Allowable : 292.2 N./mm² Passed

Occasional : 38.6, Allowable : 169.9 N./mm² Passed

Shear : 8.9, Allowable : 89.4 N./mm² Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of



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Nozzle Calcs.: Shell outlet- N Nozl: 19 9:43am Apr 21,2026

the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 30, tr = 17.3, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.64$, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 24 °C

Min Metal Temp. at Required thickness (UCS 66.1) 4 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 4 °C

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

= (Hub Thickness - Neck Thickness) / tan(30)

= 105/0.5773

= 181.8654 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1152.0000 mm

Parallel to Vessel Wall, opening length d 576.0000 mm

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 67.5000 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure] AR		103.033	103.033
Area Available in Shell A1		26.409	26.409
Area Available in Nozzle Wall A2		10.836	10.836
Area Available in Inward Nozzle A3		0.000	0.000
Area Available in Welds A4		4.000	4.000
Area Available in Pad A5		0.000	0.000
Area Available in Hub A6		141.750	141.750
TOTAL AREA AVAILABLE Atot		182.996	182.996

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: Shell outlet- N

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
Nozzle Weld 6.0000 = Min per Code 14.1400 = $0.7 \cdot Wo$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

= max(0, (A-A1+2*tn*fr1*(E1*t-tr))Sv)

= max(0, (155 - 52.8 + 2 * 127 * 1 * (1 * 27 - 20.1))128)

= 155300.69 Kgf



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Nozzle Calcs.: Shell outlet- N Nozl: 19 9:43am Apr 21,2026

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$$

$$= (153 + 4 - 0 \cdot 1) \cdot 128$$

$$= 204025.48 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (153 + 0 + 4 + (68.6)) \cdot 128$$

$$= 293382.28 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$$

$$= (153 + 0 + 4 + 0 + (68.6)) \cdot 128$$

$$= 293382.28 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 1022 \cdot 20 \cdot 0.49 \cdot 128$$

$$= 204987. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 448) \cdot (130 - 3) \cdot 0.7 \cdot 128$$

$$= 1628452. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 1022 \cdot (25 - 0) \cdot 0.74 \cdot 128$$

$$= 386966. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (204987 + 1628451) = 1833439 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (204987 + 0 + 386966 + 0) = 591953 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (204987 + 386966 + 0) = 591953 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 1833439 Kgf, must exceed W = 155300 Kgf	or W1 = 204025 Kgf
Path 2-2 = 591953 Kgf, must exceed W = 155300 Kgf	or W2 = 293382 Kgf
Path 3-3 = 591953 Kgf, must exceed W = 155300 Kgf	or W3 = 293382 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5292 KPa.

The Drop for this Nozzle is : 346.5817 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 626.5817 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 434.32 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 768.00 mm



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Nozzle Calcs.: Shell outlet- N

Nozl: 19 9:43am Apr 21,2026

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 364.80 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 768.00 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: Shell outlet- :

Diameter Basis for Cylindrical Shell	ID		
Shell Corrosion Allowance	3.0000	mm	
Shell Diameter	1100.000	mm	
Shell Thickness	30.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-387 22		
Vessel Cold S.I. Allowable	Smc 148.00	N./mm ²	
Vessel Hot S.I. Allowable	Smh 127.78	N./mm ²	
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	762.000	mm	
Nozzle Thickness	130.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-336 F22		
Nozzle Cold S.I. Allowable	SNmc 148.00	N./mm ²	
Nozzle Hot S.I. Allowable	SNmh 127.78	N./mm ²	

Using 2 * Yield for Discontinuity Stress Allowable (Div 2, 4.1.6.3), Sps.
Make sure that material properties at this temperature are not time-dependent for Material: SA-336 F22

Note:

*External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	4550.00	KPa.
Radial Load	P	-6883.56	Kgf
Circumferential Shear	Vc	4971.46	Kgf
Longitudinal Shear	Vl	4971.46	Kgf
Circumferential Moment	Mc	11468.14	Kg-m.
Longitudinal Moment	Ml	11468.14	Kg-m.
Torsional Moment	Mt	13188.37	Kg-m.
Include Axial Pressure Thrust		No	
Include Pressure Stress Indices per Div. 2		No	
Local Loads applied at end of Nozzle/Attachment		No	

*Warning - The ratio, Dn/Dv (.881) is > 0.5
check the limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Tn (7.862) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
= NozzleOD + 2 * 1.65 * $\sqrt{R_{mean}(t - ca)}$
= 1022.0 + 2 * 1.65 * $\sqrt{566.5(30.0 - 3.0)}$
= 1430.127 mm

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1133.000 mm



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Nozzle Calcs.: Shell outlet- N Nozl: 19 9:43am Apr 21,2026

Nozzle Outside Diameter (d) = 1022.000 mm
 Vessel Thickness used (T) = 27.000 mm
 Nozzle Thickness used (t) = 127.000 mm
 T / t = 0.213
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 5.843$$

Nr/P = 0.042
 Mr/P = 0.068
 M0/P = 0.022
 N0/P = 0.048
 MrD/Mc = 0.246
 NrDL/Mc = 0.085
 M0d/Mc = 0.072
 N0DL/Mc = 0.080
 MrD/Ml = 0.054
 NrDL/Ml = 0.038
 M0D/Ml = 0.017
 N0DL/Ml = 0.043

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-6	-6	6	6
Outplane Bending (ML)	-15	15	15	-15
Normal Pressure Stress	90	95	90	95
Outplane Stress Summary	85	96	129	78

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	37	-37	37	-37
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-48	48	48	-48
Inplane Pressure Stress	45	45	45	45
Inplane Stress Summary	32	54	141	-31

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	4	4	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Top	Al Top	Bu Bottom	Bl Bottom
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Nozzle Calcs.: Shell outlet- N

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	Outside	Inside	Outside	Inside
Two * Max Shear Stress	85	97	141	109

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	-12	-12	12	12
Outplane Bending (Mc)	-65	65	65	-65
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	45	45	45	45
Outplane Stress Summary	-15	90	139	-15

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	37	-37	37	-37
Inplane Membrane (Mc)	-12	-12	12	12
Inplane Bending (Mc)	-222	222	222	-222
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	90	95	90	95
Inplane Stress Summary	-102	271	368	-147

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	102	271	368	148

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-6	-6	6	6
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	14	19	14	19



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Nozzle Calcs.: Shell outlet- N Nozl: 19 9:43am Apr 21,2026

Hoop Stress Summary	12	17	25	30
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	7	7	7	7

Axial Stress Summary	5	5	11	6
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	12	17	25	30
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-12	-12	12	12
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	14	19	14	19

Hoop Stress Summary	6	11	30	36
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-8	8	8	-8
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	7	7	7	7

Axial Stress Summary	-1	12	18	0
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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Nozzle Calcs.: Shell outlet- N Nozl: 19 9:43am Apr 21,2026

(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	7	12	30	36

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	90	95	90	95	90	95	90	95
Circ. Pl (SUS)	-2	-2	10	10	-9	-9	15	15
Circ. Q (SUS)	-3	3	27	-27	-185	185	259	-259
Long. Pm (SUS)	45	45	45	45	45	45	45	45
Long. Pl (SUS)	-2	-2	8	8	-8	-8	16	16
Long. Q (SUS)	-11	11	85	-85	-53	53	77	-77
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	90.0	95.0	90.0	95.0	90.0	95.0	90.0	95.0
Pm+Pl (SUS)	88.0	93.0	100.0	105.0	81.0	86.0	105.0	110.0
Pm+Pl+Q (Total)	85.2	96.2	138.1	110.0	104.0	271.0	364.0	149.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	95.00	127.78	Passed
Pm+Pl (SUS)	110.02	191.67	Passed
Pm+Pl+Q (TOTAL)	364.04	383.34	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	14	19	14	19	14	19	14	19
Circ. Pl (SUS)	-2	-2	10	10	-8	-8	16	16
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	7	7	7	7	7	7	7	7



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Nozzle Calcs.: Shell outlet- N Nozl: 19 9:43am Apr 21,2026

Long. P1 (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	0	0	2	-2	-7	7	9	-9
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	14.0	19.0	14.0	19.0	14.0	19.0	14.0	19.0
Pm+P1 (SUS)	12.0	17.0	24.0	29.0	6.0	11.0	30.0	35.0
Pm+P1+Q (Total)	12.0	17.0	24.0	29.0	7.0	13.0	30.0	36.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	19.00	127.78	Passed
Pm+P1 (SUS)	35.00	191.67	Passed
Pm+P1+Q (TOTAL)	36.00	383.34	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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Nozzle Calcs.: V1

Nozl: 20 9:43am Apr 21,2026

Input, Nozzle Desc: V1

From: 40

Pressure for Reinforcement Calculations	P	4550.000	KPa.
Temperature for Internal Pressure	Temp	454	°C
Parent Material		SA-387 22	
Parent Allowable Stress at Temperature	Sv	127.78	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1100.00	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2347.85	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F22	
Material UNS Number		K21590	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	127.78	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		XXS	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm
Height of Beveled Transition	L'	15.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		25.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



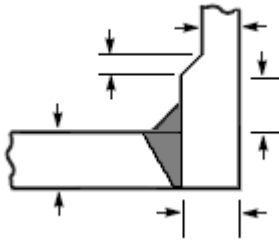
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Nozzle Calcs.: V1

Nozl: 20 9:43am Apr 21,2026



Hub Nozzle (Set-in)

Reinforcement CALCULATION, Description: V1

ASME Code, Section VIII, Div. 1, 2021, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.503 in.
Actual Thickness Used in Calculation	0.436 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4550 \cdot 553 / (128 \cdot 1 - 0.6 \cdot 4550)$
 $= 20.1225 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4550 \cdot 22.1 / (128 \cdot 1 - 0.6 \cdot 4550)$
 $= 0.8037 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot \text{Corroded Hub Thickness}$
 $= 2.5 \cdot 27$, less than the hub height, use (e-2)
 $= 67.5000 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	148.1762	mm
Parallel to Vessel Wall	Rn+tn+t	74.0881	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	62.5000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: V1.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.8037	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 23.1225	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 23.1225	mm
Wall Thickness, shell/head, external pressure	trb2	= 3.3787	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 8.0200	mm



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Nozzle Calcs.: V1

Noz1: 20 9:43am Apr 21,2026

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
 = min[8.02, max(23.1, 4.5)]
 = 8.0200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
 = max(3.8, 8.02)
 = 8.0200 mm

Available Nozzle Neck Thickness = $0.875 \times 34.286 = 30.000$ mm→OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.69, tr = 0.69, c = 3 mm, E* = 1
 Thickness Ratio = $tr \times E^*/(tg - c) = 0.1$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 28, tr = 17.3, c = 3 mm, E* = 1
 Thickness Ratio = $tr \times E^*/(tg - c) = 0.69$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	5 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 5 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $4550.00/15320.00 = 0.297$

Weld Size Calculations, Description: V1

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$14.1400 = 0.7 \times Wo$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5624 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 2.1950 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 280.1949 mm

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Nozzle Calcs.: D2

Nozl: 21 9:43am Apr 21,2026

Input, Nozzle Desc: D2

From: 40

Pressure for Reinforcement Calculations	P	4550.108	KPa.
Temperature for Internal Pressure	Temp	454	°C
Parent Material		SA-387 22	
Parent Allowable Stress at Temperature	Sv	127.78	N./mm ²
Parent Allowable Stress At Ambient	Sva	148.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1100.00	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5047.85	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-336 F22	
Material UNS Number		K21590	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	127.78	N./mm ²
Allowable Stress At Ambient	Sna	148.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		XXS	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm
Height of Beveled Transition	L'	15.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		25.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



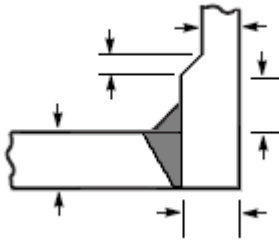
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Nozzle Calcs.: D2

Nozl: 21 9:43am Apr 21,2026



Hub Nozzle (Set-in)

Reinforcement CALCULATION, Description: D2

ASME Code, Section VIII, Div. 1, 2021, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.503 in.
Actual Thickness Used in Calculation	0.436 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4550 \cdot 553 / (128 \cdot 1 - 0.6 \cdot 4550)$
 $= 20.1230 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4550 \cdot 22.1 / (128 \cdot 1 - 0.6 \cdot 4550)$
 $= 0.8038 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
 $= 2.5 \cdot \text{Corroded Hub Thickness}$
 $= 2.5 \cdot 27$, less than the hub height, use (e-2)
 $= 67.5000 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	148.1762	mm
Parallel to Vessel Wall	Rn+tn+t	74.0881	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	62.5000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: D2.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.8038	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 23.1230	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 23.1230	mm
Wall Thickness, shell/head, external pressure	trb2	= 3.3787	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 8.0200	mm



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Nozzle Calcs.: D2

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Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.02, \max(23.1, 4.5)]$$

$$= 8.0200 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.8, 8.02)$$

$$= 8.0200 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 * 34.286 = 30.000 \text{ mm}$ —OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.69, tr = 0.69, c = 3 mm, E* = 1
Thickness Ratio = $tr * E*/(tg - c) = 0.1$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:A)

Govrn. thk, tg = 28, tr = 17.3, c = 3 mm, E* = 1
Thickness Ratio = $tr * E*/(tg - c) = 0.69$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	5 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 5 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $4550.11/15320.00 = 0.297$

Weld Size Calculations, Description: D2

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$14.1400 = 0.7 * Wo \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5624 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 2.1950 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 280.1949 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 34.76 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 74.09 mm



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ASME TS Calc: DESIGN 1

Case: 1 9:43a Apr 21,2026

Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	4550.00	KPa.
Shell Maximum Operating Pressure	Psox,max	4550.00	KPa.
Shell Minimum Operating Pressure	Psox,min	4009.00	KPa.
Shell Thickness	ts	28.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	caext	0.0000	mm
Inside Diameter of Shell	Ds	1100.000	mm
Shell Temperature for Internal Pressure	Ts	454.00	°C
Shell Material		SA-387 22	

Using 2 * Yield for Discontinuity Stress Allowable (UG-23(e)), Sps.

Make sure that material properties at this temperature are not time-dependent for Material: SA-387 22

Shell Material UNS Number		K21590	
Shell Allowable Stress at Temperature	Ss	127.79	N./mm ²
Shell Allowable Stress at Ambient		148.01	N./mm ²

Channel Description: Channel shell

Channel Type:		Cylinder	
Channel Maximum Design Pressure	Ptd,max	13820.00	KPa.
Channel Maximum Operating Pressure	Ptox,max	13820.00	KPa.
Channel Minimum Operating Pressure	Ptox,min	12499.97	KPa.
Channel Thickness	tc	64.0000	mm
Channel Corrosion Allowance	cac	3.0000	mm
Inside Diameter of Channel	Dc	1100.000	mm
Channel Design Temperature	TEMPC	400.00	°C
Channel Material		SA-387 11	

Using 2 * Yield for Discontinuity Stress Allowable (UG-23(e)), Sps.

Make sure that material properties at this temperature are not time-dependent for Material: SA-387 11

Channel Material UNS Number		K11789	
Channel Allowable Stress at Temperature	Sc	148.01	N./mm ²
Channel Allowable Stress at Ambient		148.01	N./mm ²

Tube Data:

Number of Tube Holes	Nt	750	
Tube Wall Thickness	et	2.1100	mm
Tube Outside Diameter	D	19.0500	mm
Total Straight Tube Length	Lt	3650.00	mm
Straight Tube Length (bet. inner tubsht faces) L		3444.00	mm
Design Temperature of the Tubes		454.00	°C
Tube Material		SA-213 T22	
Tube Material UNS Number		K21590	
Is this a Welded Tube		No	
Tube Material Specification used	Smls. tube		
Tube Allowable Stress at Temperature		111.78	N./mm ²
Tube Allowable Stress At Ambient		118.00	N./mm ²
Tube Yield Stress At design Temperature	Syt	180.52	N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.4000	mm
Tube Layout Pattern		Triangular	
Fillet Weld Leg	af	3.0000	mm
Groove Weld Leg	ag	3.0000	mm
Tube-Tubesheet Joint Weld Type		Full Strength	
Method for Tube-Tubesheet Jt. Allow.		UW-20	
Is the Tube Inset		No	



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ASME TS Calc: DESIGN 1

Case: 1 9:43a Apr 21,2026

Radius to Outermost Tube Hole Center	ro	524.160	mm
Largest Center-to-Center Tube Distance	U1	57.1500	mm
Length of Expanded Portion of Tube	ltx	0.0000	mm
Tube-side pass partition groove depth	hg	8.0000	mm
Area of Tube Layout	Ap	5286.8	cm ²
Perimeter of Tube Layout	Cp	2987.3	mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature	T	454.00	°C
Tubesheet Material		SA-336 F22	

Using 2 * Yield for Discontinuity Stress Allowable (UG-23(e)), Sps.
Make sure that material properties at this temperature are not
time-dependent for Material: SA-336 F22

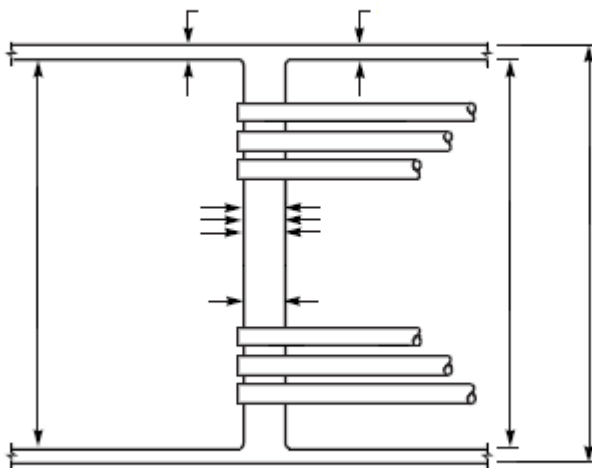
Tubesheet Material UNS Number		K21590	
Tubesheet Allowable Stress at Temperature	S	148.00	N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	148.00	N./mm ²
Thickness of Tubesheet	h	206.0000	mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000	mm
Tubesheet Corr. Allowance (Channel side)	Catc	13.0000	mm
Tubesheet Outside Diameter	A	1288.000	mm

Area of the Untubed Lanes	AL	419.1	cm ²
---------------------------	----	-------	-----------------

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	NO
Run Multiple Load Cases	YES

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX, 2021

Elasticity/Expansion Material Properties :



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ASME TS Calc: DESIGN 1

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Shell - TM-1 Material Group D

Elastic Mod. at Design Temperature	454.0 °C	0.17896E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.21097E+09 KPa.

Channel - TM-1 Material Group C

Elastic Mod. at Design Temperature	400.0 °C	0.17849E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Tubes - TM-1 Material Group D

Elastic Mod. at Tubsht. Design Temp.	454.0 °C	0.17896E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.21097E+09 KPa.

TubeSheet - TM-1 Material Group D

Elastic Mod. at Design Temperature	454.0 °C	0.17896E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.21097E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (13820 \cdot (19/2 - 0)) / (112 \cdot 1 + 0.4 \cdot 13820)$
 $= 1.1222 + 0.0000 = 1.1222 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart	CS-2	at	454.00 °C
Elastic Modulus for Material			150421952.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	3968.16	9.03	50.0000	0.0134948	74.03

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 11602 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.0779	19.05	3968.16	17.67	50.0000	0.0035217	60.32

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 60.3) / (3 \cdot 17.7) = 4550 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.1222 mm
Allowable Internal Pressure at Corroded thickness	27167.40 KPa.
Required Internal Design Pressure	13820.00 KPa.
Allowable External Pressure at Corroded thickness	11601.70 KPa.
Required External Design Pressure	4550.00 KPa.
Required Thickness due to Shell Side pressure	1.0779 mm

Detailed Results for load Case D1 corr. (Psd,min + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2021, Original Thickness :

UHX-12.5.1 Step 1:

Compute the Equivalent Outer Tube Limit Circle Diameter [Do]:
 $= 2 \cdot r_o + dt$
 $= 2 \cdot 524 + 19 = 1067 \text{ mm}$



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ASME TS Calc: DESIGN 1

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Determine the Basic Ligament Efficiency for Shear [μ]:

$$= (p - dt) / p$$

$$= (25.4 - 19) / 25.4 = 0.25$$

UHX-12.5.2 Step 2:

Compute the Ratio [ρ_s]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 1106 / 1067 = 1.04$$

Compute the Ratio [ρ_c]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 1106 / 1067 = 1.04$$

Moment on Tubesheet due to Pressures (P_s , P_t) [Mts]:

$$= D_o^2 / 16 \cdot [(\rho_s - 1) \cdot (\rho_s^2 + 1) \cdot P_s - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot P_t]$$

$$= 1067^2 / 16 \cdot [(1.04 - 1) \cdot (1.04^2 + 1) \cdot 0 -$$

$$(1.04 - 1) \cdot (1.04^2 + 1) \cdot 13820]$$

$$= -73853808.0000 \text{ KPa} \cdot \text{mm}^2$$

UHX-12.5.3 Step 3, Determination of Effective Elastic Properties :

Compute the Ratio [ρ]:

$$= l_{tx} / h = 0 / 190 = 0 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Compute the Effective Tube Hole Diameter [d^*]:

$$= \max(dt - 2tt(E_t / E) (St / S) (\rho), dt - 2tt)$$

$$= \max(19 - 2 \cdot 2.11 (178956752 / 178956752) \cdot$$

$$(112 / 148) (0), 19 - 2 \cdot 2.11)$$

$$= 19.0500 \text{ mm}$$

Compute the Effective Tube Pitch [p^*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l, 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 25.4 / \sqrt{ (1 - 4 \cdot \min(419, 4 \cdot 1067 \cdot 25.4) / (\pi \cdot 1067^2)) }$$

$$= 26.0166 \text{ mm}$$

Compute the Effective Ligament Efficiency for Bending [μ^*]:

$$= (p^* - d^*) / p^* = (26 - 19) / 26 = 0.27$$

E^*/E and ν^* for Triangular pattern from Fig. UHX-11.3.

$$h/p = 7.480315 ; \mu^* = 0.267775 ; E^* = 42821340 \text{ KPa.}$$

$$E^*/E = 0.239283 ; \nu^* = 0.379292$$

Note: As h/p (7.480) is > 2 , data values for $h/p = 2$ were used.

UHX-12.5.4 Step 4:

Compute Shell Coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / \sqrt{ (D_s + ts) ts }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (1106 + 25) 25 }$$

$$= 0.0108 \text{ 1/mm}$$

Determine Shell Coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot ts^3 / (6(1 - \nu_s^2))$$

$$= 0.011 \cdot 178956752 \cdot 25^3 / (6(1 - 0.3^2))$$

$$= 5536431104.0000 \text{ KPa} \cdot \text{mm}^2$$

Determine Shell Coefficient [λ_s]:

$$= 6 \cdot D_s \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6 \cdot 1106 \cdot 5.54E+09 / 190^3 \cdot (1 + 190 \cdot 0.011 + (190 \cdot 0.011)^2 / 2)$$

$$= 27658284.0 \text{ KPa.}$$

Determine Shell Coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot Ts) \cdot (1 - \nu_s / 2)$$



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ASME TS Calc: DESIGN 1

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$$= 1106^2 / (4 \cdot 178956752 \cdot 25) \cdot (1 - 0.3/2)$$

$$= 0.0580972880 \text{ mm/N./mm}^2$$

Calculate Parameter [ωS]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.04 \cdot 5.54E+09 \cdot 0.011 \cdot 0.058097288 (1 + 190 \cdot 0.011)$$

$$= 11004.7334 \text{ mm}^2$$

Determine Channel Coefficient [β_c]:

$$= ((12 \cdot (1 - v_c^2))^{3/4} / \sqrt{((D_c + t_c)t_c)})$$

$$= ((12 \cdot (1 - 0.3^2))^{3/4} / \sqrt{((1106 + 61)61)})$$

$$= 0.0068 \text{ 1/mm}$$

Determine Channel Coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.0068 \cdot 178489888 \cdot 61^3 / (6(1 - 0.3^2))$$

$$= 50555195392.0000 \text{ KPa} \cdot \text{mm}^2$$

Determine Channel Coefficient [λ_c]:

$$= 6 \cdot D_c \cdot k_c / h^3 \cdot (1 + h \cdot \beta_c + (h \cdot \beta_c)^2 / 2)$$

$$= 6 \cdot 1106 \cdot 50.56E+09 / (190^3) \cdot (1 + 190 \cdot 0.0068 + (190 \cdot 0.0068)^2 / 2)$$

$$= 153211104.0 \text{ KPa}$$

Determine Channel Coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot T_c) \cdot (1 - v_c/2)$$

$$= 1106^2 / (4 \cdot 178489888 \cdot 61) \cdot (1 - 0.3/2)$$

$$= 0.0238726456 \text{ mm/N./mm}^2$$

Calculate Parameter [ω_c]:

$$= p_c \cdot k_c \cdot \beta_c \cdot \delta_c (1 + h \cdot \beta_c)$$

$$= 1.04 \cdot 50.56E+09 \cdot 0.0068 \cdot 0.023872646 (1 + 190 \cdot 0.0068)$$

$$= 19551.4453 \text{ mm}^2$$

UHX-12.5.5 Step 5:

Diameter ratio [K]:

$$= A/D_o = 1288/1067 = 1.21$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.38) / 42821340 \cdot (27658284 + 153211104 + 178956752 \cdot \ln(1.21))$$

$$= 3.1092$$

UHX-12.5.6 Step 6:

Moment Acting on Unperforated Tubesheet Rim [M^*]:

$$= M_{ts} + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= -73853808 + 19551 \cdot 13820 - 11005 \cdot 0$$

$$= 196347136.0000 \text{ KPa} \cdot \text{mm}^2$$

UHX-12.5.7 Step 7:

Maximum Bending Moment acting on Periphery of Tubesheet [M_p]:

$$= ((M^*) - D_o^2/32 \cdot F(P_s - P_t)) / (1 + F)$$

$$= ((196347136) - 1067^2/32 \cdot 3.11 (0 - 13820)) / (1 + 3.11)$$

$$= 420069792.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum Bending Moment acting on Center of Tubesheet [M_o]:

$$= M_p + D_o^2/64 \cdot (3 + v^*)(P_s - P_t)$$

$$= 420069792 + 1067^2/64 \cdot (3 + 0.38)(0 - 13820)$$

$$= -411279552.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum Bending Moment acting on Tubesheet [M]:

$$= \max(\text{abs}(M_p), \text{abs}(M_o))$$



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$$= \max(\text{abs}(420069792), \text{abs}(-411279552)) \\ = 420069792.0000 \text{ KPa} \cdot \text{mm}^2$$

UHX-12.5.8 Step 8:

Tubesheet Bending Stress at Original Thickness:

$$= 6 \cdot M / (\mu \cdot (h - h_g)^2) \\ = 6 \cdot 420069792 / (0.27 \cdot (190 - 0)^2) \\ = 260.7479 \text{ N./mm}^2$$

The Allowable Tubesheet Bending Stress $[\sigma_{All}]$:

$$= 2 \cdot S = 2 \cdot 148 = 296 \text{ N./mm}^2$$

Tubesheet Bending Stress at Final Thickness $[\sigma]$:

$$= 6 \cdot M / (\mu \cdot (h - h_g)^2) \\ = 6 \cdot 423502816 / (0.27 \cdot (179 - 0)^2) \\ = 295.8076 \text{ N./mm}^2$$

Required Tubesheet Thickness, for Bending Stress $[H_{reqB}]$:

$$= h + C_{ats} + C_{atc} = 179 + 3 + 13 = 195 \text{ mm}$$

Required Tubesheet Thickness for Given Loadings (includes CA) $[H_{req}]$:

$$= \max(H_{reqB}, H_{reqS}) = \max(195, 98.6) = 195 \text{ mm}$$

UHX-12.5.9 Step 9:

$$\text{abs}(P_s - P_t) = \text{abs}(0 - 13820) = 13820 \text{ KPa.}$$

Shear Stress check $[\tau_{limit}]$:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S, 0.533 \cdot S_y) \\ = 4 \cdot 0.25 \cdot 190 / 1067 \cdot \min(0.8 \cdot 148, 0.533 \cdot 310) \\ = 21.076 \text{ N./mm}^2$$

$[P_s - P_t]$ is $\leq$ the Shear Stress limit (τ_{limit}) . Therefore, the required thickness for shear is for information only.

Minimum Required Thickness for Shear $[H_{reqS}]$:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S, 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot \text{abs}(P_s - P_t) + C_{ats} + C_{atc} \\ = 1 / (4 \cdot 0.25 \cdot 118) \cdot (4 \cdot 5287 / 2987) \cdot \text{abs}((0 - 13820)) + 16 \\ = 98.6348 \text{ mm}$$

UHX-12.5.10 Step 10:

Note: For a given Shell thickness of: 25.000mm

Min. Shell len. adjacent to the tubesheet is: 299.309 mm

Axial Shell Membrane Stress $[\sigma_{sm}]$:

$$= D_s^2 / (4 \cdot t_s(D_s + t_s)) \cdot P_s \\ = 1106^2 / (4 \cdot 25 \cdot (1106 + 25)) \cdot 0 \\ = 0.000 \text{ N./mm}^2$$

Axial Shell Bending Stress $[\sigma_{sb}]$:

$$= 6 \cdot k_s / t_s^2 \cdot (\beta S \cdot P_s + 6(1 - \nu) / (E) \cdot \\ D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))) \\ = 6 \cdot 1244667 / 25^2 \cdot [0.011 \cdot 0 + 6(1 - 0.38) / 42821340 \cdot \\ 1067 / 190^3 \cdot (1 + 190 \cdot 0.011 / 2) \cdot (420069792 + 1067^2 / 32 \cdot \\ (0 - 13820))] \\ = -104.926 \text{ N./mm}^2$$

Shell Membrane + Bending Stress $[\sigma_s]$:

$$= \text{abs}(\sigma_{sm}) + \text{abs}(\sigma_{sb}) \text{ (Should be } \leq 1.5 \cdot S_s) \\ = 104.926 \text{ N./mm}^2 \text{ (Should be } \leq 191.68)$$

Note: For a given Channel thickness of: 61.000 mm

Min. Channel len. adjacent to the tubesheet is: 467.536mm



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ASME TS Calc: DESIGN 1

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Axial Channel Membrane Stress [σ_{cm}]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c)) \cdot P_t$$

$$= 1106^2 / (4 \cdot 61 (1106 + 61)) \cdot 13820$$

$$= 59.372 \text{ N./mm}^2$$

Axial Channel Bending Stress [σ_{cb}]:

$$= 6 \cdot k_c / t_c^2 \cdot [\beta_c \cdot (\delta C \cdot P_t) - 6(1 - \nu^*) / (E^* \cdot D_o / h^3 \cdot (1 + h \cdot \beta_c / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 11365519 / 61^2 \cdot [0.0068 \cdot 0.33 - 6(1 - 0.38) / 42821340 \cdot 1067 / 190^3 \cdot (1 + 190 \cdot 0.0068 / 2) \cdot (420069792 + 1067^2 / 32 \cdot (0 - 13820))]$$

$$= 314.043 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= \text{abs}(\sigma_{cm}) + \text{abs}(\sigma_{cb}) \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 373.415 \text{ N./mm}^2 \text{ (Should be } \leq 222.01)$$

Step 11, The Cylinder-to-Tubesheet Junction is Overstressed.

Performing a simplified Elastic-Plastic calculation (option 3) to reduce the overstress condition.

Modify E_s and/or E_c and recompute from Step 4 onwards.

Modified Elasticity [E_c]:

$$= E_c \cdot \sqrt{1.5 \cdot S_c / \sigma_c}$$

$$= 178489888 \cdot \sqrt{1.5 \cdot 148 / 373}$$

$$= 137626576.0 \text{ KPa.}$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu \cdot (h - h'g)^2)$$

$$= 6 \cdot 421440064 / (0.27 (190 - 0)^2)$$

$$= 261.6 \text{ N./mm}^2 \text{ (Should be } \leq 296.0)$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.
The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [H_{reqB}]:

$$= h + C_{ats} + C_{atc} = 179 + 3 + 13 = 195 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [H_{req}]:

$$= \max(H_{reqB}, H_{reqS}) = \max(195, 98.6) = 195 \text{ mm}$$

Tube Metal Area [A_t]:

$$= \pi (d_o - t) t$$

$$= \pi (19 - 2.11) 2.11$$

$$= 1.123 \text{ cm}^2$$

Tube Weld Size Results per UW-20:

Tube Strength [F_t]:

$$= \pi \cdot t (d_o - t) S_a$$

$$= \pi \cdot 2.11 (19 - 2.11) 112$$

$$= 1279.905 \text{ Kg}$$

Groove Weld Strength [F_g]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 3 (19 + 0.67 \cdot 3) 112$$

$$= 1279.9048 \text{ Kg}$$

Weld Strength Factor [f_w]:

$$= S_a / S_w \text{ where } S_w = \min(S_a \text{ and } S_t)$$



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$$= 112/112$$

$$= 1.000$$

Design Strength Ratio [fd]:

= 1 for full strength welds

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 3 (19 + 0.67 \cdot 3) 112$$

$$= 1244.2972 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd)} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 19)^2 + 1.07 \cdot 2.11 \cdot (19 - 2.11) \cdot 1 \cdot 1)} - 0.75 \cdot 19 \right)$$

$$= 2.562 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.2810 mm

Minimum Required Groove Weld Leg [agr]: 1.2810 mm

Tube-Tubesheet Jt allowable, 1280 is >= tube strength 1280 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D1 corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	261.6 <=	296.0 N./mm ²	Ok
Tubesheet shear stress	51.5 <=	118.4 N./mm ²	Ok
Stress in Shell at Tubesheet	104.9 <=	191.7 N./mm ²	Ok
Stress in Channel at Tubesheet	373.4 <=	458.0 N./mm ²	Ok

Thickness results for loadcase D1 corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	195.113	206.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	3.000	Ok

Detailed Results for load Case D2 corr. (Psd,max + Ptd,min)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2021, Original Thickness :

UHX-12.5.1 Step 1:

Compute the Equivalent Outer Tube Limit Circle Diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 524 + 19 = 1067 \text{ mm}$$

Determine the Basic Ligament Efficiency for Shear [μ]:

$$= (p - dt) / p$$

$$= (25.4 - 19) / 25.4 = 0.25$$

UHX-12.5.2 Step 2:

Compute the Ratio [ps]:



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= D_s / D_o (Configurations a, b, c)
= $1106/1067 = 1.04$

Compute the Ratio [pc]:
= D_c / D_o (Configurations a, e, f)
= $1106/1067 = 1.04$

Moment on Tubesheet due to Pressures (Ps, Pt) [Mts]:
= $D_o^2/16 \cdot [(p_s-1) \cdot (p_s^2+1) \cdot P_s - (p_c-1) \cdot (p_c^2+1) \cdot P_t]$
= $1067^2/16 \cdot [(1.04 - 1) \cdot (1.04^2 + 1) \cdot 4550 - (1.04 - 1) \cdot (1.04^2 + 1) \cdot 0]$
= $24315092.0000 \text{ KPa} \cdot \text{mm}^2$

UHX-12.5.3 Step 3, Determination of Effective Elastic Properties :

Compute the Ratio [p]:
= $l_{tx}/h = 0/190 = 0$ (must be $0 \leq \rho \leq 1$)

Compute the Effective Tube Hole Diameter [d*]:
= $\max(d_t - 2t_t(E_t/E) (S_t/S) (\rho), d_t - 2t_t)$
= $\max(19 - 2 \cdot 2.11 (178956752/178956752) \cdot (112/148) \cdot (0), 19 - 2 \cdot 2.11)$
= 19.0500 mm

Compute the Effective Tube Pitch [p*]:
= $p / \sqrt{ (1 - 4 \cdot \min(A_l, 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$
= $25.4 / \sqrt{ (1 - 4 \cdot \min(419, 4 \cdot 1067 \cdot 25.4) / (\pi \cdot 1067^2)) }$
= 26.0166 mm

Compute the Effective Ligament Efficiency for Bending [μ^*]:
= $(p^* - d^*)/p^* = (26 - 19)/26 = 0.27$

E^*/E and ν^* for Triangular pattern from Fig. UHX-11.3.
 $h/p = 7.480315$; $\mu^* = 0.267775$; $E^* = 42821340 \text{ KPa}$.
 $E^*/E = 0.239283$; $\nu^* = 0.379292$

Note: As h/p (7.480) is > 2 , data values for $h/p = 2$ were used.

UHX-12.5.4 Step 4:

Compute Shell Coefficient [βS]:
= $((12(1 - \nu_s^2))^{1/4}) / \sqrt{ (D_s + t_s) t_s }$
= $((12(1 - 0.3^2))^{1/4}) / \sqrt{ (1106 + 25) 25 }$
= 0.0108 1/mm

Determine Shell Coefficient [ks]:
= $\beta S \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$
= $0.011 \cdot 178956752 \cdot 25^3 / (6(1 - 0.3^2))$
= $5536431104.0000 \text{ KPa} \cdot \text{mm}^2$

Determine Shell Coefficient [λS]:
= $6 \cdot D_s \cdot k_s / h^3 \cdot (1 + h \cdot \beta S + (h \cdot \beta S)^2 / 2)$
= $6 \cdot 1106 \cdot 5.54E+09 / 190^3 \cdot (1 + 190 \cdot 0.011 + (190 \cdot 0.011)^2 / 2)$
= 27658284.0 KPa .

Determine Shell Coefficient [δS]:
= $D_s^2 / (4 \cdot E_s \cdot T_s) \cdot (1 - \nu_s / 2)$
= $1106^2 / (4 \cdot 178956752 \cdot 25) \cdot (1 - 0.3/2)$
= $0.0580972880 \text{ mm/N} \cdot \text{mm}^2$

Calculate Parameter [ωS]:
= $p_s \cdot k_s \cdot \beta S \cdot \delta S (1 + h \cdot \beta S)$
= $1.04 \cdot 5.54E+09 \cdot 0.011 \cdot 0.058097288 (1 + 190 \cdot 0.011)$
= 11004.7334 mm^2



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Determine Channel Coefficient [β_c]:

$$= ((12 \cdot (1 - \nu_c^2))^{1/4}) / \sqrt{((D_c + t_c)t_c)}$$

$$= ((12 \cdot (1 - 0.3^2))^{1/4}) / \sqrt{((1106 + 61)61)}$$

$$= 0.0068 \text{ 1/mm}$$

Determine Channel Coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2))$$

$$= 0.0068 \cdot 178489888 \cdot 61^3 / (6(1 - 0.3^2))$$

$$= 50555195392.0000 \text{ KPa} \cdot \text{mm}^2$$

Determine Channel Coefficient [λ_c]:

$$= 6 \cdot D_c \cdot k_c / h^3 \cdot (1 + h \cdot \beta_c + (h \cdot \beta_c)^2 / 2)$$

$$= 6 \cdot 1106 \cdot 50.56\text{E}+09 / (190^3) \cdot (1 + 190 \cdot 0.0068 + (190 \cdot 0.0068)^2 / 2)$$

$$= 153211104.0 \text{ KPa}$$

Determine Channel Coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot T_c) \cdot (1 - \nu_c / 2)$$

$$= 1106^2 / (4 \cdot 178489888 \cdot 61) \cdot (1 - 0.3 / 2)$$

$$= 0.0238726456 \text{ mm/N} \cdot \text{mm}^2$$

Calculate Parameter [ω_c]:

$$= \rho_c \cdot k_c \cdot \beta_c \cdot \delta_c (1 + h \cdot \beta_c)$$

$$= 1.04 \cdot 50.56\text{E}+09 \cdot 0.0068 \cdot 0.023872646 (1 + 190 \cdot 0.0068)$$

$$= 19551.4453 \text{ mm}^2$$

UHX-12.5.5 Step 5:

Diameter ratio [K]:

$$= A / D_o = 1288 / 1067 = 1.21$$

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.38) / 42821340 \cdot (27658284 + 153211104 + 178956752 \cdot \ln(1.21))$$

$$= 3.1092$$

UHX-12.5.6 Step 6:

Moment Acting on Unperforated Tubesheet Rim [M^*]:

$$= M_{ts} + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= 24315092 + 19551 \cdot 0 - 11005 \cdot 4550$$

$$= -25756400.0000 \text{ KPa} \cdot \text{mm}^2$$

UHX-12.5.7 Step 7:

Maximum Bending Moment acting on Periphery of Tubesheet [M_p]:

$$= ((M^*) - D_o^2 / 32 \cdot F(P_s - P_t)) / (1 + F)$$

$$= ((-25756400) - 1067^2 / 32 \cdot 3.11 (4550 - 0)) / (1 + 3.11)$$

$$= -128837112.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum Bending Moment acting on Center of Tubesheet [M_o]:

$$= M_p + D_o^2 / 64 \cdot (3 + \nu^*)(P_s - P_t)$$

$$= -128837112 + 1067^2 / 64 \cdot (3 + 0.38)(4550 - 0)$$

$$= 144870304.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum Bending Moment acting on Tubesheet [M]:

$$= \max(\text{abs}(M_p), \text{abs}(M_o))$$

$$= \max(\text{abs}(-128837112), \text{abs}(144870304))$$

$$= 144870304.0000 \text{ KPa} \cdot \text{mm}^2$$

UHX-12.5.8 Step 8:

Tubesheet Bending Stress at Original Thickness:

$$= 6 \cdot M / (\mu^* (h - h_g)^2)$$

$$= 6 \cdot 144870304 / (0.27 (190 - 0)^2)$$



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$$= 89.9246 \text{ N./mm}^2$$

The Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S = 2 \cdot 148 = 296 \text{ N./mm}^2$$

Tubesheet Bending Stress at Final Thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 148051056 / (0.27 \cdot (106 - 0)^2) \\ &= 295.3718 \text{ N./mm}^2 \end{aligned}$$

Required Tubesheet Thickness, for Bending Stress [H_{reqB}]:

$$= h + Cats + Catc = 106 + 3 + 13 = 122 \text{ mm}$$

Required Tubesheet Thickness for Given Loadings (includes CA) [H_{req}]:

$$= \max(H_{reqB}, H_{reqS}) = \max(122, 43.2) = 122 \text{ mm}$$

UHX-12.5.9 Step 9:

$$\text{abs}(P_s - P_t) = \text{abs}(4550 - 0) = 4550 \text{ KPa.}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S, 0.533 \cdot S_y) \\ &= 4 \cdot 0.25 \cdot 190 / 1067 \cdot \min(0.8 \cdot 148, 0.533 \cdot 310) \\ &= 21.076 \text{ N./mm}^2 \end{aligned}$$

$[P_s - P_t]$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum Required Thickness for Shear [H_{reqS}]:

$$\begin{aligned} &= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S, 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot \text{abs}(P_s - P_t) + Cats + Catc \\ &= 1 / (4 \cdot 0.25 \cdot 118) \cdot (4 \cdot 5287 / 2987) \cdot \text{abs}((4550 - 0)) + 16 \\ &= 43.2061 \text{ mm} \end{aligned}$$

UHX-12.5.10 Step 10:

Note: For a given Shell thickness of: 25.000mm

Min. Shell len. adjacent to the tubesheet is: 299.309 mm

Axial Shell Membrane Stress [σ_{sm}]:

$$\begin{aligned} &= D_s^2 / (4 \cdot t_s(D_s + t_s)) \cdot P_s \\ &= 1106^2 / (4 \cdot 25 \cdot (1106 + 25)) \cdot 4550 \\ &= 49.213 \text{ N./mm}^2 \end{aligned}$$

Axial Shell Bending Stress [σ_{sb}]:

$$\begin{aligned} &= 6 \cdot k_s / t_s^2 \cdot (\beta S \cdot P_s + 6(1 - \nu) / (E) \cdot \\ &\quad D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))) \\ &= 6 \cdot 1244667 / 25^2 \cdot [0.011 \cdot 0.26 + 6(1 - 0.38) / 42821340 \cdot \\ &\quad 1067 / 190^3 \cdot (1 + 190 \cdot 0.011 / 2) \cdot (-128837112 + 1067^2 / 32 \cdot \\ &\quad (4550 - 0))] \\ &= 200.250 \text{ N./mm}^2 \end{aligned}$$

Shell Membrane + Bending Stress [σ_s]:

$$\begin{aligned} &= \text{abs}(\sigma_{sm}) + \text{abs}(\sigma_{sb}) \text{ (Should be } \leq 1.5 \cdot S_s) \\ &= 249.464 \text{ N./mm}^2 \text{ (Should be } \leq 191.68) \end{aligned}$$

Note: For a given Channel thickness of: 61.000 mm

Min. Channel len. adjacent to the tubesheet is: 467.536mm

Axial Channel Membrane Stress [σ_{cm}]:

$$\begin{aligned} &= D_c^2 / (4 \cdot t_c(D_c + t_c)) \cdot P_t \\ &= 1106^2 / (4 \cdot 61 \cdot (1106 + 61)) \cdot 0 \\ &= 0.000 \text{ N./mm}^2 \end{aligned}$$

Axial Channel Bending Stress [σ_{cb}]:

$$= 6 \cdot k_c / t_c^2 \cdot [\beta_c \cdot (\delta C \cdot P_t) - 6(1 - \nu) / (E) \cdot$$



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$$\begin{aligned} & Do/h^3 \cdot (1 + h \cdot \beta c/2) \cdot (Mp + Do^2/32 \cdot (Ps - Pt))] \\ & = 6 \cdot 11365519/61^2 \cdot [0.0068 \cdot 0 - 6(1 - 0.38)/42821340 \cdot \\ & \quad 1067/190^3 \cdot (1 + 190 \cdot 0.0068/2) \cdot (-128837112 + 1067^2/32 \cdot \\ & \quad (4550 - 0))] \\ & = -60.258 \text{ N./mm}^2 \end{aligned}$$

Channel Membrane + Bending Stress [σ_c]:

$$\begin{aligned} & = \text{abs}(\sigma_{cm}) + \text{abs}(\sigma_{cb}) \text{ (Should be } \leq 1.5 \cdot S_c) \\ & = 60.258 \text{ N./mm}^2 \text{ (Should be } \leq 222.01) \end{aligned}$$

Step 11, The Cylinder-to-Tubesheet Junction is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify Es and/or Ec and recompute from Step 4 onwards.

Modified Elasticity [Es]:

$$\begin{aligned} & = Es \cdot \sqrt{(1.5 \cdot S_s / \sigma_s)} \\ & = 178956752 \cdot \sqrt{(1.5 \cdot 128/249)} \\ & = 156866960.0 \text{ KPa.} \end{aligned}$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$\begin{aligned} & = 6 \cdot M / (\mu \cdot (h - h'g)^2) \\ & = 6 \cdot 145274464 / (0.27 (190 - 0)^2) \\ & = 90.2 \text{ N./mm}^2 \text{ (Should be } \leq 296.0) \end{aligned}$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.

The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [HreqB]:

$$= h + Cats + Catc = 106 + 3 + 13 = 122 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [Hreq]:

$$= \max(HreqB, HreqS) = \max(122, 43.2) = 122 \text{ mm}$$

Tube Metal Area [At]:

$$\begin{aligned} & = \pi (do - t) t \\ & = \pi (19 - 2.11) 2.11 \\ & = 1.123 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per UW-20:

Tube Strength [Ft]:

$$\begin{aligned} & = \pi \cdot t (do - t) Sa \\ & = \pi \cdot 2.11 (19 - 2.11) 112 \\ & = 1279.905 \text{ Kg} \end{aligned}$$

Groove Weld Strength [Fg]:

$$\begin{aligned} & = 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft) \\ & = 0.85 \cdot \pi \cdot 3 (19 + 0.67 \cdot 3) 112 \\ & = 1279.9048 \text{ Kg} \end{aligned}$$

Weld Strength Factor [fw]:

$$\begin{aligned} & = Sa / Sw \text{ where } Sw = \min(Sa \text{ and } St) \\ & = 112/112 \\ & = 1.000 \end{aligned}$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$



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$$= 0.55 \cdot \pi \cdot 3 (19 + 0.67 \cdot 3) 112$$

$$= 1244.2972 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd)} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 19)^2 + 1.07 \cdot 2.11 \cdot (19 - 2.11) \cdot 1 \cdot 1)} - 0.75 \cdot 19 \right)$$

$$= 2.562 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.2810 mm

Minimum Required Groove Weld Leg [agr]: 1.2810 mm

Tube-Tubesheet Jt allowable, 1280 is >= tube strength 1280 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D2 corr. (Psd,max + Ptd,min):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	90.2 <=	296.0 N./mm ²	Ok
Tubesheet shear stress	17.0 <=	118.4 N./mm ²	Ok
Stress in Shell at Tubesheet	249.5 <=	461.8 N./mm ²	Ok
Stress in Channel at Tubesheet	60.3 <=	222.0 N./mm ²	Ok

Thickness results for loadcase D2 corr. (Psd,max + Ptd,min):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	121.980	206.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	3.000	Ok

Detailed Results for load Case D3 corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2021, Original Thickness :

UHX-12.5.1 Step 1:

Compute the Equivalent Outer Tube Limit Circle Diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 524 + 19 = 1067 \text{ mm}$$

Determine the Basic Ligament Efficiency for Shear [μ]:

$$= (p - dt) / p$$

$$= (25.4 - 19) / 25.4 = 0.25$$

UHX-12.5.2 Step 2:

Compute the Ratio [ps]:

$$= Ds / Do \text{ (Configurations a, b, c)}$$

$$= 1106 / 1067 = 1.04$$

Compute the Ratio [pc]:

$$= Dc / Do \text{ (Configurations a, e, f)}$$

$$= 1106 / 1067 = 1.04$$

Moment on Tubesheet due to Pressures (Ps, Pt) [Mts]:



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$$\begin{aligned}
 &= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt] \\
 &= 1067^2/16 \cdot [(1.04 - 1) \cdot (1.04^2 + 1) \cdot 4550 - \\
 &\quad (1.04 - 1) \cdot (1.04^2 + 1) \cdot 13820] \\
 &= -49538708.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

UHX-12.5.3 Step 3, Determination of Effective Elastic Properties :

Compute the Ratio [ρ]:

$$= l_{tx}/h = 0/190 = 0 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Compute the Effective Tube Hole Diameter [d*]:

$$\begin{aligned}
 &= \max(dt - 2tt(EtT/E) (StT/S) (\rho), dt - 2tt) \\
 &= \max(19 - 2 \cdot 2.11 (178956752/178956752) \cdot \\
 &\quad (112/148) (0), 19 - 2 \cdot 2.11) \\
 &= 19.0500 \text{ mm}
 \end{aligned}$$

Compute the Effective Tube Pitch [p*]:

$$\begin{aligned}
 &= p / \sqrt{ (1 - 4 \cdot \min(Al, 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) } \\
 &= 25.4 / \sqrt{ (1 - 4 \cdot \min(419, 4 \cdot 1067 \cdot 25.4) / (\pi \cdot 1067^2)) } \\
 &= 26.0166 \text{ mm}
 \end{aligned}$$

Compute the Effective Ligament Efficiency for Bending [μ*]:

$$= (p^* - d^*)/p^* = (26 - 19)/26 = 0.27$$

E*/E and ν* for Triangular pattern from Fig. UHX-11.3.

$$h/p = 7.480315 ; \mu^* = 0.267775 ; E^* = 42821340 \text{ KPa.}$$

$$E^*/E = 0.239283 ; \nu^* = 0.379292$$

Note: As h/p (7.480) is > 2, data values for h/p = 2 were used.

UHX-12.5.4 Step 4:

Compute Shell Coefficient [βS]:

$$\begin{aligned}
 &= ((12(1 - \nu_s^2))^{1/4}) / \sqrt{ (Ds + ts)ts } \\
 &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ (1106 + 25) 25 } \\
 &= 0.0108 \text{ 1/mm}
 \end{aligned}$$

Determine Shell Coefficient [ks]:

$$\begin{aligned}
 &= \beta S \cdot Es \cdot ts^3 / (6(1 - \nu_s^2)) \\
 &= 0.011 \cdot 178956752 \cdot 25^3 / (6(1 - 0.3^2)) \\
 &= 5536431104.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Determine Shell Coefficient [λs]:

$$\begin{aligned}
 &= 6 \cdot Ds \cdot ks / h^3 \cdot (1 + h \cdot \beta S + (h \cdot \beta S)^2 / 2) \\
 &= 6 \cdot 1106 \cdot 5.54E+09 / 190^3 \cdot (1 + 190 \cdot 0.011 + (190 \cdot 0.011)^2 / 2) \\
 &= 27658284.0 \text{ KPa.}
 \end{aligned}$$

Determine Shell Coefficient [δS]:

$$\begin{aligned}
 &= Ds^2 / (4 \cdot Es \cdot Ts) \cdot (1 - \nu_s / 2) \\
 &= 1106^2 / (4 \cdot 178956752 \cdot 25) \cdot (1 - 0.3 / 2) \\
 &= 0.0580972880 \text{ mm/N./mm}^2
 \end{aligned}$$

Calculate Parameter [ωS]:

$$\begin{aligned}
 &= ps \cdot ks \cdot \beta S \cdot \delta S (1 + h \cdot \beta S) \\
 &= 1.04 \cdot 5.54E+09 \cdot 0.011 \cdot 0.058097288 (1 + 190 \cdot 0.011) \\
 &= 11004.7334 \text{ mm}^2
 \end{aligned}$$

Determine Channel Coefficient [βc]:

$$\begin{aligned}
 &= ((12 \cdot (1 - \nu_c^2))^{1/4}) / \sqrt{ (Dc + tc)tc } \\
 &= ((12 \cdot (1 - 0.3^2))^{1/4}) / \sqrt{ (1106 + 61) 61 } \\
 &= 0.0068 \text{ 1/mm}
 \end{aligned}$$

Determine Channel Coefficient [kc]:

$$\begin{aligned}
 &= \beta c \cdot Ec \cdot tc^3 / (6(1 - \nu_s^2)) \\
 &= 0.0068 \cdot 178489888 \cdot 61^3 / (6(1 - 0.3^2))
 \end{aligned}$$



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ASME TS Calc: DESIGN 1

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$$= 50555195392.0000 \text{ KPa} \cdot \text{mm}^2$$

Determine Channel Coefficient [λ_c]:

$$= 6 \cdot D_c \cdot k_c / h^3 \cdot (1 + h \cdot \beta_c + (h \cdot \beta_c)^2 / 2)$$

$$= 6 \cdot 1106 \cdot 50.56E+09 / (190^3) \cdot (1 + 190 \cdot 0.0068 + (190 \cdot 0.0068)^2 / 2)$$

$$= 153211104.0 \text{ KPa}$$

Determine Channel Coefficient [δ_C]:

$$= D_c^2 / (4 \cdot E_c \cdot T_c) \cdot (1 - v_c / 2)$$

$$= 1106^2 / (4 \cdot 178489888 \cdot 61) \cdot (1 - 0.3 / 2)$$

$$= 0.0238726456 \text{ mm/N} \cdot \text{mm}^2$$

Calculate Parameter [ω_C]:

$$= p_c \cdot k_c \cdot \beta_c \cdot \delta_C (1 + h \cdot \beta_c)$$

$$= 1.04 \cdot 50.56E+09 \cdot 0.0068 \cdot 0.023872646 (1 + 190 \cdot 0.0068)$$

$$= 19551.4453 \text{ mm}^2$$

UHX-12.5.5 Step 5:

Diameter ratio [K]:

$$= A / D_o = 1288 / 1067 = 1.21$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.38) / 42821340 \cdot (27658284 + 153211104 + 178956752 \cdot \ln(1.21))$$

$$= 3.1092$$

UHX-12.5.6 Step 6:

Moment Acting on Unperforated Tubesheet Rim [M*]:

$$= M_{ts} + \omega_C \cdot P_t - \omega_S \cdot P_s$$

$$= -49538708 + 19551 \cdot 13820 - 11005 \cdot 4550$$

$$= 170590720.0000 \text{ KPa} \cdot \text{mm}^2$$

UHX-12.5.7 Step 7:

Maximum Bending Moment acting on Periphery of Tubesheet [Mp]:

$$= ((M^*) - D_o^2 / 32 \cdot F (P_s - P_t)) / (1 + F)$$

$$= ((170590720) - 1067^2 / 32 \cdot 3.11 (4550 - 13820)) / (1 + 3.11)$$

$$= 291232704.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum Bending Moment acting on Center of Tubesheet [Mo]:

$$= M_p + D_o^2 / 64 \cdot (3 + v^*) (P_s - P_t)$$

$$= 291232704 + 1067^2 / 64 \cdot (3 + 0.38) (4550 - 13820)$$

$$= -266409232.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum Bending Moment acting on Tubesheet [M]:

$$= \max(\text{abs}(M_p), \text{abs}(M_o))$$

$$= \max(\text{abs}(291232704), \text{abs}(-266409232))$$

$$= 291232704.0000 \text{ KPa} \cdot \text{mm}^2$$

UHX-12.5.8 Step 8:

Tubesheet Bending Stress at Original Thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 291232704 / (0.27 (190 - 0)^2)$$

$$= 180.7755 \text{ N/mm}^2$$

The Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S = 2 \cdot 148 = 296 \text{ N/mm}^2$$

Tubesheet Bending Stress at Final Thickness [σ]:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 298020640 / (0.27 (150 - 0)^2)$$



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$$= 295.5720 \text{ N./mm}^2$$

Required Tubesheet Thickness, for Bending Stress [HreqB]:

$$= h + \text{Cats} + \text{Catc} = 150 + 3 + 13 = 166 \text{ mm}$$

Required Tubesheet Thickness for Given Loadings (includes CA) [Hreq]:

$$= \max(\text{HreqB}, \text{HreqS}) = \max(166, 71.4) = 166 \text{ mm}$$

UHX-12.5.9 Step 9:

$$\text{abs}(\text{Ps} - \text{Pt}) = \text{abs}(4550 - 13820) = 9270 \text{ KPa.}$$

Shear Stress check [τ limit]:

$$= 4 \cdot \mu \cdot h / \text{Do} \cdot \min(0.8 \cdot S, 0.533 \cdot S_y)$$

$$= 4 \cdot 0.25 \cdot 190 / 1067 \cdot \min(0.8 \cdot 148, 0.533 \cdot 310)$$

$$= 21.076 \text{ N./mm}^2$$

[$\text{Ps} - \text{Pt}$] is $\leq$ the Shear Stress limit (τ limit). Therefore, the required thickness for shear is for information only.

Minimum Required Thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S, 0.533 \cdot S_y)) \cdot (4 \cdot \text{Ap} / \text{Cp}) \cdot \text{abs}(\text{Ps} - \text{Pt}) + \text{Cats} + \text{Catc}$$

$$= 1 / (4 \cdot 0.25 \cdot 118) \cdot (4 \cdot 5287 / 2987) \cdot \text{abs}((4550 - 13820)) + 16$$

$$= 71.4287 \text{ mm}$$

UHX-12.5.10 Step 10:

Note: For a given Shell thickness of: 25.000mm

Min. Shell len. adjacent to the tubesheet is: 299.309 mm

Axial Shell Membrane Stress [σ_{sm}]:

$$= \text{Ds}^2 / (4 \cdot \text{ts}(\text{Ds} + \text{ts})) \cdot \text{Ps}$$

$$= 1106^2 / (4 \cdot 25 (1106 + 25)) \cdot 4550$$

$$= 49.213 \text{ N./mm}^2$$

Axial Shell Bending Stress [σ_{sb}]:

$$= 6 \cdot \text{ks} / \text{ts}^2 \cdot (\beta S \cdot \text{P's} + 6(1 - \nu^*) / (E \cdot$$

$$\text{Do} / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (\text{Mp} + \text{Do}^2 / 32 \cdot (\text{Ps} - \text{Pt})))$$

$$= 6 \cdot 1244667 / 25^2 \cdot [0.011 \cdot 0.26 + 6(1 - 0.38) / 42821340 \cdot$$

$$1067 / 190^3 \cdot (1 + 190 \cdot 0.011 / 2) \cdot (291232704 + 1067^2 / 32 \cdot$$

$$(4550 - 13820))]]$$

$$= 95.324 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= \text{abs}(\sigma_{sm}) + \text{abs}(\sigma_{sb}) \text{ (Should be } \leq 1.5 \cdot S_s)$$

$$= 144.537 \text{ N./mm}^2 \text{ (Should be } \leq 191.68)$$

Note: For a given Channel thickness of: 61.000 mm

Min. Channel len. adjacent to the tubesheet is: 467.536mm

Axial Channel Membrane Stress [σ_{cm}]:

$$= \text{Dc}^2 / (4 \cdot \text{tc}(\text{Dc} + \text{tc})) \cdot \text{Pt}$$

$$= 1106^2 / (4 \cdot 61 (1106 + 61)) \cdot 13820$$

$$= 59.372 \text{ N./mm}^2$$

Axial Channel Bending Stress [σ_{cb}]:

$$= 6 \cdot \text{kc} / \text{tc}^2 \cdot [\beta c \cdot (\delta C \cdot \text{Pt}) - 6(1 - \nu^*) / (E \cdot$$

$$\text{Do} / h^3 \cdot (1 + h \cdot \beta c / 2) \cdot (\text{Mp} + \text{Do}^2 / 32 \cdot (\text{Ps} - \text{Pt}))]$$

$$= 6 \cdot 11365519 / 61^2 \cdot [0.0068 \cdot 0.33 - 6(1 - 0.38) / 42821340 \cdot$$

$$1067 / 190^3 \cdot (1 + 190 \cdot 0.0068 / 2) \cdot (291232704 + 1067^2 / 32 \cdot$$

$$(4550 - 13820))]]$$

$$= 253.785 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= \text{abs}(\sigma_{cm}) + \text{abs}(\sigma_{cb}) \text{ (Should be } \leq 1.5 \cdot S_c)$$



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= 313.157 N./mm² (Should be ≤ 222.01)

Step 11, The Cylinder-to-Tubesheet Juncture is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify Es and/or Ec and recompute from Step 4 onwards.

Modified Elasticity [Ec]:

$$= E_c \cdot \sqrt{(1.5 \cdot S_c / \sigma_c)}$$

$$= 178489888 \cdot \sqrt{(1.5 \cdot 148/313)}$$

$$= 150285472.0 \text{ KPa.}$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu \cdot (h - h'g)^2)$$

$$= 6 \cdot 287609472 / (0.27 \cdot (190 - 0)^2)$$

$$= 178.5 \text{ N./mm}^2 \text{ (Should be } \leq 296.0 \text{)}$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.

The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [HreqB]:

$$= h + C_{ats} + C_{atc} = 150 + 3 + 13 = 166 \text{ mm}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [Hreq]:

$$= \max(H_{reqB}, H_{reqS}) = \max(166, 71.4) = 166 \text{ mm}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (19 - 2.11) 2.11$$

$$= 1.123 \text{ cm}^2$$

Tube Weld Size Results per UW-20:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S_a$$

$$= \pi \cdot 2.11 (19 - 2.11) 112$$

$$= 1279.905 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 3 (19 + 0.67 \cdot 3) 112$$

$$= 1279.9048 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S_a / S_w \text{ where } S_w = \min(S_a \text{ and } S_t)$$

$$= 112/112$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 3 (19 + 0.67 \cdot 3) 112$$

$$= 1244.2972 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 \cdot \sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o$$

$$= 2 \cdot \sqrt{((0.75 \cdot 19)^2 + 1.07 \cdot 2.11 \cdot (19 - 2.11) 1 \cdot 1)} - 0.75 \cdot 19$$

$$= 2.562 \text{ mm}$$



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ASME TS Calc: DESIGN 1

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Minimum Required Fillet Weld Leg [afr]: 1.2810 mm

Minimum Required Groove Weld Leg [agr]: 1.2810 mm

Tube-Tubesheet Jt allowable, 1280 is >= tube strength 1280 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	178.5 <=	296.0 N./mm ²	Ok
Tubesheet shear stress	34.5 <=	118.4 N./mm ²	Ok
Stress in Shell at Tubesheet	144.5 <=	191.7 N./mm ²	Ok
Stress in Channel at Tubesheet	313.2 <=	458.0 N./mm ²	Ok

Thickness results for loadcase D3 corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	166.312	206.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	3.000	Ok

U-Tube Tubesheet results per ASME UHX-12 2021

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	188.515 ...	239	296	47 118	Ps+Pt D1	Ok
D2uc	113.953 ...	82	296	16 118	Ps+Pt D2	Ok
D3uc	159.231 ...	167	296	32 118	Ps+Pt D3	Ok
D1c	195.113 ...	262	296	51 118	Ps+Pt-c D1	Ok
D2c	121.980 ...	90	296	17 118	Ps+Pt-c D2	Ok
D3c	166.312 ...	179	296	35 118	Ps+Pt-c D3	Ok

Max: 195.1129 ... mm 0.884 0.435 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Req'd. Thk. Shell Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc		82	192	329	458	Ok



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ASME TS Calc: DESIGN 1

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D2uc	...	...	217	462	47	222	Ok
D3uc	...	...	135	192	281	458	Ok
D1c	...	...	105	192	373	458	Ok
D2c	...	...	249	462	60	222	Ok
D3c	...	...	145	192	313	458	Ok

Max ... 0.754 0.815

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	195.113	206.000	Ok
Tube Thickness :	1.122	2.110	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	3.000	Ok

Given Shell Thickness: 28.0000 mm

Given Channel Thickness: 64.0000 mm

Min Shell length of thk, (28.000) adj. to tubesheet: 315.899 mm

Min Channel length of thk, (64.000) adj. to tubesheet: 477.594 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	15494.84	0.996	14976.69	1.000
Tubesheet Shear Stress	31775.93	1.000	31775.93	1.000
Shell Stress (Axial, Junction)	31775.93	0.522	8422.45	1.000
Tubesheet-Channel Junction	15494.84	0.996	8422.45	0.567
Tube Pressure Stress	27167.39	1.000	11601.69	1.000
Minimum MAWP	15494.84		8422.45	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	16859.78	1.000	16378.19	1.000
Tubesheet Shear Stress	34451.80	1.000	34451.81	1.000
Shell Stress (Axial, Junction)	34451.80	0.932	12964.76	1.000
Tubesheet-Channel Junction	16859.78	1.000	12964.76	0.800
Tube Pressure Stress	28678.83	1.000	19233.21	1.000
Minimum MAPnc	16859.78		12964.76	

(*) All load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):



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ASME TS Calc: DESIGN 1

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= max(tubesheet thk/4, min(tubesheet thk, shell thickness))
= max(206/4, min(206, 28))
= 51.500 mm

MDMT reduction ratio per UCS 66 (b)(1)(b):

= min(ps/Shellside MAPnc, 1)
= min(4550/12965, 1)
= 0.351

Thickness Ratio = 0.35, Temperature Reduction per Fig. UCS 66.1 = 77 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	37 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-40 °C

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):

= max(tubesheet thk/4, min(tubesheet thk, channel thickness))
= max(206/4, min(206, 64))
= 64.000 mm

MDMT reduction ratio per UCS 66 (b)(1)(b):

= min(pt/Tubeside MAPnc, 1)
= min(13820/16860, 1)
= 0.820

Thickness Ratio = 0.82, Temperature Reduction per Fig. UCS 66.1 = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	41 °C
Min Metal Temp. at Required thickness (UCS 66.1)	31 °C

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ASME TS Calc: DESIGN 2

Case: 2 9:43a Apr 21,2026

Input Echo, Tubesheet Number 2, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	4550.00	KPa.
Shell Maximum Operating Pressure	Psox,max	4550.00	KPa.
Shell Minimum Operating Pressure	Psox,min	4009.00	KPa.
Shell Thickness	ts	28.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	caext	0.0000	mm
Inside Diameter of Shell	Ds	1100.000	mm
Shell Temperature for Internal Pressure	Ts	454.00	°C
Shell Material		SA-387 22	

Using 2 * Yield for Discontinuity Stress Allowable (UG-23(e)), Sps.

Make sure that material properties at this temperature are not time-dependent for Material: SA-387 22

Shell Material UNS Number		K21590	
Shell Allowable Stress at Temperature	Ss	127.79	N./mm ²
Shell Allowable Stress at Ambient		148.01	N./mm ²

Channel Description: Channel shell

Channel Type:		Cylinder	
Channel Maximum Design Pressure	Ptd,max	13820.00	KPa.
Channel Maximum Operating Pressure	Ptox,max	13820.00	KPa.
Channel Minimum Operating Pressure	Ptox,min	12500.00	KPa.
Channel Thickness	tc	64.0000	mm
Channel Corrosion Allowance	cac	3.0000	mm
Inside Diameter of Channel	Dc	1100.000	mm
Channel Design Temperature	TEMPC	358.00	°C
Channel Material		SA-387 11	

Using 2 * Yield for Discontinuity Stress Allowable (UG-23(e)), Sps.

Make sure that material properties at this temperature are not time-dependent for Material: SA-387 11

Channel Material UNS Number		K11789	
Channel Allowable Stress at Temperature	Sc	148.01	N./mm ²
Channel Allowable Stress at Ambient		148.01	N./mm ²

Tube Data:

Number of Tube Holes	Nt	750	
Tube Wall Thickness	et	2.1100	mm
Tube Outside Diameter	D	19.0500	mm
Total Straight Tube Length	Lt	3650.00	mm
Straight Tube Length (bet. inner tubsht faces) L		3444.00	mm
Design Temperature of the Tubes		454.00	°C
Tube Material		SA-213 T22	
Tube Material UNS Number		K21590	
Is this a Welded Tube		No	
Tube Material Specification used	Smls. tube		
Tube Allowable Stress at Temperature		111.78	N./mm ²
Tube Allowable Stress At Ambient		118.00	N./mm ²
Tube Yield Stress At design Temperature	Syt	180.52	N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.4000	mm
Tube Layout Pattern		Triangular	
Fillet Weld Leg	af	3.0000	mm
Groove Weld Leg	ag	3.0000	mm
Tube-Tubesheet Joint Weld Type		Full Strength	
Method for Tube-Tubesheet Jt. Allow.		UW-20	
Is the Tube Inset		No	



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ASME TS Calc: DESIGN 2

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Radius to Outermost Tube Hole Center	ro	524.160	mm
Largest Center-to-Center Tube Distance	U1	57.1500	mm
Length of Expanded Portion of Tube	ltx	0.0000	mm
Tube-side pass partition groove depth	hg	8.0000	mm
Area of Tube Layout	Ap	5286.8	cm ²
Perimeter of Tube Layout	Cp	2987.3	mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature	T	454.00	°C
Tubesheet Material		SA-336 F22	

Using 2 * Yield for Discontinuity Stress Allowable (UG-23(e)), Sps.
Make sure that material properties at this temperature are not
time-dependent for Material: SA-336 F22

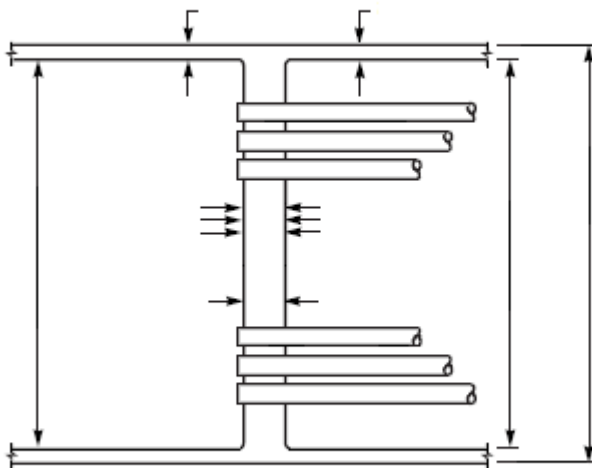
Tubesheet Material UNS Number		K21590	
Tubesheet Allowable Stress at Temperature	S	148.00	N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	148.00	N./mm ²
Thickness of Tubesheet	h	206.0000	mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000	mm
Tubesheet Corr. Allowance (Channel side)	Catc	13.0000	mm
Tubesheet Outside Diameter	A	1288.000	mm

Area of the Untubed Lanes	AL	419.1	cm ²
---------------------------	----	-------	-----------------

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	NO
Run Multiple Load Cases	YES

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX, 2021

Elasticity/Expansion Material Properties :



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ASME TS Calc: DESIGN 2

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Shell - TM-1 Material Group D

Elastic Mod. at Design Temperature	454.0 °C	0.17896E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.21097E+09 KPa.

Channel - TM-1 Material Group C

Elastic Mod. at Design Temperature	358.0 °C	0.18178E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20408E+09 KPa.

Tubes - TM-1 Material Group D

Elastic Mod. at Tubsht. Design Temp.	454.0 °C	0.17896E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.21097E+09 KPa.

TubeSheet - TM-1 Material Group D

Elastic Mod. at Design Temperature	454.0 °C	0.17896E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.21097E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (13820 \cdot (19/2 - 0)) / (112 \cdot 1 + 0.4 \cdot 13820)$
 $= 1.1222 + 0.0000 = 1.1222 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 454.00 °C
 Elastic Modulus for Material 150421952.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	3968.16	9.03	50.0000	0.0134948	74.03

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 11602 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.0779	19.05	3968.16	17.67	50.0000	0.0035217	60.32

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 60.3) / (3 \cdot 17.7) = 4550 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.1222 mm
Allowable Internal Pressure at Corroded thickness	27167.40 KPa.
Required Internal Design Pressure	13820.00 KPa.
Allowable External Pressure at Corroded thickness	11601.70 KPa.
Required External Design Pressure	4550.00 KPa.
Required Thickness due to Shell Side pressure	1.0779 mm

U-Tube Tubesheet results per ASME UHX-12 2021

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet Stresses				Case Type	Pass/Fail
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd		
D1uc	188.638	...	240	296	47	118	Ps+Pt D1	Ok
D2uc	114.013	...	82	296	16	118	Ps+Pt D2	Ok
D3uc	159.309	...	168	296	32	118	Ps+Pt D3	Ok
D1c	195.251	...	261	296	51	118	Ps+Pt-c D1	Ok
D2c	122.057	...	90	296	17	118	Ps+Pt-c D2	Ok
D3c	166.399	...	179	296	35	118	Ps+Pt-c D3	Ok



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Max: 195.2510 ... mm 0.882 0.435 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	Integral Cyl. Req'd. Thk. Shell	Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	81	192	330	473	Ok
D2uc	...	...	217	462	48	222	Ok
D3uc	...	...	136	192	282	473	Ok
D1c	...	...	104	192	375	473	Ok
D2c	...	...	249	462	61	222	Ok
D3c	...	...	145	192	314	473	Ok
Max	...	...		0.757		0.791	

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	195.251	206.000	Ok
Tube Thickness :	1.122	2.110	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	3.000	Ok

Given Shell Thickness: 28.0000 mm

Given Channel Thickness: 64.0000 mm

Min Shell length of thk, (28.000) adj. to tubesheet: 315.899 mm

Min Channel length of thk, (64.000) adj. to tubesheet: 477.594 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	15518.62	0.996	15010.57	1.000
Tubesheet Shear Stress	31775.93	1.000	31775.93	1.000
Shell Stress (Axial, Junction)	31775.93	0.517	8438.58	1.000
Tubesheet-Channel Junction	15518.62	0.996	8438.58	0.567
Tube Pressure Stress	27167.39	1.000	11601.69	1.000



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Minimum MAWP	15518.62		8438.58	
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Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	16859.77	1.000	16378.19	1.000
Tubesheet Shear Stress	34451.80	1.000	34451.81	1.000
Shell Stress (Axial, Junction)	34451.80	0.932	12964.76	1.000
Tubesheet-Channel Junction	16859.77	1.000	12964.76	0.800
Tube Pressure Stress	28678.83	1.000	19233.21	1.000
Minimum MAPnc	16859.77		12964.76	

(*) All load cases were analyzed to compute the MAWP for determining the test pressure.

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Problems/Failures Summary:

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: The net force on one of the saddles putting it in tension!**

Please review all reports carefully!

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